An Executable Observer for Observer Patch Holography: Specification, Executed Components and Ordered Joins

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Abstract

Observer Patch Holography (OPH) is made testable by its executable-model hypothesis: a fully specified local repair and memory law on the proposed carrier architecture admits continuing observers whose records have the observed quantum-field and gravitational descriptions. This paper is the base of that hypothesis. It specifies the twelve-port icosahedral carrier, the native C/E seam repair with its exactly classified family of quantum record laws, the finite observer algebra, the declared services and the scheduler, and reports the finite components executed on that specification. Uniform seam attempts have exact quantum gap $(5-\sqrt{5})/60$; a record-steered observer moves a final occupation from 1/6 to 35/264; nine source-executed measurements identify every coordinate of the record-law family. Overlapping observers read fields and event clocks from one continuing history, and one register joins quantum electric flux, charged fermions and a scalar with a feedback response certified nonzero at infinite occupation. On the refining spherical source one repair history defines a continuum operator of spectral dimension two governing Gaussian density histories read by native incident clocks; a declared radial extension gives a smooth $1+3$ free-scalar description of every fixed finite observer menu. A specified coherent memory protects exchange-history sectors; complete source factors further join fermionic propagation, Higgs-driven particle creation, native repair and one calibrated clock. Their full outcome and reference records are retained. Physical clock/energy validation, localized particle identity, an interacting chiral $3+1$ limit and realized Einstein premises are not established. Four negative controls remain: the native double-load defect decays with survival $(5/6)^n$, the static radial product fails scalar null balance, passive repair strictly attenuates coherence and has no wave sector, and the tested mass laws miss the charged hierarchy.

One theory and its executable hypothesis. OPH proposes closure: one complete history is both a timeless self-referential structure and its own internal computation, and observers inside it recover and realize its specification. The hypothesis that makes this proposal concrete and testable is executable.

> *A simple, fully specified local repair and memory law on the proposed carrier architecture admits continuing observers whose compatible records and interventions have the observed $3+1$-dimensional quantum-field description and a common classical-gravity limit. The preferred matter mechanism is localized topological defects of native overlap transport, with rest energy derived from their excess maintenance dynamics in the same physical clock and energy dictionary.*

One source process must meet every part of this statement: records, clocks, matter, energy and responsive geometry come from the same law, state and refinement. Coherence, protected codes, radial directions, fields, clocks and couplings may be specified as part of its design; their implementation and observable consequences must be established on that process. A failed specified law rejects that law and its tested interpretation; it rejects the wider proposal only when the rejected choice is proved necessary.

The tower. The papers form one tower over the executable model. Each layer states a hypothesis and proves finite theorems under displayed premises. A conclusion passes to the next layer only after that premise is proved on the same model process; until then it is a conditional implication.

0. Closure and agreement. One history reads itself into existence, and agreement on shared records defines the public world. This is the interpretation and acceptance criterion; no theorem selects a unique universe.

1. Repair mathematics. Local agreement repair reaches global consistency, and nonzero cycle holonomy is the residue that local repair cannot remove. Fixed-point, normal-form, frustration and finite event-algebra theorems hold on their stated domains.

2. Carrier and repair law. Federated twelve-port icosahedral carriers with the native C/E seam repair are the substrate. The seam rule follows from four stated requirements, the quantum record laws are classified, and the uniform native repair selects a rank-three slow-response band.

3. Executable observer. Observers are continuing native processes that prepare, read, retain and act on records. Finite complete-output executions, overlapping-observer correspondence, event clocks and checkpoint continuation are verified; one continuing observer with every service and resource accounted is the open join.

4. Clocks and fields. Modular flow of observer algebras supplies candidate local clocks, and native repair supplies a positive continuum generator of spectral dimension two on the screen. A declared radial extension gives a conditional smooth $1+3$ scalar description. Complete finite source factors join spatial CAR dynamics, particle creation, repair and a common calibrated clock. Physical validation of that clock and energy normalization, and an interacting chiral $3+1$ limit, remain open.

5. Gauge structure and matter. Gauge structure is the transport of overlap data, and stable massive matter is a localized defect whose boundary transport carries a holonomy that admitted repairs cannot erase. The compact Lie type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ follows under complete-response premises, and fixed nonzero holonomy forces positive residual mismatch. A specified coherent ordering memory protects native exchange-history sectors, and a colored extension implements their fermionic exchange. A localized physical defect, its maintenance-to-mass law and a predicted measured spectrum remain unestablished; the earlier double-load candidate is transient.

6. Gravity. Einstein dynamics is the coarse response of the same process. The null-balance and tensor-completion implication is proved under its premises; realizing those premises on one refinement tower is open, and the static radial product fails null balance.

7. Cosmology and tests. The same process must fix the vacuum, dark sector, primordial response and black-hole records, and pass independent prospective tests. The cosmology pa-

pers and the fine-structure, Koide, de Sitter and Yang–Mills supplements carry conditional branches, diagnostics and retrospective comparisons at this layer.

Completion and stopping. The aim is a consistent and complete theory: every layer realized on one process, or a proved obstruction that removes OPH or a stated branch of it. Results that no join needs leave the active chain with their provenance. A universal obstruction over a frozen admissible class, an unavoidable excluded prediction, or a contradiction among indispensable premises stops that class. A failed candidate, an unconstructed map or a resource stop has a narrower meaning. The ordered joins, retained foundations and stopping rules are recorded with the executable model in the public repository.[1]

Fixed physical boundaries. Repair counts, host runtime and entropy production are distinct from energy; the maintenance-to-energy map must be derived with one common signal speed and no per-species conversion. The modular parameter is dimensionless until a specified and implemented clock calibration relates it to duration. Stationary repair or record entropy is distinct from the fixed-volume generalized-entropy variation that the Einstein construction consumes. A massless photon makes the infinite-volume theory gapless, so sector gaps and thresholds name their state and limit; confinement precludes free massless gluons as observable particles; a quantum graviton lies outside the classical Einstein theorem.

From repair topology to a field experiment. A specified repair process admits protected quantum history sectors, fermionic exchange and an executable coupling to field and clock observables. Its ingredients are specified observer, scheduler, repair and memory laws; they need not be uniquely selected to define a mathematical model. What must be established is that the resulting operations share a carrier, retain complete quantum outputs and reproduce the intended observables. The constructions below establish several such joins, including a joint source experiment. They do not identify a complete physical theory.

Protected transport sectors. On the native 66-state binary two-load configuration complex, an exact noncommutative reduction of the face and disjoint-move relations gives $\pi_1 = \langle g \mid g^2 \rangle = \mathbb{Z}/2$. The surviving loop has odd ordered-cover parity. This is topology of the declared transport and deformation domain, not topology of a localized field defect. The complete active branch $*$-algebra on either the unextended 78-state source or its binary shell is the full matrix algebra. A Hermitian source charge commuting with every branch is therefore scalar. This excludes that charge class, not every invariant subspace, expectation-conserved observable or controlled lifetime.

A coherent ordering bit changes the algebra. If a native branch maps x to $f_h(x)$, let $p_h(x)$ be the parity of the transported order and define

$$K_h = \frac{1}{\sqrt{2}} \sum_{x \in D_h} |f_h(x)\rangle\langle x| \otimes X^{p_h(x)}, \qquad Q = I_{66} \otimes X.$$

Every branch commutes with Q, so its two sharp sectors survive arbitrary finite record-adaptive schedules. An actual odd exchange loop returns the source and applies X to the bit: the memory participates in transport. Sector preservation does not assert preservation of every coherence between the sectors. For the all-seam C or E family, effect separation, a connected movement tree

[1] https://github.com/FloatingPragma/observer-patch-holography/blob/r2058/docs/COMPLETION_PROGRAM.md

and the odd loop generate exactly $M_{66}(\mathbb{C}) \oplus M_{66}(\mathbb{C})$, with commutant $\mathrm{span}\{I, Q\}$, although the classical ordered cover is connected. The states $I_{66}/66 \otimes |\pm\rangle\langle\pm|$ have identical source and complete classical-record marginals; their distinction is quantum memory. Computational-basis reset or dephasing destroys it. Protection thus depends on the specified memory law preserving these sectors.

Literal source words implement this lift. Complete execution covers all 132 input columns on seam $(0, 5)$, and exact word audits cover all 1024 addresses of the 90 seam/current feedback words. Native outcomes, padding and source-service failures remain in the accounting. The result is a protected sector of the enlarged process; a spatially localized source realization of its charge remains an additional construction.

This same carrier also separates operator action from scalar repair cost. Writing $P_- = (I - X)/2$ and $M_h = \sum_{x \in D_h} |f_h(x)\rangle\langle x|$,

$$A_h = \pi \sum_{x \in D_h} p_h(x) |f_h(x)\rangle\langle f_h(x)| \otimes P_-, \qquad K_h = e^{-iA_h}(M_h \otimes I)/\sqrt{2}.$$

Two closed histories with identical multisets of complete branch labels and identical compiled resource totals induce respectively I and X. An additive scalar cost of those labels cannot determine this response; the displayed operator action does. Its phase is not physical energy per unit time. Nor is the reduced memory autonomous: its one-step flip probability is $26/495$, while the two-step Z multiplier differs from the squared one-step multiplier by $398/245025$. The continuing source must be retained in any proposed Hamiltonian or memory description.

Complete native occupation protection. For the bare twelve-port binary endpoint law, all 4096 occupations give the fixed algebra

$$\mathcal{A}' = \mathbb{C}^{11} \oplus M_2(\mathrm{span}\{|0\rangle, |\mathrm{full}\rangle\}).$$

Exact native effects and signed transport prove this for both C/E laws and exchange characters. A faithful weighted reference gives the full stationary family and fixed-selector limit. Every proper-port-local fixed observable is scalar. The only nontrivial passive quantum multiplicity is the global empty/full span; it has distance one and requires cross-number coherence outside fixed-total V2. Sixteen complete source histories test its preservation and local-error boundary. This classification concerns the binary endpoint instrument, not the distinct protected internal factors or outside-projector field repairs below. It identifies no physical particle or mass; the proof and complete-output checks are in `model/native_fixed_algebra/OCCUPATION.md`.

Internal states and fermionic exchange. Give each occupied port an independent internal space V. The map

$$|\{x < y\}; a, b\rangle \longmapsto c^\dagger_{x,a} c^\dagger_{y,b} |0\rangle$$

fixes the CAR convention. Passing the occupied spectator reverses the internal slots and crosses one creation operator, hence

$$K_h^V = \frac{1}{\sqrt{2}} \sum_{x \in D_h} |f_h(x)\rangle\langle x| \otimes (-\mathrm{SWAP})^{p_h(x)}.$$

The complete effects remain the original domain projectors divided by two. Independent CAR calculations check every branch for $\dim V = 2, 3$; the crossing argument applies to every finite V.

The odd loop acts as $-$SWAP, giving -1 on two equal internal states. Global Fock parity on this two-particle sector is $+1$ and cannot replace that exchange.

The ordering memory implements the same operation: for $J = -\mathrm{SWAP}$,

$$E_V\psi = (|0\rangle\psi + |1\rangle J\psi)/\sqrt{2}, \qquad (X \otimes I)E_V = E_V J.$$

Every native branch, and every common $U \otimes U$, intertwines through E_V. The negative exchange character acts on $\mathrm{Sym}^2 V$; the positive one acts on $\Lambda^2 V$. Both are compatible with identical fermions carrying independent internal states.

The finite algebra constrains but does not fix internal encodings. An untouched fiber of dimension m enlarges the repair commutant to $M_m \oplus M_m$. A separate collective-label encoding that puts all fifteen states of $\Lambda^2\mathbb{C}^5 \oplus \overline{\mathbb{C}^5}$ in the negative sector requires $m \geq 15$ and has an explicit inclusion at equality. This is not a universal memory bound for the two-particle construction. No element of the supplied $S(U(3) \times U(2))$ gauge group acts as $-I$ on the whole faithful fifteen-state module: its nonidentity involutions have signature $(7, 8)$. The smaller weak-parity encoding is consequently an internal-grading control, not an all-fermion exchange assignment.

An exact basis permutation joins that exterior module and all twelve currents to the retained chiral action. Specified two-component and three-family factors give $\dim V = 90$; neither family count nor physical Lorentz spin is derived by this basis change. On twelve ports the hard-core two-particle dimension is 534,600, whereas the complete 1080-mode two-particle space has dimension 582,660. The remaining 48,060 states are same-port collisions and must be included when propagation or interactions can reach them.

One calibrated action and its observables. The Weyl–Higgs action uses explicit charged Yukawa, Higgs-background and CKM inputs with recorded matching conventions. Recovering them from its response verifies implementation; it is not an independent mass prediction. The creation-pairing matrix $\Delta = \epsilon_{\mathrm{spin}} \otimes \overline{K}$ is antisymmetric on 90 modes. At rest,

$$B_0 = \begin{pmatrix} 0 & \Delta \\ \Delta^\dagger & 0 \end{pmatrix}, \qquad H = E_{\mathrm{vac}} + \sum_{a=1}^{90} E_a \gamma_a^\dagger \gamma_a.$$

The canonical chart retains 84 positive charged directions, six zero-energy neutrino annihilators and every particle–hole partner. Choosing the annihilator half of the zero block specifies the static reference; it does not supply a positive-energy continuation at nonzero momentum. This minimal action has massless neutrinos; a separate neutrino mass extension is not silently included.

Copying this internal system to twelve ports gives 1080 CAR modes. The static hard-core two-quasiparticle sector supports the colored construction, whereas a bare two-particle sector need not survive pairing. The compiled static convention subtracts $11E_{\mathrm{vac}}$ from the replicated vacuum scalar $12E_{\mathrm{vac}}$; the remaining common E_{vac} does not affect fixed-sector Ramsey gaps. Every extension across particle-number sectors must specify the same reference convention explicitly.

Complete source-controlled Ramsey observations recover an energy difference between equal-charge modes, with an opposite-charge partner. A separate neutral-ancilla phase probe reads the same single-particle charge, retaining all seven charge values and all pointer outcomes. Neither measures an absolute rest energy. The common electromagnetic phase cancels on the neutral pair and is not an electromagnetic-response signal. These executions retain all 128 padded internal addresses, native records and quantum output factors.

Currents are transformed with the Nambu chart, including their anomalous blocks. The neutral Higgs background has a nine-dimensional unbroken stabilizer, color and electromagnetism; three

complementary gauge directions are broken. General weak operations cannot therefore be represented solely by $U \otimes U$ within fixed quasiparticle number. A time-dependent Higgs background requires the complete CAR domain.

Spatial dynamics, clock and particle creation. A Hamiltonian cycle of the actual twelve-port seam graph supplies a finite spatial ring. With specified spacing and $c = 1$ units, its bare action is

$$h = -D \otimes \sigma_z \otimes I_{45}, \qquad D_{r,r+1} = -i/(2a), \quad D_{r+1,r} = i/(2a),$$

with the same local pairing matrix. The static symbol satisfies $E(k)^2 = \sin^2(k)/a^2 + E(0)^2$. All twelve momenta, all 2160 Nambu directions and both kinetic zeros remain. This is a one-dimensional witness, not a chiral $3 + 1$-dimensional limit. Its 1080 CAR modes retain every occupation sector, including collisions. The pairing graph splits into closed internal components of sizes one, two and six, so the largest spatial component has 72 modes. Complete four-state Fock source gates and fermionic routing determine their full Gaussian unitary maps without allocating arbitrary 2^{1080}-component vectors. A source-prepared localized excess and its bond current have complete occupation readouts, with purifiers retained. The executed finite symmetric product is specified as such; its error relative to the summed Hamiltonian is separate from its source-gate error.

Two source-prepared clock hands, each with faithful eigenvalues $1/6, 5/6$ and a retained purifier, share the reference energy $\Omega_{\mathrm{ref}} = 1/4\,\mathrm{GeV}$. Both hands and every field factor use $\lambda_j = 1 + j/4$ for the actual native current. Complementary readouts infer $\tau_{\mathrm{eff}} = \sum_h \lambda_{j_h} \Delta\tau$ on a declared unaliased interval. The inverse receives complete probabilities and error bounds, not the programmed duration or a species mass. Rational enclosures and finite-shot bounds distinguish ensemble inference from a single clock reading. The energy-unit calibration and $\hbar = h/(2\pi)$ set seconds; repair counts and host runtime do not set this clock.

The particle-changing drive follows the actual common Higgs field $H = (0, ve^{i\phi})$ with fixed gauge background. Its creation phases have opposite signs in the up and down/charged-lepton channels, according to their conjugate-Higgs tensors. In a compatible static chart it comprises 42 neutral two-mode pairs and six massless spectators per site, with

$$H_{E,\phi} = E\cos\phi\,\mathrm{diag}(-1, 0, 0, 1) + E\sin\phi\big(|00\rangle\langle 11| + |11\rangle\langle 00|\big).$$

Literal source gates retain each complete four-state space and all nonadjacent CAR signs. Across twelve sites this gives 504 pairs and 72 spectator modes. An externally driven quadratic field is not a quantized interacting gauge–Higgs theory.

Repair itself extends to every occupation sector. Binary endpoint occupancies follow the original C/E rule; nonbinary endpoints are held. A coherent extension includes that identity block in the existing zero-current Kraus operator. The binary and nonbinary output ranges are orthogonal, so the complete partial-injection effects sum to identity. On states with even occupancy at both endpoints the unread channel is exactly the identity, including coherences and references. The extension is unital before arbitrary record-conditioned controls; its faithful tracial reference has trivial modular flow. General completeness is algebraic. Literal source dilations verify the bounded collision and spectator-crossing cases, not a full 1080-mode dilation.

A separately flagged hold is retained as a control. It measures a vacuum/pair distinction that the coherent hold preserves. If A is the probability of no created pair on the repaired sites, a drive–repair–inverse echo returns $A^2 + (1 - A)^2$ for the flagged policy and one for the coherent policy on this even-endpoint domain. For the stated drive at $\tau = 0.002\,\mathrm{GeV}^{-1}$, $j = 0$ and $\phi = \pi/2$,

the flagged return is 0.63176674. Coherent collision holding does not remove backaction from the original current or E-parity records.

A joint source experiment. Selected source factors test repair, propagation, particle creation and clock readout together. Four active CAR modes on two sites carry the same charged component. A spatial superposition is source-prepared with a path reference and an opposite-charge spectator at a third site. The quasiparticle-to-bare chart is fixed by both static and driven terms; diagonalizing the static mass alone does not suffice. Each of two events applies complete native repair, routed bare hopping in that chart, and the local Higgs factors. The same current advances the clock without reset. All 16 active states, spectator states, references and clock purifiers remain, including states with three active particles.

For both C/E laws and both collision policies, all $32^2 \times 4$ outcomes are retained: 16,384 in total. Complete effects are checked on the 64-dimensional field/clock input algebra, together with the untouched reference marginal. The coherent C process has collision population 0.23404814 before the second repair and final three-particle probability 0.53649765, versus 0.36223079 for the flagged control. Their unread final-state trace distance is 0.20709787. E distinguishes the relevant odd binary occupancies, so the two policies agree on this witness within numerical admission. The composed all-outcome total-variation bound is 4.200002×10^{-7}, conditional on the full-column numerical source checks. It is not an interval-certified hardware bound. Boundary hops out of the selected sites are omitted: this experiment joins actual source factors but is not an invariant two-site restriction of the full spatial field.

Energy and modular observations on the same field. The selected four-mode factors admit energy and reference-state calculations on their complete 16-state algebra. Let H be the sum of the static hopping and Higgs terms, with its vacuum scalar retained. The executed ordered product is not silently replaced by e^{-itH}. An explicit piecewise control schedule reproduces that product, separates switching work from energy transfer during complete repair, and uses the same clock duration. The latter transfer is not assigned to heat or controller work without an apparatus model. On the spatial action, onsite terms and half of each incident bond define

$$H = \sum_x h_x, \qquad J_{xy} = i[h_x, h_y] = -J_{yx}, \qquad \frac{dh_x}{dt} = -\sum_y J_{xy} + \partial_t h_x.$$

These identities hold on all occupation sectors. Equality of full unitaries after a scalar energy-reference shift requires the corresponding global source phase; density evolution, currents and gaps do not determine that scalar. Complete, charge-preserving source instruments on all 64 apparatus input columns reconstruct energy, drive-power and ring-current expectations from separate quadratic-group settings. This is expectation readout, not a projective measurement of the summed H.

For the declared $\beta = 1\,\mathrm{GeV}^{-1}$, the same static H gives a faithful $\rho = e^{-\beta H}/Z$. A complete eight-qubit source word prepares its field–purifier state. Signed CAR restrictions and compatible Petz observations compose on every matrix unit, and a native recovery isometry retains all four regional inputs, sixteen field states and four environment states. This is a new Gibbs-input experiment with the same field instrument, not a replacement of the earlier localized input. The modular convention is

$$\rho^{is} A \rho^{-is} = e^{-i\beta s H} A e^{i\beta s H} = \tau_{-\beta s}(A);$$

the dimensionless parameter s does not supply a second physical clock.

The observation is not a regional retraction. The cross-boundary hopping has vanishing partial traces and squared Hilbert–Schmidt norm exactly 4 in the specified units. Its nonadditivity makes the Gibbs density nonproduct for $\beta > 0$, obstructing a reference-preserving UCP retraction onto the full site factor. Compatible non-idempotent Petz observations remain valid. Nor is this Gibbs density stationary under the driven, repaired history. The complete quantum entropy balance is

$$S(\sigma) - S(\rho) = \beta \operatorname{Tr}[H(\sigma - \rho)] - D(\sigma\|\rho)$$

for normalized states; retaining classical records gives an identity involving conditional relative entropies. Numerical trace drift contributes $(\operatorname{Tr}\sigma - 1)\log Z$, and the quadratic Bregman remainder subtracts $\epsilon \operatorname{Tr}\delta$ from $D(\rho + \epsilon\delta\|\rho)$. These reported entropies concern the unnormalized continuing component. First-export abort weights and output families are retained separately; their entropy is not included. A geometric cap kernel, local stress dictionary and generalized-entropy stationarity do not follow from these finite identities.

The Weyl/contact branch and its finite premise closures. The following constructions use the supplied Weyl/contact action with $c = 1/8$ and $U = 1$, separately from the calibrated Weyl–Higgs field above. Its dimer and plaquette restrictions retain their actual reference states. The charged-block service is first defined on decoupled dimers. The three-dimensional cube below instead uses bipartite eta symmetry to join a larger protected interacting sector to the complete field. The common local coefficients do not identify distinct global domains.

A complete charged internal sector. The same lower charged dimer doublets admit a sharper finite native-record attachment than the fixed-column dictionary. Retain both internal columns at each of two occupied blocks among twelve identical dimers, with the same even half-filled ground at every other block. Their carrier has dimension $66 \cdot 4 = 264$. For the declared graded whole-block repair, a movement that crosses the spectator acts internally as minus the swap; a hold acts as the identity. In the singlet–triplet basis every complete native branch is therefore

$$K_{\text{spinful}} = K_+ \otimes I_1 \ \oplus \ K_- \otimes I_3.$$

An integer certificate checks all 720 raw C/E slots and 31,680 internal input columns, including zero slots. The triplet qutrit and its untouched reference correlations survive every admitted complete history without intermediate recovery. For initially correlated spatial/internal inputs, this assertion concerns the complete internal marginal; conditioning on a record can steer it.

The raw effects separate all 66 positions. A connected transition tree and an odd exchange loop then classify the entire passive source commutant as $I_{66} \otimes (M_1 \oplus M_3)$, of complex dimension ten. Thus the three-dimensional factor is maximal under the full branch grammar; singlet–triplet coherences do not give a protected four-state factor. Positive all-seam coverage and the checked invariant maximally mixed state supply the corresponding fixed-algebra premises. This enlarged field carrier is distinct from the bare H66 source with scalar fixed algebra.

Complete extension and an exact spectral exclusion. Let P be the sum of the orthogonal two-doublet/remaining-ground block projectors. The native outcomes have total effect P; adjoining the outside outcome $I - P$ gives a trace-preserving instrument on the full 48-mode Fock space. This is a formal operator construction, with bounded checks on all 4,096 states of three dimers, not a 48-mode source execution. Every definite occupied pair has the unchanged ground on its entire exterior. With the original supplied $K_j = H_j - N_j/2$, $c = 1/8$ and $U = 1$, the same defect has

$$\Delta = \frac{\sqrt{17}}{8} - \frac{1}{16}, \qquad (K_{\text{tot}} - 12E_g I)|_P = 2\Delta I_P.$$

Its excess CAR number is $q = -1$ for the hole and $q = +1$ for the particle; native currents transport q times the defect count. All three protected triplet states have the same gap and total charge $2q$. A common scale or chemical potential cannot split them into three different masses. The triplet is an internal spin-one representation, not three generations.

The same sector is not invariant under the joined uniform four-site Weyl/contact field. For its four charged-pair columns V, exact dyadic arithmetic gives

$$\|(I - VV^*)H_{\text{uniform}}V\|_F^2 = \frac{3}{256} > 0$$

for both particle and hole pairs, whereas the decoupled sum gives zero. Indeed, with $L = (I - VV^*)H_{\text{uniform}}V$, the stronger identity $L^*L = (3P_{\text{singlet}} + P_{\text{triplet}})/512$ excludes every nonzero nonleaking input, including every triplet state. This generator leakage preserves the distinction between the supplied protecting block service and the interacting field law. No binding, topological boundary charge, physical clock, Lorentz spin, observed species or absolute mass follows; the physical particle and common-field gates remain open.

A protected interacting sector on the original three-dimensional cube. The source-bound `cube_eta_protection` packet retains the original open $2\times2\times2$ Weyl/contact action, $c = 1/8$, $U = 1$, on all $2^{16} = 65\,536$ occupation states. In the physical onsite basis define

$$\eta^+ = \sum_x (-1)^{x_1+x_2+x_3} c^\dagger_{x\uparrow} c^\dagger_{x\downarrow}, \qquad \eta^- = (\eta^+)^\dagger, \qquad \eta_z = (N-8)/2.$$

Onsite singlets are invariant under each supplied $i\sigma_i$ link; the bipartite signs cancel the hopping commutator. Thus $[H, \eta^+] = U\eta^+$, checked exactly on the complete field action. For $X = \eta^+|_{N=4} : \mathcal{H}_4 \to \mathcal{H}_6$,

$$X^\dagger X = (\eta^-|_{N=4})^\dagger(\eta^-|_{N=4}) + 4I, \qquad H_6 X = X(H_4 + UI).$$

Consequently X is injective and its polar isometry intertwines the full interacting four-particle parent with a protected candidate of dimension $\binom{16}{4} = 1820$. Its range is precisely the $\eta \geq 2$ subspace at $N = 6$.

The executed repair domain is a finite four-block service: the four x_1-directed dimers, native ports $(0, 1, 5, 7)$, and all five inherited seams $(0, 1), (0, 5), (0, 7), (1, 5), (1, 7)$. Two blocks carry the unchanged lower-hole doublet and two the unchanged dimer ground. This gives a 24-dimensional $N = 6$ support P. Its physical columns are expanded exactly as $A + \sqrt{17}B$, with integer Gaussian matrices A, B; $\eta^- A = \eta^- B = 0$, so the support has $\eta = 1$. The columns are also bound to the consumed normalized analytic dictionary: its exact normalization is $n = 136 - 32\sqrt{17}$, with $(A + \sqrt{17}B)^\dagger(A + \sqrt{17}B) = n^2 I$. Every native C/E coin and raw slot is checked against its actual graded whole-block permutation. With the complete outside outcome $Q = I - P$, all 130 slots are retained: 120 native and ten outside over the ten seam/law choices. Orthogonality gives

$$PX = 0, \qquad L_r X = 0 \quad (r \text{ native}), \qquad QX = X.$$

Every admitted complete repair is therefore exactly the identity on this sector, including arbitrary reference cross terms; interleaved field evolution remains unitary and interacting. This mechanism requires no alignment of the transverse dimer spin frames. It does not assert a twelve-block execution, arbitrary repair invariance, or maximality of the protected subspace.

Exact interacting spectral census and a local fixed-number probe. Let $v = \eta^+|0\rangle$, $v^\dagger v = 8$, and $\Pi_3 = I - vv^\dagger/8$ on the 120-dimensional $N = 2$ space. For $T = (\eta^+)^2|_{N=2}$,

$$T^\dagger T = 60I + 24vv^\dagger, \qquad H_6 T = T(H_2 + 2UI).$$

Hence $T/\sqrt{60}$ is isometric on the rank-119 $\eta = 3$ parent. The hidden-spin census is 35 singlets and 84 triplet states; this is neither Lorentz-spin identification nor a family count. Exact integer characteristic polynomials in the actual onsite Hubbard frame give, with $z = 16(E_{N=6} - 2)$,

$$p_{119}(z) = z^{57}(z^2 - 12)^{21} q(z) c_{64}(z)^3 c_{128}(z)^3,$$
$$q(z) = z^2 - 16z - 12, \qquad c_a(z) = z^3 - 16z^2 - 12z + a.$$

These are eleven distinct energies: multiplicity 57 at zero, 21 at each $\pm\sqrt{12}$, one at each quadratic root and three at each of the six cubic roots. The last twenty states have contact-shifted energies. Squarefreeness, coprimality and rational root isolation are exact; approximate displayed response weights are separate diagnostics.

The $N = 6$ reference and one-site density probe are

$$|R\rangle = \frac{(\eta^+)^3|0\rangle}{\sqrt{2016}}, \qquad H|R\rangle = 3|R\rangle, \qquad |s\rangle = \frac{4n_0 - 3}{\sqrt{15}}|R\rangle.$$

Here $\langle n_0\rangle_R = 3/4$, $\mathrm{Var}_R(n_0) = 15/16$; $|s\rangle$ is an $\eta = 3$ singlet with mean energy 3 and variance 3/112. Its normalized two-particle parent is $(8|\uparrow\downarrow\rangle_0 - v)/\sqrt{56}$. The exact parent Krylov Gram determinant is $1488395072912184839188800532906554156908544 > 0$. Together with the annihilator $qc_{64}c_{128}$ this proves positive response weight at all eight interacting energies and zero weight at the three free-energy values.

The local number-preserving phase $F_0 = e^{i\pi n_0/2}$ satisfies

$$F_0|R\rangle = \frac{|R\rangle - \sqrt{15}|s\rangle}{4}, \qquad \mathrm{Tr}_0(F_0|R\rangle\langle R|F_0^\dagger) = \mathrm{Tr}_0\,|R\rangle\langle R|.$$

Thus the entire exterior reduced state is unchanged initially, while the global pure-state trace distance is $\sqrt{15}/4$ and the disturbed state has energy variance 45/1792. This is a finite local unitary disturbance of an entangled reference, not a boundary charge or an asymptotic localized particle. Both components remain protected. A fixed-$N = 6$ pointer Ramsey construction using controlled F_0, common field evolution and controlled $F_0^\dagger$ has

$$C(t) = \langle R|F_0^\dagger e^{-it(H-3)} F_0|R\rangle = \frac{1}{16} + \frac{15}{16}\sum_{j=1}^{8} w_j e^{-it(z_j/16-1)}, \qquad w_j > 0, \quad \sum_j w_j = 1.$$

No vacuum/charged-sector phase reference is needed in this protocol. The specified H and reference energy fix the frequency origin; these finite spectral gaps are not physical masses.

Native preparation and the remaining detector premise. The actual source prepares the 56 signed three-doublon amplitudes using an eight-bit, 5379-pulse amplitude tree on the even modes, with the odd modes blank. Eight native CNOT words copy into the odd modes in 24 pulses, followed by the three-pulse one-site phase including its scalar phase. The compact tree evaluation is an exact spectator blank isometry; all 65 536 rows are retained for copying and the phase. The independent exact trigonometric and literal-port certificate bounds the complete source-state error by 8.48×10^{-13}, including every source export. All 130 following repair records are retained

without success normalization. The native seventeen-bit time-dependent detector and native field continuation are not executed here: the fixed-number response above is analytic, while the reference preparation and local kick are executed. Hardware precision, growing-volume localization, deformed excitation transport, a continuum limit and empirical species identification are not established. The earlier small-carrier leakage and transverse hidden-spin obstruction remain valid; the larger eta sector supplies a different common protection mechanism.

The preceding preparation packet stops before the time-dependent detector. The following extension executes that detector and field continuation on the same cube, closing this finite readout premise.

An internal observer reads the protected response. The `protected_particle_readout` construction attaches the preceding cube spectrum to complete native pointer records. Its identified object is the eight-dimensional cyclic response generated by the centered local density probe, inside the 119-state $\eta = 3$ sector and the larger 1820-dimensional protected space. It is a neutral, fixed-$N = 6$ response; these three dimensions are not interchangeable particle counts. The construction uses Ramsey interference to read a correlation function, a standard measurement principle, and supplies its actual source implementation for this field.[2]

The apparatus retains sixteen field bits and one pointer, hence all $2^{17} = 131\,072$ field/pointer occupation rows. The unchanged source prepares $|R\rangle$ and a pointer $|+\rangle$, then applies controlled $F_0 = \exp(i\pi n_0/2)$. There are two field segments of duration $1/2$: the first is followed by complete C repair on all five admitted seams, and the second by complete E repair on those seams. Controlled $F_0^\dagger$, a pointer phase $\phi \in \{0, \pi/2\}$, and the inverse preparation rotation precede the complete six-outcome pointer export. Every source first-failure record and all 130 repair records remain present. An outside repair outcome continues; a native repair outcome or source failure terminates in its own retained record. No probability is conditioned on survival.

With exact field evolution, each repair acts identically on both ideal arms. Immediately before the final pointer rotation their field vectors are

$$b_0 = e^{-itH}|R\rangle, \qquad b_1 = e^{i\phi}F_0^\dagger e^{-itH}F_0|R\rangle.$$

Consequently the unconditional ideal pointer-zero probability is

$$p_0(t,\phi) = \tfrac{1}{2}\{1 + \mathrm{Re}[e^{i\phi}C(t)]\},$$
$$C(t) = \frac{1}{16} + \frac{15}{16}\sum_{j=1}^{8} w_j e^{-it\omega_j}, \qquad \omega_j = z_j/16 - 1.$$

Thus the two settings read $\mathrm{Re}\, C = 2p_0(t,0) - 1$ and $\mathrm{Im}\, C = 1 - 2p_0(t,\pi/2)$. The phase $3t$ follows from the reference eigenenergy; it is not an adjustable frequency fit. A common scalar added to H, or a common number chemical potential on this fixed number domain, cancels between the arms. Absolute energy zero is therefore not determined by this experiment.

Complete source and field error, without a sector projection. The executor composes success and every first-export failure map of each local word on at most three work bits, then applies these maps to the full field/pointer array. This is finite block contraction of the same native word, not replacement by an ideal gate. Independent tests compare the contracted maps with direct

[2] For many-body Ramsey correlation measurements, see Knap et al., *Phys. Rev. Lett.* 111, 147205 (2013), https://arxiv.org/abs/1307.0006.

source execution, including order-one failure controls. The exact $N = 6$ restriction is used only by the independent original-H comparator, not by the native execution.

Each half-time field segment uses 32 symmetric product steps. The actual ordered support list has 24 hopping terms of norm 1/16 and eight contact terms of norm one. Counting only possibly nonzero nested commutators gives the all-Fock coefficient $K = 629/256$. The two segments therefore contribute

$$\varepsilon_{\mathrm{split}} = 2\,\frac{629}{256}\,\frac{(1/2)^3}{32^2} = \frac{629}{1048576}.$$

Independent rational trigonometric and literal-port bounds add less than 8.56×10^{-11} for the preparation, controlled pointer operations, and both complete source field words. The complete output-vector error is consequently $\varepsilon < 6.00 \times 10^{-4}$. Every halted record is part of this norm. Complete repair and export preserve the comparison by isometric extension or contraction; the all-input field bounds also hold with untouched references. The initial preparation bound consumes the specified initialized blank.

For normalized complete outputs, half trace distance is at most their vector distance. Each unconditional pointer probability therefore differs from its ideal value by at most ε, and each reconstructed quadrature by at most 2ε. Rational root, residue and trigonometric enclosures at $t = 1$ imply, for the actual specified source,

$$0.993649 < p_0(1, 0) < 0.994850, \qquad 0.501379 < p_0(1, \pi/2) < 0.502580.$$

Both intervals exclude the corresponding no-response values, one and one half. Numerical source residuals have their separate declared admission; they are not substituted for these proved source bounds. The executed construction uses 681 283 native pulses per setting. It gives pointer-zero probabilities approximately 0.9942497011 and 0.5019798894, versus the independent original-H values 0.9942494663 and 0.5019797169. The complete output-vector discrepancy is approximately 2.27×10^{-6}, below the rigorous bound.

The imaginary fringe also distinguishes the contact interaction from the same free cube. Set $U = 0$ while retaining the hopping, reference, local probe and ideal repair schedule. Bipartite occupation parity $\Gamma_{\mathrm{sub}} = (-1)^{\sum_{x\,\mathrm{odd}} n_x}$ satisfies

$$\Gamma_{\mathrm{sub}} H_0 \Gamma_{\mathrm{sub}} = -H_0, \qquad H_0|R\rangle = 0, \qquad \Gamma_{\mathrm{sub}} F_0|R\rangle = F_0|R\rangle.$$

The free characteristic function is consequently real at every time, so $p_{0,\mathrm{free}}(t, \pi/2) = 1/2$ exactly. Both free arms remain in the protected sector. The certified interacting interval therefore excludes this specified contact-free response. This is discrimination between two internal dynamics, not a fit of a physical particle mass.

For independent reset repetitions, Hoeffding's inequality and a union bound give simultaneous sampling error at most 1/2048 in these two probabilities using 2^{24} trials per setting, except with probability less than $4(3/8)^8$.[3] The sampling allowance separates both fringes from no response. Independent resets are the sampling-service premise; a seeded numerical sample is an illustration of that model, not laboratory data. The recorded PCG64 illustration, with seed 20261101, gives respectively 16 680 454 and 8 423 380 pointer-zero outcomes out of 2^{24} trials; the complementary count includes every other complete outcome.

[3] The bounded-independent-sum inequality is imported from Hoeffding, *J. Amer. Statist. Assoc.* 58 (1963), 13–30, https://doi.org/10.1080/01621459.1963.10500830.

What eight-line resolution would additionally require. The executed two settings attach the known protected spectral measure to an internally accessible record. They do not reconstruct eight unknown frequencies or individually resolve the eight lines. There is, however, a certified prospective experiment on the source-specified support. At times $t_n = 39n$, $n = 0, \dots, 7$, use both pointer quadratures and set

$$y_n = \frac{16C(39n) - 1}{15}, \qquad V_{nj} = e^{-i39n\omega_j}, \qquad y = Vw.$$

Exact interval sine bounds in the Vandermonde product show

$$|\det V| = \prod_{j<k} 2\left|\sin\frac{39(\omega_k - \omega_j)}{2}\right| > 1400.$$

Hadamard's inequality bounds every 7×7 cofactor by $7^{7/2} < 343(8/3)$, so

$$\|V^{-1}\|_\infty < \frac{2744}{525}, \qquad \|\widehat{w} - w\|_\infty < \frac{43904}{2625}\, e,$$

where e bounds the total error of each pointer probability. The latter includes both quadratures and removal of the known elastic atom. Finite precision evaluation of an implemented inverse would need its own error allowance.

A sufficient prospective budget is systematic probability error $1/65536$ and sampling error $1/32768$ at each of the sixteen settings. With $9 \cdot 2^{29}$ independent trials per setting, the simultaneous failure probability is less than $32(3/8)^9 < 1/100$ and the weight error is less than $343/448000$. Twice that error is smaller than the least certified weight, so every line is separated from zero. The total is 77 309 411 328 trials and the largest model time is 273. This long-time accuracy and sampling schedule is not executed here. It identifies weights on the supplied support, not an arbitrary unknown spectrum from finitely many samples.

The finite result is an internally observable protected excitation response of the admitted simulator, with complete-history and finite-time errors. It does not require an external particle name or an SI calibration. It also does not establish that these neutral modes are single odd fermions, that the reference is a vacuum ground state, or that finite spectral lines are relativistic rest masses. Transported excitations, many-body continuum dynamics and comparison with nature remain separate constructions.

The one-time readout above leaves its long-time reconstruction prospective. The following certified accelerated calculation closes that known-support weight-resolution premise while retaining a bound to the full native word.

Full weight resolution by certified accelerated emulation. The preceding sixteen-setting proposal can be executed as a certified emulation of the same source word. The result resolves all eight weights on the source-specified support; it does not infer arbitrary unknown frequencies or identify physical species. The observable remains the eight-dimensional neutral cyclic response of the excited reference, with elastic weight $1/16$.

The actual chronological field word places the two disjoint spin hoppings of each spatial edge consecutively. They commute, and their sum is the complete Weyl edge commuting with physical η^+. The complete contact layer satisfies $D_6T = TD_2 + 2T$, where $T = (\eta^+)^2$. Thus every ideal Strang step has the exact identity

$$S_6(h)T = e^{-2ih}TS_2(h), \qquad S_6(h)|R\rangle = e^{-3ih}|R\rangle.$$

The complete protected repairs act identically on these ideal states. This is a property of the grouped ideal word, not an exact projection identity for its rounded native implementation.

The actual Hubbard-frame singlet chart has dimension36, retaining all35 $\eta = 3$ singlets and the $\eta = 4$ parent. If U is the spatial forward half-hop matrix and $W = \mathrm{Sym}^2 U$ on normalized unordered pairs, the step is $W^T \exp(-ihD)W$. Transpose reverses the hopping order because every individual Hubbard hopping matrix is symmetric. Integer generator checks bind this chart to every original edge and the contact layer. Outward interval square roots supply the distinct-site normalization.

Use the common step $h = 39/2^{19}$. Time $39j$ takes $j2^{19}$ steps, with half before C repair and half before E repair. At time273 this is 3 670 016 steps. Their 38 755 368 960 field pulses are represented by the repeated source word; they are not individually executed. Instead, outward160-bit step entries are rounded to128-bit dyadics and powered by exact Gaussian-integer matrix multiplication. For approximate unitaries with errors a, b, one rounded product has error at most $a+b+ab+36\,2^{-128}$. The longest power needs24 products and has operator error $249561087/2^{128} < 7.34 \times 10^{-31}$.

Independent literal-source templates bound the complete 10 560-pulse step error by 1.323294×10^{-12}. Telescoping includes all native first-export failures, and the full-Fock splitting error is $(629/256)t^3/n^2$. With preparation, pointer operations, complete repairs, certified arithmetic and exact binary64 probability-conversion error, every setting has systematic probability error below $8.568121 \times 10^{-6} < 1/65536$. Individual long-word failure weights are not materialized; their record addresses and complete-output norm bound are retained. No survival normalization is used.

The emulator draws counts from a binary64 proxy to the certified powered ideal-word probabilities, with PCG64 seed20261102. These are surrogate counts, not exact samples of the unmaterialized native failure instrument. There are $9\,2^{29}$ independent reset trials at each of sixteen settings, or77 309 411 328 overall. Zero is contrasted with all other complete outcomes. The stated independent binomial service gives simultaneous sampling error 1/32768 except with probability below $32(3/8)^9 < .01$. The same conditional recovery theorem applies to genuine native counts under that sampling event and the proved native systematic bounds.

Only the source frequency intervals enter the inverse. Rational Lagrange polynomials give a96-bit matrix B for $V_{nj} = e^{-i39n\omega_j}$; outward arithmetic certifies $\|B\|_\infty < 2$ and $\|BV - I\|_\infty < 10^{-18}$. Counts determine $\widehat{y}$ and $\widehat{w} = \mathrm{Re}(B\widehat{y})$. Since the true source weights form a probability vector, the conditional error is bounded by

$$\|\widehat{w} - w\|_\infty \le \|B\|_\infty \frac{16}{5}(\varepsilon_{\text{systematic}} + \varepsilon_{\text{sampling}}) + \|BV - I\|_\infty < 0.000258393828.$$

In ascending frequency order the recovered weights are

$$(.002457894673,\ .005641397395,\ .005890881572,\ .007131698528,$$
$$.004528321839,\ .421578515341,\ .415796456292,\ .136974834359).$$

Every resulting interval is strictly positive and contains the independent exact residue, checked after inference. This closes weight resolution on the given support in the certified emulator. It does not turn $E_H - 3$ into an absolute rest mass, identify an unknown support, or establish physical particle species or native hardware execution.

Comparing the actual interacting regional clocks. The complete four-mode Weyl/contact Gibbs source supplies a useful finite alternative to assuming regional modular covariance. Let $W_S(s) = \exp(isH)$ be its state modular unitary at $\beta = 1$, and let $W_A(s) = \rho_A^{-is}$ be the embedded unitary of an actual CAR-reduced reference, with fixed scalar gauges understood. The relative operations

$$C_{AB}(s) = W_A(s)W_B(s)^*, \qquad C_{AB}(s)C_{BD}(s) = C_{AD}(s)$$

compare these clocks on every input and untouched reference. They obey the time-cocycle identity $C_{AS}(s+t) = C_{AS}(s)W_S(s)C_{AS}(t)W_S(s)^*$. The existing source words implement each global-to-regional correction at $s = 1/2$ with 2,826 pulses and ideal operator splitting bound 11/32768; the regional-to-regional comparison needs twenty pulses. All sixteen Fock columns and first export failures remain present. Such corrections are active: the first changes the common Gibbs reference by half trace distance about 0.00444. They do not assert that both regional flows preserve the same global state.

The stronger identification fails on this reference. For a linear modular intertwiner, an eigenoperator coefficient satisfies $(\nu_j - \omega_i)L_{ji} = 0$. The actual dimer at speed 1/8, contact $U = 1$, $\beta = 1$ and $\mu = 0$ has three positive regional frequencies, approximately 0.0001258906, 0.9988218345 and 0.9989477251. An exact source characteristic polynomial and nine partial-trace moment identities determine its reduced spectral weights. Rational bounds on the square root, exponential and logarithm then separate all three frequencies from every global energy difference by more than 10^{-4}, in all 243 comparisons. Consequently every same-parameter linear intertwiner has range in the six-dimensional regional modular centralizer and cannot transmit a moving local clock hand. The group and coefficient implications are Lean checked; the concrete spectral and interval attachment is checked separately by exact arithmetic. This excludes a strong identification on one finite reference, while the explicit relative operations provide a functioning comparison. The supplied action, coherent services, reference preparation and dimensionless parameter remain visible inputs. No physical duration, common energy calibration, geometric modular flow or Einstein premise is discharged.

Controlled regional modular energy. The same complete-Fock Weyl/contact action supplies a quantitative regional modular attachment on its four-site open plaquette. All 256 field states and all sixteen states of the two-site region are retained. For the normalized signed partial trace E, put $H_c = H - I$ and $X(\beta) = E(e^{-\beta H_c})$. Exact Gaussian-integer and rational arithmetic constructs the first eight coefficients of $-\log X$. In particular, with $M_j = E(H_c^j)$,

$$C_1 = M_1, \qquad C_2 = \tfrac{1}{2}(M_1^2 - M_2), \qquad C_3 = \tfrac{1}{6}M_3 - \tfrac{1}{4}(M_1M_2 + M_2M_1) + \tfrac{1}{3}M_1^3.$$

For $0 \le x = \beta B < \log 2$, where $B \ge \|H_c\|$, the positive coefficient majorant $f(x) = -\log(2 - e^x) = \sum a_j x^j$ gives

$$\left\| -\log X - \sum_{j=1}^{p} \beta^j C_j \right\| \le f(x) - \sum_{j=1}^{p} a_j x^j.$$

This follows from the norm-convergent noncommutative logarithm series, without replacing the reduced reference by a local Gibbs ansatz. At the declared $\beta = 1/32$ and exact bound $B = 3$, an outward rational evaluation gives eighth-order error below 1.917×10^{-9}. Hence every Hermitian trace-zero regional variation obeys the entropy-tangent error bound $1.917 \times 10^{-9}\|\delta\|_1$. The normalization scalar in $K_A = -\log X + \log \operatorname{Tr}_A X$ cancels only on that tangent space.

The first nonscalar boundary correction is cubic and does not commute with the bare regional energy. Removing the contact interaction removes its interaction-dependent terms; removing the crossing bonds removes all nonscalar corrections. A full-versus-empty regional matrix-element witness and the same rational tail prove distance greater than 2.7×10^{-8} from $\beta C_1 + aI$ for every scalar a on this instance. This supplies both a corrected finite modular generator and a certified boundary to its bare-energy identification. The source preparation below supplies this plaquette reference to a separate controlled error. The action, temperature and lattice remain declared; the

norm bound can grow with volume. Geometric cap weights, null stress, fixed-volume entropy–area stationarity, a physical scaling limit and positive Newton normalization remain additional obligations.

A separately prepared interacting Gibbs reference. On the complete 256-state plaquette at $\beta = 1/32$, the degree-eight Gibbs ensemble retains all 256 preparation records and every export abort. Independent outward arithmetic certifies its literal-source preparation error below 1.700×10^{-12} on the initialized input. The actual regional success logarithm, including the exact normalization scalar, is within 2×10^{-9} of the preceding P_8 comparison. The all-input continuation has a separate source and splitting bound. The proof in `model/certified_reference/README.md` supplies finite Gibbs preparation; it does not assert invariance under the complete repair. The following stationary reference has a different modular generator.

The actual stationary source weight. One identical literal native $Y(-\pi/3)$ request and complete dephasing export per occupation mode prepare $\sigma_M = \operatorname{diag}(a, b)^{\otimes M} = a^M e^{\gamma N}$, where $a = |U_{00}|^2$, $b = |U_{10}|^2$ for the actual represented source unitary and $\gamma = \log(b/a)$. The exact one-mode error is below 1.894×10^{-16}, proving $a, b > 0$ and $b > a$; abort blocks remain separate. Number conservation and complete unital repair preserve this weight exactly. Tracing complete outside outputs before regional compression gives a k-mode logarithm error at most $k\delta/(1/4 - \delta)$, with $\delta = (2 + \epsilon)\epsilon$. This charge clock is trivial within fixed number. On the earlier six-site $N = 4, S = 1$ active sector, every common stationary modular generator stays at distance at least $|\beta|\sqrt{13/48}$ from βK, even allowing scalar and protected-spin terms. Thus the stationary reference and interacting Gibbs energy remain distinct. The exact source and reference proofs are in `model/continuing_reference/README.md`; the following construction closes its earlier restricted scheduler premise.

A stationary finite controller for native feedback. The reference can continue under an explicit record-adaptive scheduler. Use two classical working bits (c, d); $c = 0$ selects the complete native C repair on seam $(0, 1)$, and $c = 1$ the complete E repair on $(0, 5)$. Append every raw record r, including zero slots and the outside outcome $Q = I - P$, and update

$$(c, d) \longmapsto (d \,\mathrm{XOR}\, f(r), c), \qquad f(r) = \text{coin XOR (raw mod 2)}, \quad f(Q) = 0.$$

At output (u, v) every incoming term uses the same instrument v. Completeness and unitality of the unchanged repair therefore give

$$\sum_r L_r^{(v)} \frac{\sigma_M}{4} L_r^{(v)*} = \frac{\sigma_M}{4}, \qquad \Omega = \sigma_M \otimes I_4/4, \quad \sigma_M = \operatorname{diag}(a, b)^{\otimes M}.$$

Here a, b are the actual literal-source weights, so the successful block has trace $(a + b)^M$. All initial export-failure sectors are retained and frozen, emitting a bypass record. Separate unitary layers selected by the current working memory and generated by the original number-conserving field Hamiltonian also preserve this marginal. The controller is a declared reversible classical law on $\mathbb{C}^4$, not a claimed quantum pulse circuit.

Two unchanged native events on an explicitly initialized binary spectator, occupied at ports $2, 3$ and repaired on $(0, 1)$, leave it fixed and supply two exact fair coins. Their retained records seed $I_4/4$ without an angle-error assumption. This uses the supplied simulator basis initialization; no hidden interacting-field preparation of that spectator is asserted. The working memory remains two bits, while the archive grows by at most five bits per tick, besides the two seed records and all

6^M source readouts. Stationarity traces these records; it never normalizes the successful field block or declares the growing archive stationary. The modular generator remains $-\gamma N - M \log a + \log 4$, $\gamma = \log(b/a)$, and every complete step intertwines the charge flow $e^{it\gamma N}$. No physical energy-temperature identification follows. Earlier local literal-source bounds remain uniform in volume and ideal adaptive history length; implemented per-step errors must be added.

A complete finite mixing certificate and its boundary. On the invariant three-position highest-spin repair corner, with no joined Hamiltonian step inserted, the field/controller algebra is $\bigoplus_4 M_3$. Each E outcome has rank at most one. Exact integer enumeration finds at least one eight-tick history encountering E for every one of the 12^2 input/output population pairs, with common probability denominator 2^8. The corresponding rank-one Kraus histories prove

$$\Phi^8 = \frac{3}{64}\mathcal{D} + \frac{61}{64}\mathcal{R}, \qquad \mathcal{D}(X) = \mathrm{Tr}(X) I_{12}/12,$$

where $\mathcal{R}$ is completely positive, trace preserving and unital. Thus $I_{12}/12$ is the unique stationary density on this CQ algebra, and the archive-traced error contracts by $(61/64)^{\lfloor n/8 \rfloor}$, including arbitrary untouched quantum references and all field coherences. This is neither whole-Fock mixing nor mixing with interleaved Hamiltonian evolution. Indeed the full retained history recovers the initial corner basis state and memory. As a negative control, the recordwise reversible one-bit policy $c \mapsto c \,\mathrm{XOR}\, f(r)$ sends the uniform six-state input to $(3, 1, 2, 1, 3, 2)/12$, a trace distance $1/6$ from uniform. The two-register incoming-row balance is a substantive premise, not a consequence of memory reversibility alone.

Exact energy Ward cells under coarse-graining. The same complete-Fock Weyl/contact action supplies a conserved energy column on its standard three-dimensional lattice. With the declared half-bond convention $H = \sum_x h_x$, its range-two currents are $J_{xy} = i[h_x, h_y]$. Distributing each current equally over all shortest nearest-neighbour paths preserves its incidence $\delta_x - \delta_y$: intermediate vertices cancel on every path. The resulting oriented face current $F_{x,i}$ therefore satisfies $\dot{h}_x + \sum_i (F_{x,i} - F_{x-e_i,i}) = 0$ exactly. The rational Clifford certificate checks the actual $3 \times 3 \times 3$ action at $c = 1/8, U = 1$: all 351 site pairs, 81 nonzero currents, 54 faces and 27 local equations, on the complete 54-mode algebra. Its coverage of all 2^{54} Fock states is a symbolic operator identity, not execution of that many amplitudes. Independent complete small-Fock matrices attach all three hopping axes and the interaction to the existing source action.

For a finite time cell, the exact flux is $I_{x,i}(n) = \int_{t_n}^{t_{n+1}} F_{x,i}(t)\, dt$. Hence

$$h_x(t_{n+1}) - h_x(t_n) + \sum_{i=1}^{3} \big(I_{x,i}(n) - I_{x-e_i,i}(n)\big) = 0.$$

For any normalized input and untouched reference, setting $T_{00}(n, x) = \langle h_x(t_n) \rangle$ and $T_{i0}(n, x) = \langle I_{x-e_i,i}(n) \rangle$ supplies the $j = 0$ forward-difference Ward column of the finite tensor interface. These are cell energy and integrated flux, not pointwise continuum stress. Summing spatial blocks cancels internal faces, so $D_{\mathrm{coarse}} P_{\mathrm{faces}} = P_{\mathrm{sites}} D_{\mathrm{fine}}$. Successive block maps compose exactly; successive time cells telescope. All 27 rectangular width choices, including ragged blocks, pass the exact source certificate. Generic finite-sum divergence, continuity, block composition and time telescoping are also Lean-checked; the concrete CAR source binding remains the exact rational certificate.

The existing local source bounds control the approximation uniformly in ambient volume. With $A = U + 3c$, one has $\|h_x\| \le A$ and $\|F_{x,i}\| \le B = 14A^2$: a face receives absolute shortest-path weight at most seven. Its support fits a radius-three cube. If L_h, L_F are the existing normalized locality

tails and ϵ is the patch unitary error, an endpoint energy error is at most $A\min(2, L_h + 2\epsilon)$ and an integrated-face error at most $tB\min(2, L_F + 2\epsilon)$. Replacing the integral by m midpoint samples adds the explicit bound $t^2 B\, 343(U + 6c)/(2m)$; endpoint sampling is not exact. A cell residual is bounded by two endpoint errors plus six face errors, with supplied readout errors added separately. The receipt evaluates these resource bounds without claiming to execute the large sample count. Input preparation, hardware errors and controller/failure-space energy remain separate. This closes a finite same-action energy-column attachment, not the three momentum columns, a symmetric Lorentz tensor, geometric modular weights, area stationarity, a continuum scaling theorem or the full set of Einstein premises.

Local momentum balance retains its force. On the open Weyl/contact box, the three local central-difference momenta obey exact balances with kinetic face stress and quartic contact force. All spatial blocks and time cells retain that force. On the 3^3 box at $U = 1$ its total-force Gram matrix is $(9/4)I_3$, so an internal closed divergence cannot absorb it. Conserved number-preserving quadratic charges of the connected action reduce to number and hidden spin. These exact results in `model/lattice_momentum_balance/README.md` concern local quadratic candidates. The following periodic construction supplies conserved global momenta with a separate locality boundary.

Global periodic momentum and the local question. On a declared periodic box, graded translations T_j commute with the full interacting action. Their principal Hermitian logarithms P_j are exactly conserved. They differ from additive one-body momentum by integer many-body winding. On odd-prime rings every number-preserving translation logarithm has nonzero one-particle spin-trace kernels at all nonzero separations. For the principal branch, every range-R density sum obeys $\|P_j - A_R\| \geq 1/(R+1)$ when $L > 2(R+1)$. These full-space statements, proved in `model/periodic_momentum/README.md`, leave controlled restricted state domains available. The following bounds quantify one such domain and retain its actual interaction leakage.

A collision-suppressed local Ward bound. The same periodic Weyl/contact action supplies a controlled local approximation beyond its nonlocal conserved momentum. Let B be a spin-complete set of spatial Fourier modes, $f = |B|/V$, and let Ω restrict the initial state to that band and $N \leq N_{\max}$. Write $D = \sum_x n_{x\uparrow} n_{x\downarrow}$ and $d = \lfloor N_{\max}/2 \rfloor$. Band annihilators have norm $\sqrt{f}$, so $\Omega D \Omega \leq f \min(N_\uparrow, N_\downarrow) \leq fd$. Since $D^2 \leq dD$ on this number sector,

$$\|UD\Omega\| \leq Ud\sqrt{f}, \qquad \|(I - \Omega)e^{-itH}\Omega\| \leq \min\{1, |t|Ud\sqrt{f}\}.$$

The second bound uses free invariance of Ω and Duhamel's formula; the full interacting action is never projected. For two particles, the exact contact principal angles are $\lambda_K = |B \cap (K - B)|/V$, giving escape norm squared $\max_K \lambda_K(1 - \lambda_K)$ before multiplication by U^2. Thus projecting the outgoing contact back into B would miss its leading square-root leakage.

Let $s_j(x)$ be the local sine-momentum density. Write $\Pi_{xy,j}$ for the antisymmetric kinetic stress across a nearest-neighbor face. The exact identity is $i[H, s_j(x)] + \sum_y \Pi_{xy,j} = F_j(x)$, with $F_j(x) = i[UD, s_j(x)]$. Normal ordering its quartic contact terms gives $\|F_j(x)\| \leq U$ and $\|F_j(x)\Omega\| \leq 4Uf$. For preparation at zero and $0 \leq t_0 \leq t_1$, the actual integrated Ward residual therefore obeys

$$\left\| \left[s_j(x, t_1) - s_j(x, t_0) + \sum_y \int_{t_0}^{t_1} \Pi_{xy,j}(t)\, dt \right] \Omega \right\|$$
$$\leq \min\left\{ U(t_1 - t_0),\ 4Uf(t_1 - t_0) + \frac{U^2 d\sqrt{f}}{2}(t_1^2 - t_0^2) \right\}.$$

The preparation clock is not restarted in later cells. Spatial blocks cancel their internal faces and sum the site bounds; time cells telescope. The complete 54-mode 3^3 source certificate checks all 81 local equations, 81 kinetic faces per column and 27 rectangular coarse partitions per column. Finite-sum transport of the errors is also Lean-checked. This is symbolic full-Fock coverage, not execution of all its amplitudes.

The two-particle total sine momentum has multiplier $2\sin(K_j/2)\cos(k_j - K_j/2)$. The uniform contact vector gives the sharp complete-fiber force norm $\|i[UD, S_j]\| = \sqrt{2}U|\sin(K_j/2)|$. If initially $|k_j| \le \kappa$ and $2\kappa < \pi$, total momentum stays within $|K_j| \le 2\kappa$ even after relative-momentum scattering. Combining this conserved domain with the sine remainder gives

$$\|(P_j - S_j)e^{-itH}\Omega\| \le \frac{\kappa^3}{3} + 4\kappa \min\{1, |t|U\sqrt{f}\}.$$

For a 64^3 counting example with $B = \{-1, 0, 1\}^3$, $N_{\max} = 2$, $U = 1$ and $t = 1/8$, the certified leakage is below 0.00127, the local Ward defect below 0.000132, and the momentum error after division by κ below 0.0083. The separate eight-site exact-Hamiltonian diagnostic retains a nonzero unforced local defect, so the contact force has not been discarded. These are fixed-number dilute bounds. The critical band-edge scaling $f \sim a^3$, $U \sim a^{-2}$ instead makes $U\sqrt{f} \sim a^{-1/2}$, for which this estimate supplies no vanishing leakage. A finite-density or critically scaled local stress limit, Lorentz symmetry, physical units and the remaining Einstein premises remain separate. Proofs, finite domains and replay are in `model/dilute_momentum_ward/README.md`.

Interacting lattice scattering and its zero threshold. The same Weyl/contact action admits an on-shell two-fermion scattering theorem. On the physical exchange subspace of $L^2(\mathbb{T}^3, \mathbb{C}^4)$ at total momentum K, its free multiplication operator $H_{0,K}$ is bounded self-adjoint and the contact perturbation is $U|d\rangle\langle d|$, with $\|d\| = 1$. The difference is trace class with norm U. The Kato–Rosenblum theorem therefore gives complete wave operators on the absolutely continuous subspaces and a unitary scattering operator.[4] With $g_+(E) = a(E) - i\pi\rho_0(E)$, its contact-channel eigenvalue is

$$S_c(E) = \frac{1 - Ua(E) - i\pi U\rho_0(E)}{1 - Ua(E) + i\pi U\rho_0(E)}.$$

The orthogonal reducing complement of the contact-generated subspace has identity scattering. Point spectrum, including the isolated pair and the free dark zero modes, remains outside the incoming scattering domain. Completeness here concerns two-body absolutely continuous scattering; it is not many-body or chiral-continuum completeness.

At $K = 0$, the singlet resolvent is $z/(z^2 - 4c^2|\sin k|^2)$. Coarea with all eight sine-map preimages gives, for $0 < |E| < 2c$,

$$\rho_0(E) = \frac{E^2}{4\pi^2 c^3} \sum_{n \ge 0} A_n \left(\frac{E}{2c}\right)^{2n}, \qquad A_0 = 1, \quad A_1 = \tfrac{1}{2}, \quad A_2 = \tfrac{11}{40}, \quad 0 < A_n \le 1.$$

The coefficients are exact sphere moments of the binomial Jacobian. After degree M, the dimensionless remainder is at most $r^{2M+2}/(1 - r^2)$, $r = |E|/(2c)$; rational arithmetic and a Machin bound for π give certified density intervals. Local analyticity and evenness give $g_+(E) \to 0$ at zero. Hence $\rho_U(E) = \rho_0(E)/|1 - Ug_+(E)|^2 \sim E^2/(4\pi^2 c^3)$. This supplies nontrivial scattering

[4] The imported trace-class and finite-rank scattering results are reviewed in https://arxiv.org/abs/1711.00528 and https://arxiv.org/abs/0902.3568; the attachment here verifies their premises for the stated Weyl/contact exchange fiber.

and a gapless contact spectral support on one interacting lattice action while retaining its isolated pair. Zero model energy is not an identified physical vacuum threshold, so no massless physical particle is assigned. The source-bound proof and imported scattering theorems are recorded in `model/contact_scattering/ONSHELL.md`.

Interacting edge scattering and a moving bound pair. The same two-body source family has a controlled moving bound band. Set $c_a = \sqrt{3}/a^2$ and retain the repulsive coupling

$$U_a^{-1} = \frac{a^2 W_0}{2} - \frac{a^3 \gamma}{\pi}, \qquad W_0 = \left\langle \frac{1}{\sum_j \cos^2 k_j} \right\rangle, \qquad \gamma > 0.$$

The normalized full-torus integral satisfies $2/3 \le W_0 < 5$; $a\gamma \le 1$ makes the coupling positive. The single global comparison is $J_a = 6/a^2 - H_a$, explicitly shifting energy and reversing time. Neither γ nor this clock/energy choice is inferred from physical data.

At zero total momentum the threshold estimate is

$$\left| \frac{g_0(6/a^2 + z) - a^2 W_0/2}{a^3} + \frac{\sqrt{z}}{\pi} \right| \le 19a|z|$$

for real $|a^2 z| \le 1/8$, using the upper boundary square root; the exterior constant improves to 11. Coarea for all eight valleys, an inner ball of radius $1/2$ and its complementary measure give the respective Green remainders 18η and $36|\eta|$ before rescaling. Thus, for $0 \le s \le S$, $a^2 S \le 1/8$ and $836aS \le 7\gamma$,

$$S_J(s) \longrightarrow \frac{\gamma - i\sqrt{s}}{\gamma + i\sqrt{s}}, \qquad \sup_{[0,S]} |S_J - S_{J,\lim}| \le \frac{1672aS}{7\gamma}.$$

This is the earlier nonidentity scalar contact-channel limit; its orthogonal free channels remain present. Lattice contact renormalization is an established mechanism, not a new particle identification; compare Koerber–Berkowitz–Luu (2019) and Winkler et al. (2006).

Nonzero total momentum has a controlled pole. Write $K = aP$, $q = k - K/2$, $d_j = \sin q_j \cos(K_j/2)$, $e_j = \cos q_j \sin(K_j/2)$ and $R^2 = \sum_j \cos^2(K_j/2)$. For $|K_j| \le \pi/2$, concavity gives the exact complete free edge $B_a(K) = 2c_a R$, attained at all eight $q_j = \pm\pi/2$. The four-spin singlet resolvent is

$$G_K(q, E) = \frac{E(E^2 - 4c_a^2 e^2)}{E^4 - 4c_a^2(d^2 + e^2)E^2 + 16c_a^4 (d \cdot e)^2}.$$

At $E = B_a(K) + z$, put $t = E/(2c_a)$, $\eta = t^2 - R^2$ and $V = \sum_j \cos K_j \cos^2 q_j$. Equivalently,

$$g_K(E) = \frac{t}{2c_a} \left\langle \frac{1 - e^2/t^2}{V + \eta + (d \cdot e)^2/t^2} \right\rangle.$$

Let $\kappa = \max |K_j| \le 1/2$ and $h = a^2 z \le 1/8$, with $z \ge 0$. Writing $Q = \sum_j \cos^2 q_j$, the denominator lies between $(1 - \kappa^2/2)(Q + h + h^2/12)$ and $Q + h + h^2/12$; $e^2 \le \kappa^2 Q/4$ and $(d \cdot e)^2/t^2 \le e^2$. The complete-torus integral and prefactor therefore give

$$|g_K(B_a(K) + z) - g_0(6/a^2 + z)| \le 3a^2 \kappa^2.$$

Combining this with the exterior threshold estimate proves

$$\left| \frac{g_K(B_a(K) + z) - a^2 W_0/2}{a^3} + \frac{\sqrt{z}}{\pi} \right| \le a(11z + 3P_{\max}^2), \qquad |P_j| \le P_{\max}.$$

Under $256a(\gamma^2 + P_{\max}^2) \le \gamma$, the pole denominator has opposite signs at $z = \gamma^2/4$ and $z = 4\gamma^2$. Strict monotonicity of the exterior Stieltjes transform supplies exactly one upper pole $E_a(K) = B_a(K) + \delta_a(K)$, uniformly on that momentum cube:

$$\frac{\gamma^2}{4} \le \delta_a(K) \le 4\gamma^2, \qquad |\delta_a(K) - \gamma^2| \le \frac{22}{7} a\gamma(132\gamma^2 + 9P_{\max}^2).$$

This does not invoke the older, stronger sufficient condition $U > B_a$.

An explicit bound-band wavepacket comparison. The same global J_a has bound energy $\epsilon_a(P) = 6/a^2 - B_a(aP) - \delta_a(aP)$. Rationalizing B_a and bounding the sine remainder gives

$$\sup_{|P_j| \le P_{\max}} \left| \epsilon_a(P) - \left(\frac{|P|^2}{4} - \gamma^2 \right) \right| \le \mathcal{E}_a := \frac{a^2 P_{\max}^4}{4} + \frac{22}{7} a\gamma(132\gamma^2 + 9P_{\max}^2).$$

In particular, subtracting $B_a(K)$ separately at each momentum would incorrectly remove the center-of-mass kinetic term.

Let $R_K = (E_a(K) - H_{0,K})^{-1}$ and $\psi_{a,K} = R_K d / \langle d, R_K^2 d \rangle^{1/2}$. Its positive contact overlap fixes its phase, and the pole equation makes it a normalized, measurable, exchange-antisymmetric eigenvector. With consistent measures $dK/(2\pi)^3$ and $dP/(2\pi)^3$, the formula

$$(W_a f)(K, k) = a^{-3/2} f(K/a) \psi_{a,K}(k)$$

defines an isometry on packets supported in the stated cube. Its range reduces the two-body source action; the remaining source states are retained. If h_{pair} multiplies by $|P|^2/4 - \gamma^2$, then for arbitrary untouched references,

$$\| e^{-itJ_a} W_a - W_a e^{-ith_{\text{pair}}} \| \le \min\{2, |t| \mathcal{E}_a\}.$$

This proves coherent moving-pair propagation with an explicit rational spacing schedule. The limiting inertial mass is twice that of a single free edge in these model coordinates; neither an SI mass nor a relativistic rest energy is identified. The exact bounds, independent four-spin and torus checks, and wrong-subtraction/time-orientation controls are retained in `model/moving_bound_pair/README.md`.

The result concerns one composite sector. Nonzero-K on-shell scattering, a physical vacuum/detector map, many-pair interactions, relativistic massive/massless sectors and Einstein dynamics remain separate. The earlier fixed-zero $c = v/a$ result also remains valid: $\int_{-M}^{M} |S - I|^2 dE \le 1024 M^2/(63c)$ for arbitrary real finite $U(a)$ and $M \le c/4$, so every fixed energy-normalized packet becomes transparent while tuned shrinking resonances can persist. Its scaling and energy comparison differ explicitly from the bound-band construction.

Identified model energies and the physical particle boundary. The model energies and inertial mass are identifiable. The cube has eight exactly characterized interacting energies, each reached by the local probe with positive weight. The moving pair has $\epsilon(P) = |P|^2/4 - \gamma^2$ for the reflected generator $J_a = 6/a^2 - H_a$, hence binding energy γ^2 and inertial mass 2 in those model units. If physical energy and length units $E_{\text{u}}, L_{\text{u}}$ are independently attached and $p_{\text{phys}} = \hbar P / L_{\text{u}}$, then $M = 2\hbar^2/(E_{\text{u}} L_{\text{u}}^2)$ and $E_{\text{bind}} = E_{\text{u}} \gamma^2$, with $E_{\text{u}} T_{\text{u}} = \hbar$ for the common clock. These conversion formulas do not select the units or fix an absolute rest-energy offset. Binding energy is not total rest mass. For the cube, the following rounded displays have exact rational energy and weight intervals in the receipt, with the original contact unit $U = 1$:

H energy	$E-3$	Probe weight
1.816498920	−1.183501080	0.002465366
1.860537501	−1.139462499	0.005644624
1.955137632	−1.044862368	0.005881219
2.108692846	−0.891307154	0.007137664
2.167639734	−0.832360266	0.004528123
3.015861346	+0.015861346	0.421577939
3.030769653	+0.030769653	0.415789141
3.044862368	+0.044862368	0.136975924

Weights refer to the normalized centered density probe; the phase-kicked target has 15/16 times these weights plus an elastic weight 1/16 at energy 3. The lowest displayed line is not the $N=6$ ground: $-3\sqrt{3}/8 \le E_{0,N=6} \le -\sqrt{3}/8$, by the free one-particle bound and a polarized zero-contact variational state. These are finite energy levels and signed transitions, not eight rest masses. The physical species assignment count remains zero. On the cube the density probe, its phase-kicked reference and all spectral components remain in $N=6$: the transition operator is number-neutral and even under $\Gamma=(-1)^N$. An isometric dictionary intertwining this grading cannot map that response to a single odd fermion, since an odd creation operator changes parity. This does not exclude other odd model excitations or a fermion–antifermion response; even parity alone does not establish Bose exchange or Lorentz spin. Hidden spin and model number have not been identified with physical spin and electric charge. The physical target conventions are the PDG electron and quark quantum numbers and relativistic kinematics.[5]

The supplied cube reference also fails the vacuum ground-state spectral condition for its own H. Its positive density-probe spectral measure has five negative and three positive frequencies $\omega = z/16 - 1$; all eight weights are strictly positive by the exact Krylov certificate. In particular $z_- = 8 - 2\sqrt{19}$ has weight $(1 - 4/\sqrt{19})/14 > 0$ and gives an $N=6$ eigenenergy $5/2 - \sqrt{19}/8 < 2$, below the reference energy 3. A number-preserving unitary swapping these two states lowers the energy, so the reference is not passive for the full field unitary algebra. This is a mathematical witness, not an executed extraction protocol: the actual local phase preserves the mean energy. These are frequencies of a positive greater correlation, not the two-sided frequencies of a retarded commutator. A scalar or common number chemical potential leaves the gaps unchanged, and time reversal leaves mixed signs.

The moving-pair limit $\epsilon(P) = |P|^2/4 - \gamma^2$ is a genuine controlled nonrelativistic band. For any fixed affine energy map $E = \alpha\epsilon + \beta$, $\alpha \neq 0$, and fixed invertible linear momentum map $p = LP$, the residual $E^2 - c^2|p|^2 - m^2c^4$ has nonzero quartic coefficient $\alpha^2/16$ along a coordinate line. It cannot vanish on an open momentum set. Thus these maps do not turn this limit into an exact massive or massless Lorentz shell. A low-momentum approximation remains possible: after matching constant and quadratic terms, the difference between $E_0 + Ar^2$ and $\sqrt{E_0^2 + 2E_0Ar^2}$ is $A^2r^4/(E_0 + Ar^2 + \sqrt{E_0^2 + 2E_0Ar^2})$, for rest energy $E_0 > 0$ and $A > 0$.

`model/particle_identification` binds these scoped tests and the remaining vacuum, gauge-charge, spin, clock, detector and mass-shell requirements. The earlier dimer electron mass was a calibration input, not an independent mass prediction; its retrospective equality comparison does not identify the new cube or pair modes. No mass fitting, new eligible physical comparison, or whole-program exclusion is added.

The preceding identity audit leaves a relativistic construction open. The following extension

[5] https://pdg.lbl.gov/2025/listings/rpp2025-list-electron.pdf; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-quark-model.pdf; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-kinematics.pdf.

constructs a massless finite-particle limit on the growing lattice. Its carrier and reference remain distinct from the fixed protected cube, and its limiting interaction vanishes.

A controlled massless relativistic sector of the unchanged action. The same Weyl/contact family admits a simultaneous eight-valley continuum comparison on every fixed finite-particle momentum domain. Set

$$H_a = d\Gamma(h_a) + U(a)D, \qquad h_a(k) = -\frac{v}{a}\sum_{j=1}^{3}\sigma_j \sin k_j, \qquad D = \sum_x n_{x\uparrow}n_{x\downarrow},$$

with fixed $v > 0$ and finite $U(a) \geq 0$. This declares the scaling $c = v/a$; it adds no mass term, energy subtraction or time reversal. For $\nu = \pi b$, $b \in \{0,1\}^3$, let $C = [-\Lambda, \Lambda]^3$ and $a\Lambda < \pi/2$. With both Fourier measures normalized by $(2\pi)^{-3}$,

$$(W_a f)(\nu + ap) = a^{-3/2} f_b(p), \qquad W_a : \bigoplus_b L^2(C, \mathbb{C}^2) \longrightarrow L^2(\mathbb{T}^3, \mathbb{C}^2)$$

is an isometry, with zero output outside the eight disjoint patches. There is no additional factor $1/\sqrt{8}$. Its exterior powers give an isometry $W_a^{\leq N}$ on all particle numbers through a fixed N, with the vacuum and every valley retained.

Writing $s_j = (-1)^{b_j}$ and $\chi_b = -s_1 s_2 s_3$, the limiting blocks are

$$h_b(p) = -v\sum_j s_j \sigma_j p_j, \qquad E_\pm(p) = \pm v|p|, \qquad E^2 - v^2|p|^2 = 0.$$

All eight continuum masses are zero. Four blocks have each chirality. The proper frame $R_b = -\chi_b \operatorname{diag}(s)$ has a constant $SU(2)$ lift and spin $S_{b,j} = R_{b,jj}\sigma_j/2$, so $h_b = 2\chi_b v S_b \cdot p$. The positive-energy helicity is $\chi_b/2$; the negative-energy helicity is $-\chi_b/2$. At $p = 0$ the two spin states are degenerate and helicity is undefined. In these frames an opposite pair has $\alpha_j = \operatorname{diag}(-\sigma_j, \sigma_j)$. With off-diagonal identity β, the matrices $\gamma = (\beta, \beta\alpha_1, \beta\alpha_2, \beta\alpha_3)$ satisfy the Clifford relations for signature $(1,-1,-1,-1)$, checked exactly. Thus the limiting differential equation is massless Weyl/Dirac in coordinate $x^0 = vt$, with the Clifford relations verified.[6] The fixed cutoff and the lattice regulator are not Lorentz invariant; neither the valley pairing nor this spin frame selects physical species.

The full interacting evolution has an explicit comparison bound. The sine remainder gives, uniformly on C,

$$\|h_a(\nu + ap) - h_b(p)\| \leq \frac{va^2}{6}\sum_j |p_j|^3 \leq \frac{va^2\Lambda^3}{2}.$$

For $a\Lambda \leq 1$ and $p \neq 0$, both spectral-projector errors are at most $a^2\Lambda^2/6$. The squared-energy defect lies between $-v^2a^2\sum_j p_j^4/3$ and zero. No projector at the degenerate node is assigned.

Let Ω_a project onto the initial Fock band and $d = \lfloor N/2 \rfloor$. On the infinite lattice the site-diagonal band weight is exactly $f_a = 8(a\Lambda/\pi)^3$. The positive countable partial sums defining D are bounded by d, hence converge strongly and satisfy $D^2 \leq dD$. On band states each annihilation has norm at most $\sqrt{f_a}$ and preserves the band for the remaining particles. Summing its squared norms

[6] For the standard Weyl equation and helicity convention, see D. Tong, *Quantum Field Theory*, Sections 4.4 and 4.7, https://www.damtp.cam.ac.uk/user/tong/qft/qfthtml/S4.html.

gives $\Omega_a D\Omega_a \le f_a d$, separately on each up/down number sector, and therefore $\|D\Omega_a\| \le d\sqrt{f_a}$. This proves the infinite-volume attachment of the finite dilute-band estimate without substituting a continuum fraction for a finite count.

Duhamel's formula yields

$$\|e^{-itH_a} W_a^{\le N} - W_a^{\le N} e^{-itd\Gamma(h_{\mathrm{W}})}\| \le \min\{2, |t|\epsilon_a\},$$
$$\epsilon_a = \frac{Nva^2\Lambda^3}{2} + |U(a)|d\sqrt{8(a\Lambda/\pi)^3}.$$

The left evolution is the full source Hamiltonian: no band projection is inserted and every outgoing contact state remains present. The estimate also holds with arbitrary untouched references; it bounds half trace distance and individual probability errors, capped by one. For fixed N, Λ, v and bounded times it tends to zero whenever $U(a) = o(a^{-3/2})$. At $N \le 1$ the contact is exactly zero for every $U(a)$. Thus the proved limit is free; the critical pair scaling $U \sim a^{-2}$ and finite-density limits are outside this conclusion. For $N = 2$, $\Lambda = v = U = t = 1$ and $a = 1/64$, the complete-vector bound is less than 0.001237 and the projector bound is $1/24576$.

The finite source check retains all 128 one-particle Fourier columns on a periodic 4^3 cube. An eight-valley two-particle state there has exact $\|D\psi\|^2 = 1/8$ and $\|(1-\Omega)D\psi\|^2 = 7/64$, so the contact escape is real. Propagation on all 8,128 two-particle states at $t = 1/8$ gives a numerical vector difference about 0.0441641, below the rigorous $\sqrt{1/8}/8$. These matrix diagnostics and exact source identities do not constitute an additional native preparation or detector execution.

Resulting energies and the physical boundary. At $v = 1$, $a = 1/64$, every valley has the following energies; both signs are retained, and only their common positive magnitude is displayed. The receipt gives rational enclosures for the rounded lattice values.

p	lattice $\lvert E_a\rvert$	continuum $\lvert E\rvert$	continuum mass
$(0,0,0)$	0	0	0
$(1,0,0)$	0.999959310393	1	0
$(1,1,0)$	1.414156018578	$\sqrt{2}$	0
$(1,1,1)$	1.731980331101	$\sqrt{3}$	0

These are original-generator model energies, without an SI calibration. The finite-spacing mass-shell defect is a regulator error, not a tachyonic mass. This growing-lattice branch is distinct from the protected eight-site cube: neither its repair invariance nor the identity of its eight response lines is transferred to the valleys. The empty Fock state is not a ground state: at finite spacing, a one-particle mode at $(\pi/2, 0, 0)$ has energy $-c$, with zero contact for every U. This compares different number sectors; it is not a number-preserving extraction protocol from the empty sector. A filled vacuum, its interacting stability and a physical antiparticle dictionary remain unconstructed. A single 2×2 constant Hermitian mass anticommuting with all three kinetic Pauli matrices is exactly zero; this limited obstruction does not exclude valley coupling or many-body mass generation. The package `model/relativistic_sector_audit` therefore adds a controlled massless relativistic dynamical sector, not massive physical species, a photon assignment, an interacting relativistic vacuum theory or Einstein dynamics.

The preceding continuum construction does not itself execute a detector. The following finite periodic construction attaches model flavor and energy labels to a source-compiled occupation readout, and compares the resulting massless sector with observed particle data.

A detector identifies the model's propagating Weyl flavors. The massless sector has a concrete model-particle readout. On the periodic 8^3 lattice at $c = 1/8$, retain all 1,024 physical CAR modes of the original Weyl/contact action. For each $b \in \{0,1\}^3$, choose momentum labels $\ell_b = (4b_x, 4b_y, 4b_z + 1)$, with $k_b = \pi\ell_b/4$. For component eigenvalue $r = \pm 1$ of σ_z, the normalized mode is

$$\psi_{b,r}(x,s) = \frac{e^{ik_b\cdot x}}{\sqrt{512}}\delta_{s,(1-r)/2}, \qquad H\psi_{b,r} = E_{b,r}\psi_{b,r}, \qquad E_{b,r} = -r(-1)^{b_z}\frac{\sqrt{2}}{16}.$$

The contact term is exactly zero on the entire one-particle sector for every supplied $U \geq 0$. These sixteen modes are orthonormal, checked in Gaussian integers with amplitudes $(A+\sqrt{2}B)/32$. Their rank-one projectors satisfy $H\Pi_{b,r} = E_{b,r}\Pi_{b,r}$; the remaining one-particle projector commutes with H. Thus a label gives an actual source-generator energy, not a fitted interpretation of a peak. All modes have $N = 1$, odd CAR parity and fermionic exchange statistics from the supplied CAR algebra.

The operational species are eight Weyl flavors indexed by measured valley momentum. Their common energy magnitude is $\sqrt{2}/16 \simeq 0.0883883476483$ in original model units; it alone does not distinguish the flavors. With $\chi_b = -(-1)^{b_x+b_y+b_z}$, the positive-energy branch has cone helicity $\chi_b/2$ and the negative branch has the opposite helicity.

flavor b	momentum labels ℓ_b	positive-branch component	positive cone helicity
000	$(0,0,1)$	1	$-1/2$
001	$(0,0,5)$	0	$+1/2$
010	$(0,4,1)$	1	$+1/2$
011	$(0,4,5)$	0	$-1/2$
100	$(4,0,1)$	1	$+1/2$
101	$(4,0,5)$	0	$-1/2$
110	$(4,4,1)$	1	$-1/2$
111	$(4,4,5)$	0	$+1/2$

Here helicity uses the explicit proper valley frame $R_b = -\chi_b\,\mathrm{diag}((-1)^{b_j})$; it is $(-1)^{b_x+b_y}r/2$ on this family. The raw detector measures momentum and CAR component, so bare $\sigma_z/2$ is not silently identified with this cone spin or with physical particle spin. The total directional current $J_z(k) = -c\cos k_z\,\sigma_z$ has eigenvalue $E_{b,r}$ here, with nonzero current $\pm\sqrt{2}/8192$ on every forward z bond. Current is stationary on these selected modes; $[H, J_z] \neq 0$ at generic momenta. The witness is propagating flux, even for a uniform-density plane wave.

The Fourier readout is compiled with complete native errors. The detector applies the fermionic extension of $F[k,x] = e^{-2\pi ik\cdot x/8}/\sqrt{512}$ separately to each component, then copies all 1,024 occupations to fresh pointers. Its output includes every binary occupation string, the vacuum, all other one-particle addresses, multiple occupations and every export abort. On the chosen sixteen effects it measures the energies above; the other 1,008 one-particle records remain present without that energy assignment. On higher-number inputs these same records measure mode occupations, not the total interacting energy.

An explicit three-axis FFT uses 4,608 two-mode butterflies and 768 bit-reversal transpositions. Each butterfly is a mode phase followed by the native hopping word with coefficient $-i$ and duration $\pi/4$. All nonadjacent operations use graded FSWAP routes, including their double-occupation sign. Exact arithmetic in $\mathbb{Z}[\zeta_8]/(\zeta_8^4+1)$ proves the 64 entries and 64 Gram sums of the one-axis transform; tensoring gives the full mode map. One transform requests 1,260,800 adjacent FSWAPs

and 32,909,824 native pulses. Native excitation from the vacuum, inverse-transform preparation, forward-transform readout and all occupation pointers total 65,822,721 formal pulses. Original-H propagation is a separate specified model-law service, not an uncounted synthesized field word.

The complete literal-source error for preparation and detection is

$$\epsilon_{\text{source}} < 8.31 \times 10^{-9}.$$

The estimate retains all first-export failures and is valid with untouched references. It does not assume that literal native maps are exactly Gaussian or number preserving. Unitary original-H propagation on the full Fock space does not amplify preparation error. Therefore, at every model time, a prepared catalog member produces its correct native label with unconditional probability at least $1 - \epsilon_{\text{source}}$; all wrong, outside, vacuum, multiple-occupation and abort outputs together have probability at most ϵ_{source}.

The replay executes the ideal Gaussian comparison on all 1,024 one-particle rows and all sixteen inputs. At time eight its complete output-vector discrepancy from the exact spectral prediction is about 1.94×10^{-15}, separate from the rigorous native bound. A seeded 65,536-record illustration retains all sixteen label counts and every zero outside count for the exact uniform input mixture. These are model-simulated counts, not a literal 2^{1024}-row native execution or laboratory data. No survival normalization is used.

Mass readout and comparison with measured particles. For the periodic family $L = 8 \cdot 2^m$, $a = 2^{-m}$ and $c = (1/8)/a$, the model period stays eight and the momentum offset stays $p_z = \pi/4$. The exact branch energies satisfy

$$E_{\pm}(a) = \pm \frac{\sin(a\pi/4)}{8a} \longrightarrow \pm \frac{\pi}{32}, \qquad m_{\text{continuum}} = 0.$$

This is a direct finite-torus consequence of the original spectrum. The finite-spacing mass-shell defect is a regulator correction, not a tachyonic mass. No SI scale is assigned.

A separate retrospective comparison uses fixed PDG 2025 targets, which never enter the source or detector calculation:

physical target	measured mass in MeV/c^2	present continuum mass
electron	$0.51099895000 \pm 0.00000000015$	0
muon	$105.6583755 \pm 0.0000023$	0
tau	1776.93 ± 0.09	0

No common positive unit conversion maps zero to these positive masses; equal finite-momentum energies are not a rest-mass hierarchy.[7] The photon has spin one, so direct spin-preserving identification with these cone-helicity-1/2 modes also fails. Three such massless flavors give zero mass-squared differences, outside the nonzero ranges of the specified three-neutrino fit in PDG Table 14.7.[8] A single massless neutrino or a composite photon in another sector is not excluded. The model's Weyl flavors are operationally identified; direct identifications with measured physical species remain zero. The empty reference is not a ground state, and the negative branch is retained rather than renamed as a physical antiparticle. Interacting vacuum stability, mass generation, physical charges and the empirical dictionary remain additional work.

[7] PDG 2025 listings: https://pdg.lbl.gov/2025/listings/rpp2025-list-electron.pdf; https://pdg.lbl.gov/2025/listings/rpp2025-list-muon.pdf; https://pdg.lbl.gov/2025/listings/rpp2025-list-tau.pdf.

[8] The photon listing is https://pdg.lbl.gov/2025/listings/rpp2025-list-photon.pdf. The selected normal-ordering fit gives $\Delta m^2_{21} \in [6.92, 8.05] \times 10^{-5}\,\text{eV}^2/c^4$ and $\Delta m^2_{32} \in [2.388, 2.531] \times 10^{-3}\,\text{eV}^2/c^4$; these are reported 3σ fit ranges, not exact bounds: https://pdg.lbl.gov/2025/reviews/rpp2025-rev-neutrino-mixing.pdf.

The missing ground-reference and mass-generation steps admit further constructions with explicit domains. The original cube has a global interacting ground, and its one-dimensional restriction has an interaction-generated canonical charge gap. A specified refinement of that restriction also permits a controlled relativistic mass calculation. The following results retain the reference and coupling choices rather than combining these distinct domains into an assumed physical sector.

A global interacting ground on the original cube. The unchanged open $2 \times 2 \times 2$ Weyl/contact source, with $c = 1/8$ and $U = 1$, admits a certified comparison across its entire 65 536-dimensional Fock space:

$$E_G = -\frac{\sqrt{3}}{4}, \qquad N_G = 4, \qquad \dim\ker(H - E_G) = 5.$$

The five states form one hidden-spin $S = 2$ multiplet. No chemical potential, energy recentering, mass term, or replacement Hamiltonian is introduced. This is a finite interacting ground reference, not a claim of a unique or Lorentz-invariant infinite-volume vacuum.

The existing onsite unitary frame sends the original hopping to the spin-independent signed cube adjacency $A/16$, where

$$A = \begin{pmatrix} 0 & B \\ B^T & 0 \end{pmatrix}, \quad BB^T = 3I_4, \quad K = \begin{pmatrix} \sqrt{3}I_4 \\ -B^T \end{pmatrix}, \quad K^*K = 6I_4.$$

Filling the four negative orbitals $K/\sqrt{6}$ with hidden spin up gives an explicit normalized state G. Its occupation coefficients belong to $\mathbb{Q}(\sqrt{3})$, it has no doublons, and the original physical-frame eigenvalue equation is checked exactly. With $\lambda = \sqrt{3}/16$, the $N = 4$ bound $H \geq -4\lambda$ is saturated precisely by negative-band states annihilated by every local pair annihilator. On the full 70-dimensional four-particle negative-band space, the stacked contact map has spin-weight block dimensions $1, 16, 36, 16, 1$ and ranks $0, 15, 35, 15, 0$. Nonzero integer minors modulo101 certify the rank lower bounds; five explicit spin-lowered vectors certify the matching upper bounds. This proves the complete five-dimensional ground kernel in that number sector.

For $N \leq 3$, positivity gives $H \geq -N\lambda > -4\lambda$. For $N \geq 9$, the occupancy bound $D \geq N - 8$ gives $H \geq -N\lambda + N - 8 > 0$. For $N = 5, 6, 7, 8$, hidden $SU(2)$ symmetry ensures that every eigenvalue occurs in the balanced spin weight. Those block dimensions are $1568, 3136, 3920, 4900$. A numerical factorization proposes an integer matrix Q; only the following subsequently checked integer identity enters the proof:

$$H = 2^{-40}QQ^T - \frac{27}{64}I + R, \qquad \|R\|_2 \leq \|R\|_\infty < \frac{1}{512}.$$

All integer product and row-sum overflow bounds are checked before evaluation. Since QQ^T is positive semidefinite for every proposal, the argument requires no trusted floating eigenvalue, factorization error estimate, or nonzero-pivot premise. The exact comparison

$$\frac{\sqrt{3}}{4} > \frac{441}{1024} > \frac{217}{512}$$

then proves that every $N \neq 4$ lies more than $7/1024$ above E_G. This last bound concerns charged sectors; no numerical lower bound for neutral excitations inside $N = 4$ is asserted. Any density matrix supported on this finite ground space is passive for the stated H.

Positive charged spectral data above this ground. At the origin the onsite frame is the identity, so the probes below are the original physical CAR modes. Their exact local spectral data are

probe or component	N	weight	$\omega = E - E_G$
$c_\uparrow G$	3	1/2	λ
$c_\uparrow^\dagger G$	5	1/2	λ
$c_\downarrow G$	3	0	absent
$S = 5/2$ component of $c_\downarrow^\dagger G$	5	1/10	λ
$\eta = 5/2$ component of $c_\downarrow^\dagger G$	5	1/10	$1 + \lambda$

Thus the resolved model gaps are approximately 0.108253175473 and 1.108253175473 in source energy units. They are finite-volume gaps, not relativistic rest masses. The remaining down-addition weight is 4/5, with strictly positive spectral support and nonzero variance.

More explicitly, set $v = c_\downarrow^\dagger G$, with $\|v\| = 1$. The exact vectors

$$v_s = \frac{1}{5} S^- S^+ v, \qquad v_\eta = \frac{1}{5} \eta^+ \eta^- v, \qquad v_r = v - v_s - v_\eta$$

are mutually orthogonal and have squared norms $1/10, 1/10, 4/5$. The first two obey the original Hamiltonian eigenvalue equations at -3λ and $1 - 3\lambda$. Consequently the complete greater addition measure satisfies

$$\mu_\downarrow = \frac{1}{10}\delta_\lambda + \frac{1}{10}\delta_{1+\lambda} + \frac{4}{5}\mu_r, \qquad \operatorname{supp} \mu_r \subset (7/1024, \infty).$$

The displayed 1/10 values are exact symmetry-resolved contributions; without further resolution of μ_r, coincident residual atoms are not excluded. All moments below are checked over $\mathbb{Q}(\sqrt{3})$:

$$\mathbb{E}\omega = \frac{1}{2}, \qquad \mathbb{E}\omega^2 = \frac{131}{256}, \qquad \operatorname{Var}(\omega) = \frac{67}{256},$$

$$\mathbb{E}_r\omega = \frac{1}{2} - \frac{\sqrt{3}}{64}, \qquad \mathbb{E}_r\omega^2 = \frac{131}{256} - \frac{\sqrt{3}}{64}, \qquad \operatorname{Var}_r(\omega) = \frac{1069}{4096} > 0.$$

The contact expectation in v is 1/2; deleting that interaction changes its mean gap from 1/2 to zero. This channel therefore probes the actual contact interaction. Addition and removal carry model number changes $+1$ and -1 and odd CAR grading relative to G; electric charge and physical Lorentz spin do not follow from these labels.

Attachment to the declared repair process. The existing four-dimer repair completion has24 active columns, all entirely in $N = 6$. Its complete130-outcome dictionary therefore acts on $N = 3 \oplus N = 4 \oplus N = 5$ as zero for every active Kraus outcome and as identity for the outside projector. This6748-dimensional statement retains cross-number coherences and arbitrary reference systems. Since the original H preserves number, it also survives every finite interleaving with the ordinary field evolution. This is passive outside-service protection, not active correction or autonomous preparation of the ground.

The new finite ground removes the negative-frequency objection that applied to the earlier excited reference. It does not retroactively turn that earlier preparation into ground-state preparation. A native source and detector for G, the unresolved charged spectrum, a controlled growing-volume interacting vacuum, relativistic massive poles, physical dictionaries and units, and the geometric Einstein premises remain separate obligations. The exact certificate and negative controls are retained in `model/interacting_vacuum`.

An interaction-generated charge scale in the original one-dimensional law. There is a positive many-body charge threshold without inserting a Dirac mass. On the unchanged periodic Weyl/contact geometry $(L,1,1)$, $L=4m$, the onsite frame $G_x=(-1)^x(i\sigma_x)^x$ satisfies $G_x(i\sigma_x)G_{x+1}^\dagger=-I$ and has period four. Its CAR lift fixes local doublons, so the original action is exactly the repulsive Hubbard chain with $t=c/2=1/16$ and $U=1$. Four exact local frame identities, source-matrix checks through $L=16$, and an independent 256-state full-Fock witness bind this join. Unlike the earlier logarithmic-reference alternative Hamiltonian, no coupling is reassigned.

For the *half-filled canonical ground sector*, the imported Lieb–Wu charge-gap theorem gives

$$\Delta_c=\lim_{L\to\infty}[E_L(L+1)+E_L(L-1)-2E_L(L)]=U-4t+8t\sum_{k\ge1}\binom{1/2}{k}(2t/U)^{2k-1}\eta(2k-1).$$

Here the series is used only for $2t/U<1$. Twelve binomial terms and 96-term rational Euler bounds for each Dirichlet eta value enclose $\Delta_c=0.7715517157380375\ldots$ with width below $1.67\,10^{-25}$. The binomial tail is bounded by $8t|\binom{1/2}{K+1}|(2t/U)^{2K+1}/[1-(2t/U)^2]$; the nonnegative Euler coefficients give tail 2^{-96}. Floating quadrature is an independent check only.

Particle–hole symmetry fixes the charge costs for the original H:

$$E_+=\frac{U+\Delta_c}{2}=0.885775857869\ldots,\qquad E_-=\frac{\Delta_c-U}{2}=-0.114224142131\ldots.$$

Thus half filling is a specified canonical reference, not the original $\mu=0$ grand-canonical vacuum. The symmetric threshold $\Delta_c/2=0.3857758578690188\ldots$ uses the explicitly centered $K=H-(U/2)N$; Δ_c itself is invariant under any common $-\mu N$. The spin sector remains gapless, contact vanishes in the one-particle sector, and no preparation or complete-repair preservation of this many-body reference is established by the gap theorem.

The fixed-interaction continuum boundary. The equivalent positive integral for the charge gap and the elementary K_1 integral bounds imply, for every $t,U>0$,

$$\Delta_0\le\Delta_c\le\Delta_0\frac{1+3/(8A)}{1-e^{-2A}},\qquad A=\frac{2\pi t}{U},\qquad \Delta_0=\frac{8}{\pi}\sqrt{Ut}\,e^{-A}.$$

Consequently fixed U with $t=v/(2a)$ has $\Delta_c\to0$ as $a\to0$: this lattice gap does not itself supply a nonzero continuum mass. A sufficient additional running family is

$$a(A)=\frac{4ve^{-A}}{M\sqrt{2\pi A}},\qquad t(A)=\frac{v}{2a(A)},\qquad U(A)=\frac{\pi v}{a(A)A},$$

for a *chosen* $M,v>0$. It gives $\Delta_0/2=M$ exactly and $1\le\Delta_c/(2M)\le[1+3/(8A)]/[1-e^{-2A}]\to1$. This is a quantitative charge-threshold limit, not a derivation of M from the fixed source, a complete relativistic scaling theory, or a $3+1$-dimensional massive species. The receipt `model/mass_generation_audit/results.json` keeps these reference, coupling-family, and dimensional premises explicit.

The preceding charge-threshold result alone does not prove a mass shell. The next construction supplies energy and momentum together, with a controlled relativistic limit along an explicitly chosen coupling family.

A controlled relativistic charge mass on an explicit running family. The preceding charge gap can be promoted to a relativistic dispersion *in one spatial dimension on a specified coupling trajectory*. The source-bound half-filled Hubbard reference and centered generator $K = H - (U/2)N$ remain explicit. The staggered particle–hole map $d_{x,\downarrow} \mapsto (-1)^x d^\dagger_{x,\downarrow}$, with $d_{x,\uparrow}$ unchanged, preserves hopping and reverses the centered interaction. A complete 256-state CAR witness checks this duality and the translation identity $PTP^\dagger = -(-1)^{N_\downarrow}T$. Consequently the attractive spin dispersion becomes the repulsive charge dispersion, with the appropriate momentum-carrier shift. Here p is relative to that low-energy branch; the onsite source frame also carries its specified uniform internal rotation. The even-ring SO(4) selection rules are retained: an elementary thermodynamic quasiparticle is not automatically an isolated finite-ring state or a local-CAR pole.

Writing $A = 2\pi t/U$, $\theta = A\sin k$, $|k| < \pi/2$, and $v = 2ta$, the imported exact dressed dispersion has the source-normalized form

$$E = \frac{8t}{\pi} \sum_{\substack{m \geq 1 \\ m \mod d}} \frac{K_1(mA)}{m} \cosh(m\theta), \qquad vp = \frac{8t}{\pi} \sum_{\substack{m \geq 1 \\ m \mod d}} \frac{K_0(mA)}{m} \sinh(m\theta).$$

On $a(A) = 4ve^{-A}/[M\sqrt{2\pi A}]$, $t = v/(2a)$, $U = \pi v/(aA)$, the leading Bessel coefficient is exactly M. For $A \geq 1$ and $|\theta| \leq \Theta < A$, put

$$q = e^{-2(A-\Theta)}, \qquad r_E = \frac{1 + 3/(8A)}{1-q} - 1, \qquad r_p = \max\{1/(8A), q/(1-q)\}.$$

The Gamma-integral tangent bounds $1 - 1/(8z) \leq K_0(z)/b(z) \leq 1$ and $1 \leq K_1(z)/b(z) \leq 1 + 3/(8z)$, with $b(z) = \sqrt{\pi/(2z)}e^{-z}$, and the positive odd-mode tail prove

$$|E/M - \cosh\theta| \leq r_E \cosh\Theta, \qquad |vp/M - \sinh\theta| \leq r_p \sinh\Theta,$$

$$\left|(E^2 - v^2p^2)/M^2 - 1\right| \leq \cosh^2\Theta(2r_E + r_E^2) + \sinh^2\Theta(2r_p + r_p^2) \longrightarrow 0.$$

At $\Theta = 1$, the last bound is below 0.034 for $A = 64$ and 0.017 for $A = 128$. Thus the limiting charge-doublet dispersion has $E = M\cosh\theta$, $p = (M/v)\sinh\theta$, and the calculated rest mass is $m = M/v^2$. Exact rational finite-parameter enclosures and independent Bessel/integral controls are retained in `model/hubbard_relativistic_sector/results.json`.

For a concrete trajectory through the original point, take $A_0 = \pi/8$, $v = 1/8$ and $M_0 = e^{-\pi/8}/\pi$. Then $a(A_0) = 1$, $t(A_0) = 1/16$, $U(A_0) = 1$, and

$$M_0 = 0.2149329913552644\ldots, \qquad m_0 = \frac{64e^{-\pi/8}}{\pi} = 13.75571144673692\ldots \quad \text{in model mass units.}$$

This is the mass of the *chosen running trajectory*, not a unique mass selected by the original bare point. Indeed, for any $M > 0$, multiplying $a_M(A)$ by $1 + (M/M_0 - 1)e^{-(A-A_0)}$ gives the same starting point and limiting rest energy M. The initial lattice half-gap is different, and the fixed-U continuum family has zero limiting charge mass. No native ground preparation, repair-invariant charged pole, $3+1$-dimensional massive species, or empirical particle-mass identification follows from this one-dimensional result.

The original three-dimensional action also permits a growing-volume ground construction at fixed spacing. This closes ground-state existence and positive excitation energy without selecting the one-dimensional canonical reference or its coupling continuation. Its local response is then connected to the existing geometry theorem interfaces.

An original-action infinite-volume ground representation at fixed spacing. The full three-dimensional Weyl/contact action admits a translation-invariant, gauge-invariant ground state at its original zero chemical potential, without choosing a particle-number sector. This is a compactness existence theorem, not a computed or natively prepared state. Keep fixed finite $c > 0$, $U > 0$ and the complete quasi-local CAR algebra with two modes per site. The earlier proved full-Fock locality estimate supplies the strongly continuous dynamics τ_t. On each even-sided cube Λ_L of side $2L$, take the normalized projector onto the absolute minimum of H_{Λ_L}, extend this even state by the exterior vacuum, and average its translates over all sites of Λ_L. A diagonal subsequence of the bounded local matrix entries defines a state ω; positivity, normalization and CAR compatibility pass to the limit. The vanishing boundary-to-volume ratio makes ω translation invariant.

For a local A, including an odd charged probe, the local derivation $[H, A]$ contains only interaction terms touching its support. Whenever the support and its nearest neighbors lie inside the box,

$$\omega_L(A^*[H, A]) = \operatorname{Tr} \rho_L A^*(H_{\Lambda_L} - E_L)A \geq 0.$$

Bad translates occupy a vanishing fraction. Their real-part lower defect and imaginary magnitude are each bounded by $2\|A\|^2|\operatorname{supp} A|(U + 6c)$, so $\omega(A^*[H, A]) \geq 0$ in the limit. This uses the full ground minimum and would fail for a merely canonical ground with an unrestricted charged probe.

For a radius-R buffer, let $B_R(t)$ denote the existing volume-uniform local-dynamics boundary bound. Comparing both finite and infinite dynamics to this intermediate cube gives, for $\|A\| \leq 1$,

$$\left|\nu_L(\tau_t(A)) - \nu_L(A)\right| \leq \min\Big\{2,\, 2B_R(t) + 2\left[1 - (1 - R/L)^3\right]\Big\}.$$

First $L \to \infty$, then $R \to \infty$, establishes stationarity. The same comparison applies to two-point correlations. Every finite full-ground correlation has a positive spectral measure supported on $[0, \infty)$; compact-time convergence and bounded correlations pass all Schwartz Fourier tests supported on negative energies to the limit. The GNS implementation therefore satisfies

$$U_t\pi(A)\Omega = \pi(\tau_t(A))\Omega, \qquad U_t = e^{itL_\omega}, \qquad L_\omega \geq 0, \quad L_\omega\Omega = 0.$$

The generator measures excitation energy in this representation; no chemical potential has been inserted into the source action.

The state can be chosen at nonzero density. Tiling each even box by the certified four-particle cube vector gives contact zero, energy $-2\sqrt{3}c$ per eight-site tile, and zero expectation for every cross-tile hopping. Hence $E_L/|\Lambda_L| \leq -\sqrt{3}c/4$. Cartesian difference operators commute and the Pauli mixed terms cancel, giving $h^2 \leq 3c^2 I$ and $H \geq -\sqrt{3}cN$. Thus every such full ground has density at least $1/4$. The exact real-bipartite particle–hole identity $E(2V - N) = E(N) + U(V - N)$ excludes $N > V$ from an absolute minimum when $U > 0$. Spatial averaging before the subsequence passes both bounds:

$$\frac{1}{4} \leq \omega(n_{0\uparrow} + n_{0\downarrow}) \leq 1.$$

Half-filling and a particular density are not assumed or concluded. The original ground energy per site also has a limit e_∞. For every even cube side ℓ, tiling larger boxes gives the computable bracket, conditional on its finite full-ground energy,

$$\frac{E_\ell}{\ell^3} - \frac{3c}{\ell} \leq e_\infty \leq \frac{E_\ell}{\ell^3}, \qquad -\sqrt{3}c \leq e_\infty \leq -\frac{\sqrt{3}c}{4}.$$

The lower surface term counts intertile hopping norms; a product of gauge-invariant tile grounds gives the upper bound. Untiled boundary terms vanish per volume, and the bracket width tends

to zero. For the constructed averaged ground, $\omega(h_0) = e_\infty$, where h_0 assigns half of each incident bond to the origin. This supplies an energy-density observable, not a state-convergence rate or rest mass.

The contact interaction remains represented:

$$[H_U - H_0, a_0^* a_2] = U a_0^* a_2 (n_1 - n_3) \neq 0.$$

Every unital GNS representation is injective on each full local matrix algebra, so this quartic operator cannot disappear in representation. This faithfulness of the representation does not assert faithfulness of the ground-state functional. The claim is a nonzero represented interaction derivation, not a scattering or connected-vacuum-correlator theorem.

An exact source countercontrol prevents simply substituting the free filled sea. For a pure Slater projector with $P_{xx} = I_2/2$,

$$\mathrm{Var}(D) = \frac{V}{16} + \sum_{x \neq y} |\det P_{xy}|^2 \geq \frac{V}{16}.$$

If it is also a kinetic eigenstate, its interacting energy variance is $U^2 \mathrm{Var}(D) > 0$. On the original cube, the complete 3844-term negative-band filling has $N = 8$, $\langle D \rangle = 2$ and $\mathrm{Var}(H) = 2/3$ at $U = 1$. The exact checks and proof are retained in `model/thermodynamic_ground/results.json`. For local CAR probes, the same subsequence also supplies the analytic retarded response

$$G_{AB}(z) = -i \int_0^\infty e^{izt} \omega(\{\tau_t(A), B^*\})\, dt, \qquad \mathrm{Im}\, z > 0.$$

Each time tail after T is at most $2\|A\|\|B\|e^{-\eta T}/\eta$ for $\mathrm{Im}\, z \geq \eta > 0$. Compact-time locality and weak-star convergence, followed by this tail bound, give uniform convergence on compact subsets of the upper half-plane. The unknown finite-state convergence error is retained; no real-axis pole, local inverse or metric is inferred. The result closes fixed-spacing ground-state existence and the positive-energy representation for the actual interacting lattice. It supplies no selected ground functional, convergence rate, native preparation, repair stationarity, uniform gap, Lorentz continuum, physical particle assignment or Einstein dynamics.

Exact matter response before a geometric identification. The unchanged open $2 \times 2 \times 2$ Weyl/contact cube at $c = 1/8$, $U = 1$ supplies a state-dependent retarded response, including its exact eliminated-state memory. In the existing onsite hidden-spin frame, write $h = A/16$, where A is the real signed cube adjacency and $A^2 = 3I$. The retained full-Fock ground theorem gives the normalized highest-weight reference G, with four occupied negative-band up orbitals, $E_G = -\sqrt{3}/4$, and a strict gap $\delta = 7/1024$ to every different particle-number sector. The second exact stationary reference is $G_+ = \sqrt{2}\, c_{0\uparrow}^\dagger G$, of energy $-3\sqrt{3}/16$. No chemical potential, fitted mass, or new geometry law is inserted. Physical CAR responses are obtained by the existing onsite unitary frame; hidden spin is not assigned a physical Lorentz or electric-charge meaning.

For either reference ψ of energy E, the complete eight-site fermionic response, for $\mathrm{Im}\, z > 0$, is

$$\begin{aligned} \mathcal{G}_{xy}(z) &= -i \int_0^\infty e^{izt} \langle \{c_x(t), c_y^\dagger\} \rangle_\psi \, dt \\ &= \langle c_x [z - (H - E)]^{-1} c_y^\dagger \rangle_\psi + \langle c_y^\dagger [z + (H - E)]^{-1} c_x \rangle_\psi. \end{aligned}$$

There is an exact isometric dilation

$$\begin{aligned} V_y &= c_y^\dagger \psi \oplus \overline{c_y \psi}, & V^\dagger V &= I, \\ L &= (H_{N+1} - E) \oplus -\overline{(H_{N-1} - E)}, & \mathcal{G}(z) &= V^\dagger (z - L)^{-1} V. \end{aligned}$$

The conjugate removal space fixes the matrix indices; the present source arrays happen to be real. With $P = VV^\dagger$, $Q = I - P$, define $M_1 = V^\dagger LV$, $W = QLV$ and $B = QLQ|_{\operatorname{ran} Q}$. Exact block elimination gives

$$\mathcal{G}(z) = \left[zI - M_1 - \Sigma(z)\right]^{-1}, \qquad \Sigma(z) = W^\dagger (z - B)^{-1} W, \qquad W^\dagger W = M_2 - M_1^2.$$

Both L and B are self-adjoint; consequently $-\operatorname{Im} \mathcal{G}(z) > 0$ and $-\operatorname{Im} \Sigma(z) \geq 0$ in the upper half-plane. The eight observed columns do not replace the source Hilbert space: their exact complement remains in the memory kernel.

A computed change of response at fixed microscopic geometry. For the up channel of both references, the full cyclic identity $LV = Vh$ gives $\mathcal{G}_\uparrow(z) = (z - h)^{-1}$ and $W = 0$. For G, the up dilation retains all 56 addition and 56 removal states. The ground down channel has no removal component and retains all 560 states with $N_\uparrow = 4$, $N_\downarrow = 1$. Exact arithmetic over $\mathbb{Q}(\sqrt{3})$ gives

$$M_{1,\downarrow} = h + \tfrac{1}{2} I, \qquad W_\downarrow^\dagger W_\downarrow = \tfrac{1}{4} I.$$

Thus a static Hartree term alone does not reproduce this interacting response. Indeed $W_x = (n_{x\uparrow} - 1/2) c_{x\downarrow}^\dagger G$; orthogonal down locations and up occupation variance $1/4$ give the displayed Gram matrix directly.

For G_+, the down channel retains its entire 448-state invariant block. In lexicographic order of the eight cube sites, its exact matrices are

$$M_{1,\downarrow}^+ - h = \operatorname{diag}(1, \tfrac{2}{3}, \tfrac{2}{3}, \tfrac{1}{2}, \tfrac{2}{3}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}),$$
$$(W_\downarrow^+)^\dagger W_\downarrow^+ = \operatorname{diag}(0, \tfrac{2}{9}, \tfrac{2}{9}, \tfrac{1}{4}, \tfrac{2}{9}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}).$$

The first-moment change is exactly the added orbital density, $(1/2, 1/6, 1/6, 0, 1/6, 0, 0, 0)$. Removing the contact term in the explicit $U = 0$ control removes both this Hartree term and the ground down memory, giving $(z - h)^{-1}$. That comparison does not declare G a full free-Fock ground. The excited reference receives no positive-gap assertion.

The complete local charged spectral measure. For the original physical origin probe $v = c_{0\downarrow}^\dagger G$, the retained calculation resolves all 78 distinct positive spectral lines, including the earlier unresolved weight $4/5$. A monic degree-78 polynomial p over $\mathbb{Q}(\sqrt{3})$ is admitted by the exact source equation $p(16H)v = 0$. A nonzero 78×78 Hankel determinant modulo 97, with $\sqrt{3} \mapsto 10$, proves the matching cyclic rank lower bound. Exact moments determine a numerator q such that

$$\langle v, (z - 16H)^{-1} v \rangle = \frac{q(z)}{1296\, p(z)}.$$

The denominator 1296 is the squared norm of the inherited integer state numerator. Modular reconstruction and a numerical eigensolver only propose coefficients and brackets. Integer annihilation and rank checks, 78 disjoint strict sign changes, and outward 1024-bit interval arithmetic certify every root and its strictly positive residue $q(x)/(1296 p'(x))$. Each charged gap is $(x + 4\sqrt{3})/16$.

The smallest gap is approximately 0.029950938342053226, with weight 0.3130113772155167; the largest is approximately 1.8360744654423855, with weight $2.711810625844545 \times 10^{-10}$. The receipt retains every gap and weight in rational intervals rounded outward to 24 decimal places. Polynomial substitution proves that the two lines $\sqrt{3}/16$ and $1 + \sqrt{3}/16$ each have total weight exactly $1/10$. The remaining 76 lines have total weight $4/5$, leaving zero unresolved weight for this local probe. This is the complete local cyclic measure, not the entire 560-state spectrum, a native detector execution, an infinite-volume pole theorem, or a relativistic mass assignment.

Controlled low-frequency coefficients. For the interacting ground down channel, put $K = H_{4,1} - E_G$. The full-ground certificate gives $K > \delta I$. Also, $D \le 1$ in this block, the kinetic norm is at most 5λ, and $-E_G = 4\lambda$, with $\lambda = \sqrt{3}/16$; hence $K \le 1 + 9\lambda < 2$. The same bounds hold for its compression B. For $|z| \le r < \delta$ the exact memory has the convergent expansion

$$\Sigma(z) = -\sum_{n=0}^{p} z^n C_n + R_p(z), \qquad C_n = W^\dagger B^{-n-1} W \ge 0,$$
$$\|R_p(z)\| \le \frac{r^{p+1}}{4\delta^{p+1}(\delta - r)}.$$

These are finite gapped frequency coefficients, not a spatial derivative expansion or a relativistic cone. All retarded moments $M_0, \dots, M_4$ are also computed exactly; the corresponding large-frequency remainder is $2^{p+1}/[R^{p+1}(R-2)]$ for $|z| \ge R > 2$.

The static response is computed with an exact residual certificate. For $C = V^\dagger K^{-1} V = -\mathcal{G}_\downarrow(0)$,

$$\|C - (12.9058381883 I - 4.9026412320 A)\| < 10^{-7},$$
$$\|\Sigma(0) - (-0.3633623064 I - 0.0105943755 A)\| < 10^{-6}.$$

A numerical solve only proposes a matrix with denominator 2^{40}. Integer arithmetic encloses its full $\mathbb{Q}(\sqrt{3})$ residual; division by the proved δ bounds the solution error. The exact isometry then bounds C, and $C > I/2$ bounds its inverse. An independent rational inverse residual certifies $\Sigma(0) = C^{-1} - M_1$. No numerical tolerance admits the theorem, and exact equality to an affine adjacency polynomial is not inferred.

The existing geometry interface and its remaining join. The same-action thermodynamic-ground construction separately supplies analytic local responses along its ground subsequence on compact subsets of $\operatorname{Im} z > 0$. The cube calculation does not supply a rate of convergence to that state, and its finite gap is not transferred to infinite volume. The existing `wave_density_metric` constructor takes a complete local action density $\mathcal{C} = \mu \operatorname{diag}(-w, a)$, with $\mu, w > 0$ and $a > 0$, and recovers $g = \sqrt{-\det \mathcal{C}}\, \mathcal{C}^{-1}$. The existing `source_tetrad` interface in the common matter join then constructs the common coframe and the scalar, Weyl, Dirac and Maxwell principal-symbol identities for its supplied density. The present result supplies an actual matter-dependent inverse response from which such coefficients must be derived. It does not substitute a Hartree diagonal for a spatial principal symbol, identify the cube with the distinct screen source, or complete the common action-normalization join. Fixed microscopic geometry is compatible with this emergent-response route; a new fundamental metric register is not required. Common local limits, modular and area response, full stress, and the same-family Einstein balance premises remain to be proved. No new physical species, native response detector, or completed G1–G6 premise is claimed.

The finite ground and its charged probe can be prepared and read by the declared observer service. The full matrix spectrum extends the local probe calculation. Separately, the infinite ground admits a spatially local inverse response with nonzero interaction memory at complex frequencies separated from the real axis. These constructions supply additional inputs to the existing field and geometry interfaces.

Native preparation of the same ground and its charged response. The original cube ground and its local charged spectral measure have an explicit native source preparation and readout. The exact highest-weight ground G, with $N = 4$ and $E_G = -\sqrt{3}/4$, has 62 nonzero hidden-up occupation amplitudes $(a + b\sqrt{3})/36$, where a, b are integers and $ab = 0$. Its squared

amplitudes are therefore exact integers divided by 1296. These source-derived weights and signs determine an eight-bit binary amplitude tree; no fitted spectrum or unknown-state oracle is used. The existing compiler emits 5,379 literal native pulses. An independent rational interval check consumes every event, inverts the Walsh layers, and verifies the sine and cosine of each conditional rotation against the exact signed square-root ratios. Disjoint control blocks give a maximum error per level, and telescoping the levels and complete native load/export maps bounds the full initialized preparation, including aborts.

Eight blank down modes are included by an exact isometric embedding into all 65,536 Fock rows. A further 184 native pulses apply the inverse onsite frame. Each local frame word is checked on its full four-state Fock space, including vacuum and doublon phases, using $e^{is\pi P/4} = (I + isP)/\sqrt{2}$ and Gaussian-integer products. Its literal-angle and native-instrument errors are separately enclosed. The resulting complete preparation-state error is below 10^{-10}; the retained exact bound is approximately 8.67×10^{-13}.

At the origin the hidden and original frames coincide, and the down mode is empty in G. Thus

$$\gamma_1 = Z_0 X_1 = c_1 + c_1^\dagger, \qquad \gamma_1 G = c_1^\dagger G, \qquad \|\gamma_1 G\| = 1.$$

The Jordan–Wigner sign is essential: replacing $Z_0 X_1$ by X_1 produces an orthogonal probe because the exact up occupation is $1/2$. This unitary creation preserves the entire initial exterior CAR state: disjoint even observables commute, and exterior odd expectations vanish in both definite-parity states. The origin-down occupation changes from zero to one and total number from four to five. Its mean added energy is $1/2$ with variance $67/256$; this is a localized initial defect with a conserved number flag, not a persistent localization or mass-eigenstate claim. A 27-pulse pointer word prepares $(|0\rangle G + |1\rangle \gamma_1 G)/\sqrt{2}$. Its controlled Majorana is the product of commuting Pauli rotations in $\exp[i\pi(I - Z_{\text{pointer}})(I - \gamma_1)/4]$. This explicitly supplied native qubit service maintains a phase relation between the $N = 4$ and $N = 5$ arms; it does not derive a physical fermion-parity reference from passive repair.

Both arms evolve under the same original Hamiltonian. With a second controlled Majorana and pointer analyzer of phase ϕ,

$$C(t) = \langle \gamma_1 G, e^{-it(H - E_G)} \gamma_1 G \rangle,$$
$$P_0(\phi, t) = \tfrac{1}{2} \left[1 + \mathrm{Re}(e^{i\phi} C(t)) \right].$$

The ground-energy phase is supplied by the evolving reference arm; no separate scalar offset is inserted into the source law. The two executed settings have $t = 1$ and $\phi = 0, \pi/2$. Each setting uses two half-time segments, each with 32 symmetric splitting steps and 337,920 native pulses, followed by a 29-pulse analyzer. Including its ground preparation and pointer, a complete setting uses 681,459 pulses on the full 17-bit field/pointer register.

Protection and complete output accounting. The five C repair seams follow the first time segment, and the five E seams follow the second. The complete repair theorem acts as identity on $N4 \oplus N5$, including cross-number coherences: all native branches vanish there, and the outside branch is identity. This is passive protection during the waits. Preparation and control words are not claimed to remain in this subspace at every intermediate pulse. The final inverse Majorana can create an $N = 6$ component after the charged particle has moved; no repair is applied after that analyzer.

Every source first-export failure is retained as its own terminal record. The numerical execution also retains all 130 repair outcome slots, including off-subspace outputs from control error, and all

six final pointer outcomes. A native repair outcome terminates with that record; continuation uses the outside branch without renormalization. The lossless archive retains the prepared full ground, both full final states and every first-failure weight table. Individual failure vectors remain addressable from their actual source prefixes.

Connection to the resolved spectrum. All field and pointer words are consumed against the exact source templates. Their complete operator errors, the preparation error and the full-Fock Strang errors of both time segments give a complete output error below 3/5000 for either setting. Contractivity through the complete repair and readout channels preserves the telescoping bound. The same bound controls each unconditioned pointer probability. Independently, the 78 certified spectral intervals evaluate

$$C(t) = \sum_{j=1}^{78} w_j e^{-i\Delta E_j t}$$

with rational sine and cosine enclosures. Adding the source error gives certified pointer intervals, checked against the literal execution and an independent exponential of the original Hamiltonian. The $\pi/2$ outcome is separated from $1/2$; the $U = 0$ comparator has the purely real response $\cos(\sqrt{3}t/16)$. Zero-time native runs check the analyzers as well. These two settings measure the charged characteristic function. They do not independently resolve all 78 lines, select an autonomous ground, prepare a thermodynamic vacuum, establish laboratory units, or turn finite energy gaps into relativistic particle masses.

The complete finite matrix spectrum and its exact spatial algebra. The 78-line local down-addition result extends to all eight source columns $V_x = c_{x\downarrow}^\dagger G$. On the entire 560-state invariant sector, exact arithmetic proves $q(16H)V = 0$ for the same monic degree-78 polynomial q and verifies all 78 matrix moments in $\operatorname{span}\{I, A\}$, where $A = A^{\mathsf{T}}$, $A^2 = 3I$ and $\operatorname{tr} A = 0$. Polynomial division of the resolvent then gives

$$\mathcal{G}(z) = \frac{16\big[p_I(x)I + p_A(x)A\big]}{1296q(x)}, \qquad x = 16z - 4\sqrt{3}.$$

The numerator coefficients are exact convolutions of q and those source moments. This is a rational resolvent identity, rather than a truncated large-frequency expansion.

A second exact polynomial calculation gives $q \mid (p_I^2 - 3p_A^2)$. At each simple root x_i, define $\alpha_i = p_I(x_i)/(1296q'(x_i))$ and $\beta_i = p_A(x_i)/(1296q'(x_i))$. The earlier interval certificates give $\alpha_i > 0$; the new identity gives $3\beta_i^2 = \alpha_i^2$, and strict interval signs determine $s_i = \operatorname{sign}\beta_i$. Therefore

$$R_i = \alpha_i I + \beta_i A = 2\alpha_i P_{s_i}, \qquad P_\pm = \tfrac{1}{2}(I \pm A/\sqrt{3}).$$

Both $P_\pm$ have rank four. There are 40 plus-band lines and 38 minus-band lines; each band's $2\alpha_i$ weights sum to one. The rank of $E_i V$ equals that of its Gram matrix R_i, so the reducing cyclic space generated by all eight probes has dimension $78 \cdot 4 = 312$. The remaining 248 source dimensions are orthogonal to these probes and their time translates; their spectrum is not claimed.

Writing $\mathcal{G} = FI + JA$, both band functions have strictly negative imaginary part for $\operatorname{Im} z > 0$. Hence

$$\mathcal{G}^{-1} = \frac{FI - JA}{F^2 - 3J^2}, \qquad \Sigma = zI - (A/16 + I/2) - \mathcal{G}^{-1}.$$

The full finite inverse response and memory are exactly onsite plus the original signed nearest-neighbor adjacency. This special cube algebra does not supply an infinite-volume spatial limit or a Lorentz metric.

The certified common spectral support lies strictly between $g = 299/10000$ and $b = 37/20$. On the reducing cyclic space, both $K = H - E_G$ and its compression perpendicular to the observed columns obey these bounds. The exact coupling $W = KV - VM_1$ lies in that same space, so the orthogonal unobserved sector contributes zero to the Schur memory. Using $W^\dagger W = I/4$, its order-p Taylor remainder satisfies

$$\|R_p\| \le \frac{r^{p+1}}{4g^{p+1}(g-r)}, \qquad |z| \le r < g.$$

This improves the earlier conservative full-sector disk without assigning the stronger gap to the unresolved sector or to infinite volume. Evaluation at $x = -4\sqrt{3}$ computes $-\mathcal{G}(0)$ and $\Sigma(0)$ exactly over $\mathbb{Q}(\sqrt{3})$, including the earlier bounded coefficients 12.9058381883 and -4.9026412320. No floating linear solve, native detector execution, physical species assignment or relativistic rest-mass inference is used in this extension.

Excluding a complete finite-gap mass dictionary. The full support gives an exact test of assigning these visible single-charge gaps to charged-lepton rest energies under one positive unit factor. Its largest-to-smallest ratio is below 62, including the positive up-channel line. The retained retrospective PDG 2025 benchmark has muon/electron ratio above 206 and tau/electron ratio above 3477, even in the declared central-value plus/minus reported-uncertainty boxes. If a source ratio is at most R and the target ratio at least $Q > R$, simultaneous relative errors at most $\epsilon < 1$ require

$$\frac{1-\epsilon}{1+\epsilon} \le \frac{R}{Q}, \qquad \epsilon \ge \frac{Q-R}{Q+R}.$$

The exact rational calculation therefore forces an error above 54 percent for at least one member of an electron/muon assignment, and above 96 percent for an electron/tau assignment, for every common positive scale and every choice of visible lines. The target values do not enter the source law or spectrum. This rejects the stated finite-gap mass dictionary, not a continuum, an unobserved or composite sector, or OPH as a whole. The finite gaps are not thereby proved to be relativistic rest masses, and no eligible prospective comparison is added.

A local inverse response on the interacting infinite ground. The original zero-chemical-potential Weyl/contact action on $\mathbb{Z}^3$ admits a controlled spatial inverse response on the earlier constructed thermodynamic ground, at fixed spacing and away from real frequency. This does not extrapolate the special two-eigenvalue cube identity. Let ω be the translation-averaged full-ground limit, L_ω its positive GNS generator, and

$$L = L_\omega \oplus -\overline{L_\omega}, \qquad V_{x\sigma} = c^\dagger_{x\sigma}\Omega \oplus \overline{c_{x\sigma}\Omega}.$$

The CAR give an isometry V from $\ell^2(\mathbb{Z}^3;\mathbb{C}^2)$. Define J in the same way from $b_{x\sigma} = n_{x\bar\sigma}c_{x\sigma}$ and write $\rho_{x\sigma\tau} = \omega(c^\dagger_{x\tau}c_{x\sigma})$. Exact onsite CAR identities, with vanishing different-site anticommutators, give

$$R = \bigoplus_x (\mathrm{tr}(\rho_x)I - \rho_x), \qquad V^\dagger J = J^\dagger V = J^\dagger J = R,$$
$$LV = Vh + UJ, \qquad 0 \le R \le I, \quad \|LV\| \le \sqrt{3}c + U.$$

Finite-support approximants and closedness of L extend the last identity to the entire one-particle input space. With $P = VV^\dagger$, $Q = I - P$, the bounded off-diagonal coupling is $W = QLV =$

$U(J-VR)$. Subtracting $WV^\dagger + VW^\dagger$ from L proves self-adjointness of its compressed complement $T = QLQ|_{Q\,\mathrm{Dom}\,L}$. The exact infinite-space Schur identity is therefore

$$\begin{aligned} \mathcal{G}(z) &= V^\dagger (z-L)^{-1} V = [zI - h - UR - \Sigma(z)]^{-1}, \\ \Sigma(z) &= W^\dagger (z-T)^{-1} W, \qquad\qquad W^\dagger W = U^2(R-R^2). \end{aligned}$$

In particular $\|\Sigma(z)\| \le U^2/(4\,\mathrm{Im}\,z)$.

This ground has strictly nonzero memory for $U > 0$. The source hidden-spin frame obeys $R_{x+t} = \pm R_t R_x$, so translations normalize its global SU(2) symmetry. Every full-box ground projector and the exterior vacuum are invariant; translated averages and weak-star limits retain the symmetry. Hence $\rho_x = rI$, with the earlier density theorem giving $1/8 \le r \le 1/2$, and

$$\frac{7U^2}{64} I \le W^\dagger W = U^2 r(1-r) I \le \frac{U^2}{4} I.$$

The spectral theorem gives $iy\Sigma(iy) \to W^\dagger W$ strongly as $y \to \infty$. Thus the same-mode response cannot be that of a number-conserving static quadratic Hamiltonian with zero memory. Neither a scattering theorem nor faithfulness of the local ground-state density matrices follows.

A quantitative connected four-point witness. The same ground has a non-Gaussian multi-time field, a stronger statement than nonzero two-point memory. Fix $a = c_{x\sigma}$ and $n = n_{x\bar{\sigma}}$, and set

$$A(t) = \{\tau_t(a), a^\dagger\}, \qquad D(t) = A(t) - \omega(A(t))I, \qquad F(t) = \omega(D(t)^\dagger D(t)).$$

The local commutator gives $D(0) = 0$ and $D'(0) = -iU(n - rI)$. Since $n^2 = n$, $F(t)/t^2 \to U^2 r(1-r) \ge 7U^2/64$. An explicit remainder uses $M = 9c^2 + 12cU + U^2$ and $\delta = [H, \cdot]$:

$$\|\delta^2(a)\| \le 3c(3c+U) + U(6c+3c+U) = M, \qquad \|D(t) + itU(n - rI)\| \le 2Mt^2.$$

Thus, for $U > 0$,

$$F(t) \ge \frac{U^2 t^2}{64} > 0, \qquad 0 < |t| \le \frac{U}{16M}.$$

For odd zero-mean operators define the connected fermionic function

$$\begin{aligned} \kappa_4(B_1, B_2, B_3, B_4) = {} & \omega(B_1B_2B_3B_4) - \omega(B_1B_2)\omega(B_3B_4) \\ & + \omega(B_1B_3)\omega(B_2B_4) - \omega(B_1B_4)\omega(B_2B_3). \end{aligned}$$

With $(B_1, B_2, B_3, B_4) = (a, \tau_t(a)^\dagger, \tau_t(a), a^\dagger)$, $F(t)$ is the sum of κ_4 on words $1234, 1243, 2134, 2143$. All crossed contractions, including anomalous ones, cancel pairwise. At least one connected function therefore has magnitude $U^2t^2/256$ or greater on the displayed interval. This excludes a quasifree representation of the complete multi-time field by linear fermionic fields, including pairing and auxiliary Gaussian modes. Indeed any quadratic evolution on an enlarged CAR algebra preserves linearity of an observed mode, making its anticommutator scalar and $D(t) = 0$ in every reference state. Thus even a non-Gaussian reference cannot realize this witness through such a linear quadratic embedding. It does not exclude nonlinear encodings or establish continuum scattering.

The full ordered Euclidean hierarchy. The same positive ground generator $L_\omega \geq 0$ defines, for ordered real times $s_1 \leq \cdots \leq s_n$ and bounded local CAR operators,

$$S(A_1, s_1; \ldots; A_n, s_n) = \langle \Omega, A_1 e^{-(s_2-s_1)L_\omega} A_2 \cdots e^{-(s_n-s_{n-1})L_\omega} A_n \Omega \rangle, \qquad |S| \leq \prod_j \|A_j\|.$$

Every finite approximant uses one common translated full-ground state and its positive generator $H_\Lambda - E_\Lambda$ for the whole word, with zero exterior Hamiltonian, before averaging over translations. For each positive gap the spectral theorem gives

$$e^{-sL} = \int_{\mathbb{R}} \frac{s}{\pi(s^2+t^2)} e^{itL}\, dt, \qquad L \geq 0.$$

The product integral is dominated by $\prod_j \|A_j\|$. Its real-time integrand is the expectation of $A_1 \tau_{t_1}(A_2) \tau_{t_1+t_2}(A_3) \cdots$. Local-dynamics bounds, telescoping the product, and the same weak-star ground subsequence give uniform convergence on compact time boxes. The omitted product-measure tail is at most $\sum_j 2s_j/(\pi T_j)$, capped by one; a finite-versus-limit comparison costs twice this tail times the product of operator norms. Dominated convergence therefore supplies every ordered Euclidean n-point limit on this one ground, without another subsequence or a state-convergence rate. Zero gaps are handled by the identity operator.

For a positive-time word $F = (A_1, t_1; \ldots; A_n, t_n)$ define

$$\Psi_F = e^{-t_1 L_\omega} A_1 e^{-(t_2-t_1)L_\omega} A_2 \cdots e^{-(t_n-t_{n-1})L_\omega} A_n \Omega.$$

The reflected list is $(A_n^*, -t_n; \ldots; A_1^*, -t_1)$, in that order. Concatenation with a second positive-time word gives $S(\theta F, G) = \langle \Psi_F, \Psi_G \rangle$; hence every reflected correlation matrix is positive semidefinite. No CAR factors are reordered. Exact rational two-mode controls check the Gram identity, all 31 principal minors of a five-word example, and a negative odd-word cross entry. This proves ordered time-reflection positivity at fixed spacing. It does not establish $O(4)$ covariance, bosonic permutation symmetry, a spatial continuum, or the full relativistic Euclidean axioms.

Uniform inverse and spatial decay bounds. Fix $\operatorname{Im} z \geq \eta > 0$, $|z| \leq Z$, with $Z \geq \eta$. For a unit vector $v = Vf$, Cauchy–Schwarz applied to $\langle (\bar{z} - L)v, (z-L)^{-1}v \rangle = 1$ yields

$$-\operatorname{Im} \mathcal{G}(z) \geq B_0^{-1} I, \qquad \|\mathcal{G}(z)^{-1}\| \leq B_0, \qquad B_0 = \frac{(Z + 2c + U)^2}{\eta}.$$

Both $\mathcal{G}$ and its adjoint are bounded below; its range is therefore closed and dense, which proves surjectivity in infinite volume. No determinant or finite-volume gap is used.

Let $d = \|x - y\|_\infty$. The original graded locality bound gives

$$\|\mathcal{G}_{xy}(z)\| \leq A(1+d)^2 e^{-2\mu d},$$
$$\mu = \min\{1/8, \eta/(384c)\}, \qquad A = 192c/\eta^2 + 8/\eta.$$

For $d \geq 2$, approximate the evolved odd operator in the radius $d-1$ cube, whose anticommutator with the remote mode is zero. Split the Laplace integral at $(d-1)/(192c)$ and use the previous full-CAR boundary estimate on the first part and the uniform norm bound on its tail. This proves the displayed constants without assuming spatial clustering of ω.

Put $q = (1+\mu)^{-1}$ and $T_n(q) = \sum_{d \geq 1} d^n q^d$. These are rational functions with positive Eulerian coefficients. Define

$$\widehat{S} = 24T_5(q) + 48T_4(q) + 26T_3(q) + 4T_2(q) + 2T_1(q),$$
$$\alpha = \min\{\mu, (2B_0 A \widehat{S})^{-1}\} > 0.$$

For any bounded real one-Lipschitz function f, the block Schur test gives

$$\|e^{\alpha f}\mathcal{G}e^{-\alpha f}-\mathcal{G}\|\le\alpha A\widehat{S}\le(2B_0)^{-1}.$$

A Neumann series bounds the conjugated inverse by $2B_0$. Choosing $f(u)=\min\{\|u-y\|_\infty,R\}$ with $R\ge d$ proves

$$\begin{aligned}\|(\mathcal{G}^{-1})_{xy}(z)\|&\le 2B_0e^{-\alpha d},\\ \|\Sigma_{xy}(z)\|&\le(Z+U+2B_0+2c)e^{-\alpha d}.\end{aligned}$$

Only bounded weights were used. The constants are conservative bounds, not estimates of a physical correlation length.

The controlled spatial symbol and its geometric boundary. Translation invariance and exponential summability give an analytic two-by-two inverse-response symbol $K(k,z)$ for $\|\operatorname{Im}k\|_1<\alpha$. For real momentum increment p, its total-degree-n Taylor remainder is bounded by

$$\frac{2B_0\|p\|_1^{n+1}}{(n+1)!}\sum_{x\in\mathbb{Z}^3}\|x\|_\infty^{n+1}e^{-\alpha\|x\|_\infty}.$$

The shell count $24d^2+2$ and the same rational T_n functions give explicit upper bounds on this convergent sum. A frequency circle of radius $\eta/2$ gives, for $|w|<\eta/2$, the additional remainder $B_*(2|w|/\eta)^{n+1}/(1-2|w|/\eta)$, where B_* is the above inverse bound evaluated at $(\eta/2,Z+\eta/2)$.

These estimates supply actual local spatial coefficients on the interacting infinite ground, with a controlled complex-frequency domain. They do not compute the ground functional or those coefficients, and deteriorate as $\eta\to0$. A real hyperbolic principal symbol, common matter coefficients and full action normalization are required before applying the existing `wave_density_metric` and `source_tetrad` interfaces. The joint modular, area, stress and Einstein balance requirements remain separate.

For a refinement $c=v/a$ at fixed $v,U,Z,\eta>0$, the retained estimates have $\alpha=O(a^9)$ and therefore physical momentum width $\alpha/a=O(a^8)$; the connected-witness time window is $O(a^2)$. These shrinking bounds do not prove uniform continuum control or a no-go theorem for the action.

The full ground-family spectrum and its internal spin. The complete five-dimensional $N=4$ ground space closes a remaining finite spectral ambiguity. Work in the exact hidden-spin frame, with G_0 its spin-two highest-weight ground, $\lambda=\sqrt{3}/16$, and $V_{\uparrow,x}=c_{x\uparrow}^\dagger G_0$, $V_{\downarrow,x}=c_{x\downarrow}^\dagger G_0$. The actual 56×560 raising matrix $T=S_+$ obeys

$$TV_\downarrow=V_\uparrow,\qquad TT^\dagger=5I,\qquad H_{5,0}T=TH_{4,1}.$$

The full spatial Gram is $V_\uparrow^\dagger V_\uparrow=P_+$, and $(H-E_G)V_\uparrow=\lambda V_\uparrow$. At magnetic weight $3/2$,

$$V_{\text{high}}=T^\dagger V_\uparrow/5,\qquad V_{\text{low}}=V_\downarrow-V_{\text{high}},\qquad TV_{\text{low}}=0,\qquad V_{\text{high}}^\dagger V_{\text{high}}=P_+/5.$$

The retained residue at λ is exactly $P_+/5$. Positivity therefore leaves no low-spin contribution at that line: it has total hidden spin $5/2$, while the other 77 visible lines have total hidden spin $3/2$.

This conclusion covers every ground vector. Put $G_k=S_-^kG_0$, with $\|G_k\|^2=k!4!/(4-k)!$, $0\le k\le4$. Exact occupation-space CAR give, for every site,

$$\begin{aligned}c_\downarrow^\dagger G_k&=S_-^{k+1}V_\uparrow/5+S_-^kV_{\text{low}},\\ c_\uparrow^\dagger G_k&=(5-k)S_-^kV_\uparrow/5-kS_-^{k-1}V_{\text{low}}.\end{aligned}$$

The second term vanishes at $k = 0$. All eighty creator columns are checked on the original source; the Hamiltonian commutes with every ladder. No coherent ground superposition or linear CAR probe can introduce another addition frequency.

The full $N = 5$ space has spin multiplicities $56, 504, 1008$ at $S = 5/2, 3/2, 1/2$, respectively. Four reachable copies occur at λ, and four at each of the other 77 lines. Thus the complete reachable dimension is

$$4 \cdot 6 + 77 \cdot 4 \cdot 4 = 1256 \quad \text{inside} \quad \binom{16}{5} = 4368.$$

The dark complement has dimensions $312, 784, 2016$ in those three spin sectors. These are dimensions of source subspaces, not full energy degeneracies; their dark eigenvalues remain uncomputed. All annihilator probes similarly generate sixteen $N = 3$, spin-3/2 states with positive removal cost λ, leaving 544 dark dimensions.

The invariant mixed-reference energy readout. For $\rho_G = P_G/5$, the exact addition and retarded matrices are

$$\overline{\mathcal{G}}_{\mathrm{add}}(z) = \frac{1}{2}\left[\mathcal{G}_{\downarrow}(z) + \frac{P_+}{z-\lambda}\right] \otimes I_2,$$
$$\overline{\mathcal{G}}_{\mathrm{ret}}(z) = \frac{1}{2}\left[\mathcal{G}_{\downarrow}(z) + \frac{P_+}{z-\lambda} + \frac{P_-}{z+\lambda}\right] \otimes I_2.$$

At each site and fixed spin, the addition line λ has weight $3/10$, and each other line has half its previous weight. The total addition weight is $3/4$; conditional weights are therefore $2/5$ at λ and $2w_i/3$ elsewhere. Removal has weight $1/4$ at the signed retarded pole $-\lambda$. All 78 positive lines survive. The 79 signed poles have rank-eight residues, giving a minimal mixed-state response realization of dimension 632. This realization keeps sixteen observed columns and differs from the physical reachable range that keeps all five ground inputs distinct.

Direct full-source Gram matrices verify every spin and spatial entry of the zeroth and first moments, including

$$\overline{M}_{0,\mathrm{ret}} = I_{16}, \qquad \overline{M}_{1,\mathrm{ret}} = (A \otimes I_2)/16 + I_{16}/4.$$

The actual physical-frame block is $g_{xy} R_x^\dagger R_y$; intersite spin coherence is retained, while local blocks are scalar. The same positive-energy envelope and finite charged-lepton ratio exclusion hold for every ground choice and linear single-CAR probe. These total internal spins include the spin-two background. They do not identify intrinsic particle or Lorentz spin, a measured species, or a relativistic mass. The result does not exclude composite probes or a continuum spectrum, and does not execute native preparation or complete finite-shot spectroscopy of the mixed reference.

Complete finite-resolution energy readout. For a unit one-particle probe f, the same ground defines the probability measure on signed addition and removal energies

$$\mu_f(B) = \langle Vf, 1_B(L) Vf \rangle.$$

The complete Kraus pair $c(f)^\dagger, c(f)$ labels addition and removal; each branch has a nonnegative physical excitation energy. Only the response-frequency variable assigns a minus sign to removal. The exact identities $LV = Vh + UJ$ and $V^*J = J^*J = rI$ give the operator moments

$$M_0 = I, \qquad M_1 = h + UrI, \qquad M_2 = h^2 + 2Urh + U^2 rI.$$

The second-moment identity is a quadratic-form statement and does not require $Vf \in \operatorname{Dom} L^2$. For a local mode at $c = 1/8, U = 1$, the mean lies in $[1/8, 1/2]$, the second moment in $[19/128, 67/128]$, and the variance in $[17/128, 35/128]$. Thus $\mu_f(|E| \geq R) \leq 67/(128R^2)$. More generally, $\|hf\| \leq K$ gives a second moment at most $K^2 + UK + U^2/2$, useful for uniformly controlled physical smearing.

Choose $M \geq 2$ and $\tau > 0$. The positive effects

$$K_k(E) = \left| \frac{1}{M} \sum_{j=0}^{M-1} e^{-ij(\tau E - 2\pi k/M)} \right|^2, \qquad q_k = \int K_k(E)\, d\mu_f(E), \quad 0 \leq k < M,$$

sum exactly to one by finite Fourier orthogonality. Writing $\phi_f(t) = \int e^{-itE}\, d\mu_f(E)$, their complete probabilities are

$$q_k = \frac{1}{M} + \frac{2}{M} \sum_{n=1}^{M-1} \left(1 - \frac{n}{M}\right) \operatorname{Re}(e^{2\pi ikn/M} \phi_f(n\tau)).$$

The principal energy centers are modulo $2\pi/\tau$ and are readout bins, not asserted eigenvalues. For $0 < \tau\Delta \leq 3$, circular error has probability at most $\min\{1, 10/(M\tau^2\Delta^2)\}$, improved to $22/(7M\tau\Delta - 44)$ when its denominator is positive. This follows by summing the two inverse-square grid tails. A second-moment bound C adds an alias term at most $C/(3/\tau - \Delta)^2$. Consequently the probability of a real energy error at least Δ is controlled, without a pointwise spectral density or an isolated-pole assumption.

The complete frozen 78-line cube measure gives certified distributions for $M = 8, 16, 64$ at $\tau = 1$. At $\Delta = 1/2$, direct positive-kernel and Fourier interval calculations bound the respective error probabilities by 0.100284517428, 0.055193158232, and 0.020209031434. All outcomes are retained. The earlier executed native time-one quadratures check the first harmonic; higher harmonics here are source calculations, not new native long-time or finite-shot measurements.

Along the existing full-ground subsequence, the finite positive spectral measures converge weakly. The uniform fixed-spacing spatial kernel bound also promotes entrywise convergence to strong convergence of the compressed resolvents on the common one-particle space. No identification of the varying GNS Hilbert spaces is assumed. An explicit finite readout error adds the local-dynamics, native-control, and statistical errors to the unspecified finite-ground state discrepancy; the last term is never set to zero. Sharp interval probabilities require zero limiting mass at their endpoints. These results do not supply physical particle identifications, relativistic rest masses, or an interacting continuum.

An all-valley positive-energy continuum field net. There is a spacing-limit construction beyond fixed-lattice response. Take $c(a) = v/a$, fixed $v > 0$, and $0 \leq U(a) \leq U_{\max} < \infty$. At each spacing retain the full original Weyl/contact dynamics and a translation-invariant full ground. On centered cells $C_{a,n} = a(n + [-1/2, 1/2)^3)$ define

$$(J_a f)(n) = a^{-3/2} \int_{C_{a,n}} f(x)\, dx, \qquad P_a = J_a^* J_a.$$

Cellwise Cauchy–Schwarz and coordinate averaging give $\|J_a\| \leq 1$ and $\|P_a f - f\| \leq a \sum_j \|\partial_j f\|$ for $f \in H^1$. Thus $P_a \to I$ strongly. For every $b \in \{0,1\}^3$ set $M_b u(n) = (-1)^{b \cdot n} u(n)$ and $C_{a,b}(f) = c(M_b J_a f)$, with the conventional antilinear annihilator. All eight valleys evolve under the original full Hamiltonian. Writing $D_{a,j} f = [f(\cdot + ae_j) - f(\cdot - ae_j)]/(2a)$ gives

$$h_a M_b J_a f = M_b J_a \left[iv \sum_j (-1)^{b_j} \sigma_j D_{a,j} f \right], \qquad \|h_a M_b J_a f\| \leq v \sum_j \|\partial_j f\|.$$

For $b \neq d$, shifting one lattice step in a coordinate where they differ reverses the cross-valley pairing. Subtracting the shifted pairing proves

$$|\langle M_b J_a f, M_d J_a g\rangle| \leq \frac{a}{2}\left(\|\partial_j f\|\|g\| + \|f\|\|\partial_j g\|\right).$$

The limiting equal-time fields therefore have the full continuum CAR over $L^2(\mathbb{R}^3;\mathbb{C}^2)\otimes\mathbb{C}^8$.

The necessary time estimate holds in the full CAR operator norm. For $B(u)=\sum_{x\sigma}\overline{u_{x\sigma}}n_{x\bar{\sigma}}c_{x\sigma}$, the onsite anticommutator is positive and obeys $\{B(u_x),B(u_x)^*\}\leq\|u_x\|^2 I$. Different-site odd anticommutators vanish, hence $\|B(u)\|\leq\|u\|$. The exact contact commutator consequently gives

$$\|\tau_{a,t}(C_{a,b}(f))-\tau_{a,s}(C_{a,b}(f))\|\leq|t-s|\left(v\sum_j\|\partial_j f\|+U_{\max}\|f\|\right).$$

This avoids an extensive interaction-norm estimate.

A diagonal subsequence of spacings makes every word in a countable dense set of smooth smears, rational times and all eight valleys converge simultaneously. The displayed time and norm estimates extend the limits to all L^2 smears and real times. Polynomial positivity, norm bounds and the CAR pass to the limit and construct bounded fields in one GNS representation. Common time shifts are strongly continuous. Nearest *even* lattice translations approximate every physical translation while keeping all valley phases equal to one, so continuous spatial translations also act on the limiting net. For every multi-time polynomial X, the correlations $\omega_a(X_a^*\tau_{a,t}(X_a))$ have nonnegative energy support and converge uniformly on compact time intervals. Testing against smooth negative-energy spectral functions and using dominated convergence therefore proves a nonnegative generator on the entire limiting GNS space, including all polynomial-field vectors.

Graded locality also survives with an explicit bound. Suppose f is supported in $[-r,r]^3$, g outside $[-R,R]^3$, and put $\delta=(R-r)/2-48vT>0$. For $0<a\leq\min\{1,(R-r)/8\}$ and $|t|\leq T$, the original full-CAR bound, with lattice radii $\lceil r/a\rceil+1$ and $\lfloor R/a\rfloor-1$, yields

$$\begin{aligned}\|\{\tau_{a,t}(C_{a,b}(f)),C_{a,d}(g)^{\#}\}\|&\leq 192vT(2r+5)^3(2R+1)^2\|f\|\|g\|a^{-6}e^{-\delta/a}\\&\leq 192\cdot 7!vT(2r+5)^3(2R+1)^2\|f\|\|g\|\frac{a}{\delta^7}.\end{aligned}$$

Here $\#$ allows an adjoint or no adjoint. Inserting this norm estimate between arbitrary polynomial vectors makes the limiting anticommutator the zero operator. A finite partition of unity extends the result to separated compact supports. The continuum net thus has exact graded locality at sup-norm distance greater than $96v|t-s|$. This is a conservative propagation cone, not a Lorentz metric.

There is a further uniform response bound on every physical momentum window $|p|\leq P$ around any retained valley. These windows reduce the translation-invariant response and have kinetic norm at most vP. The exact Schur formula gives

$$\|\mathcal{G}_a(z)^{-1}\|_P\leq|z|+vP+\frac{U_{\max}}{2}+\frac{U_{\max}^2}{4\operatorname{Im}z},\qquad -\operatorname{Im}\mathcal{G}_a(z)\geq\frac{\operatorname{Im}z}{(|z|+vP+U_{\max})^2}I_P.$$

These bounds are uniform in spacing away from real frequency. The cell-average map differs from the exact band isometry by at most $a^2P^2/24$, so the two conventions have the same limiting correlations. Weak convergence alone does not allow interchanging the inverse and limit.

A norm-controlled free limit and the remaining interacting step. A Duhamel comparison with the free lattice evolution uses only $\|B(u)\| \le \|u\|$ and gives an error at most $|U(a)t|\|u\|$. The central-difference remainder then gives, for $f \in H^3$ and $h_b = iv\sum_j (-1)^{b_j}\sigma_j\partial_j$,

$$\|\tau_{a,t}(C_{a,b}(f)) - C_{a,b}(e^{ith_b}f)\| \le |t|\left(U(a)\|f\| + \frac{va^2}{6}\sum_j \|\partial_j^3 f\|\right).$$

If $U(a) \to 0$, the limiting dynamics is exactly the free eight-valley Weyl dynamics. Positive energy forces positive-band annihilators and negative-band creators to annihilate the ground. The one-particle zero-energy kernel is zero in L^2; spectral-band approximation and the CAR recurrence therefore uniquely determine the filled-sea quasifree state and all its moments. This is a controlled free continuum, not the required interacting one.

For bounded positive limiting U, the construction gives the full positive-energy continuum field net, but it has not proved that a connected interaction survives, selected a unique limiting state, or derived Lorentz covariance. Its propagation cone and positive energy supply no common physical stress, geometric modular flow, renormalized area stationarity or completed G1–G6 Einstein package.

Ground occupation and energy constraints under refinement. The original zero-chemical-potential ground satisfies a useful uniform constraint before any continuum particle identification. With normalized Brillouin measure, let $\rho(k)$ be its one-body covariance and $P_\pm(k)$ the spectral projectors of the original Weyl hopping. The free filled sea has onsite covariance $I/2$ and double occupation $1/4$. Its finite-box restrictions give the variational bound

$$0 \le e_{\text{kin}} - e_{\text{free}} + Ud = e_U - e_{\text{free}} \le U/4, \qquad d = \omega(n_\uparrow n_\downarrow).$$

Here $e_{\text{free}} = -c\int \sqrt{\sum_j \sin^2 k_j}$; no free reference is substituted for the interacting state. Set

$$w(k) = \operatorname{tr}(P_+\rho) + \operatorname{tr}(P_-(I-\rho)), \qquad D = \int w(k).$$

Then $0 \le w \le 2$ and the kinetic excess is $c\int \sqrt{\sum_j \sin^2 k_j}\, w(k)$. The region with $\sqrt{\sum_j \sin^2 k_j} \le \delta$ has normalized volume at most δ^3 for $0 < \delta \le 1$, since $2\arcsin\delta/\pi \le \delta$. Splitting this integral proves

$$D \le \min\{2, 2\delta^3 + U/(4c\delta)\}, \qquad |n-1| \le D.$$

For $U \le 24c$, choosing $\delta = (U/(24c))^{1/4}$ gives $D \le 8(U/(24c))^{3/4}$. Thus the density tends to one per site, and the per-spin density r tends to $1/2$, when $U/c \to 0$. At $v = 1/8$, $U = 1$, $a = 1/65536$, $c = v/a$ and $\delta = 1/16$, the exact bound is $1023/1024 \le n \le 1$.

The same defect controls every fixed microscopic local word. The free-sea annihilation parts $c(P_+f)$ and $c^\dagger(P_-f)$ of a unit site-basis field have interacting GNS norm at most $\sqrt{D}$. Moving the first annihilation part to the right by CAR gives the free Wick recursion, with two boundary errors at most $\sqrt{D}$. Hence for an even word of length m,

$$|\omega(A_1\cdots A_m) - \omega_{\text{free}}(A_1\cdots A_m)| \le C_m\sqrt{D}, \qquad C_0 = 0, \quad C_m = 2 + (m-1)C_{m-2}.$$

In particular $C_4 = 8$ and $d \to 1/4$. This controls fixed microscopic local observables; shrinking momentum pockets can carry a full normalized continuum packet, so no Gaussian continuum conclusion follows.

For cell-averaged physical H^1 packets and $c = v/a$, bounded $U(a) \to U_0$ fixes the first spectral moment to $\langle f, h_{\mathrm{W}} f\rangle + (U_0/2)\|f\|^2$ in every weak spectral limit. The uniform second-moment bound supplies the required first-moment tail control. The scalar term is retained without a chemical-potential shift. Second moments need not pass: the positive measures $(1 - n^{-2})\delta_0 + n^{-2}\delta_n$ converge weakly to δ_0 while their variances tend to one. Thus even a nonzero memory Gram need not survive the limiting response. These constraints supply ground and energy inputs, not relativistic rest masses or an Einstein balance. The proof and exact controls are in `model/continuum_ground_constraints`.

A complete native finite-clock energy measurement. The original zero-chemical-potential cube has a finite native preparation of its invariant ground reference $P_G/5$. Five azimuths $2\pi j/5$ and three polar cosines $-\sqrt{3/5}, 0, \sqrt{3/5}$, with polar weights $5/18, 4/9, 5/18$, twirl the spin-two highest-weight ground exactly. The azimuth sum cancels every nonzero magnetic difference through four, and the three-node quadrature integrates the remaining degree-four binomial weights to $1/5$. A native four-bit selector has weights $[5, 8, 5]$ repeated five times, followed by zero, divided by 90. Its fifteen conditional full-Fock spin rotations and the original inverse frame are executed; all selector and export outcomes are kept. The complete prepared-state error is below 10^{-9}.

At the origin, a native occupation-copy followed by $\gamma_1 = Z_0 X_1 = c_1 + c_1^\dagger$ gives the complete charge instrument with Kraus maps $c_1^\dagger, c_1$. Both branches are retained. Two clock bits control original-action evolutions for model times one and two, with the independently known phase from $E_G = -\sqrt{3}/4$. The positive Fourier transform produces

$$K_k(E) = \left| \frac{1}{4} \sum_{n=0}^{3} e^{-in(E - 2\pi k/4)} \right|^2, \qquad \sum_{k=0}^{3} K_k(E) = 1.$$

This is a complete positive energy measurement. Both physical charge branches have nonnegative excitation cost; principal bin centers are defined modulo 2π, and the removal tag retains the sign needed for the retarded frequency. For the invariant ground, the local addition measure is $\mu_\downarrow/2 + \delta_\lambda/4$ and the removal measure is $\delta_\lambda/4$, where $\lambda = \sqrt{3}/16$. The complete frozen 78-line measure therefore determines every joint charge/bin probability without fitting.

All eighteen-bit clock rows and thirty classical selector/charge columns are propagated through the actual native local instruments. The exact column decomposition used for computation discards no quantum coherence or outcome. Controlled hopping and contact words use at most three local work bits; no abstract controlled Hamiltonian is supplied as a primitive. With sixteen and thirty-two Strang steps, the full-Fock split errors sum to $1887/65536$. Including every preparation, control, scalar-phase and Fourier error gives complete state and total-variation error below $3/100$. A longest successful history contains 612666 native pulses. All 3240 joint selector/tag/clock slots, the selector stops and every first-export failure are retained. The independent original-Hamiltonian exponential is a comparator only.

This construction executes coherent histories through model time three and closes the stated finite native preparation-and-energy-POVM task. Four broad bins do not separately resolve seventy-eight lines. It neither identifies physical species nor converts a finite excitation energy into a relativistic rest mass. It also does not prepare the separate centered-action reference, establish infinite-ground state convergence, or supply the remaining geometric premises.

A CPT constraint on the unchanged charged fields and clock. For a real, even, unit scalar Schwartz packet ϕ, average the signed addition/removal measure over all sixteen continuum field components. Writing this probability measure as ν_ϕ, spin trace cancels the Weyl first moment and

its kinetic cross term. The exact finite-spacing sum rules, $r_a \to 1/2$, and weak lower semicontinuity give

$$\int E\, d\nu_\phi(E) = \mu = U_0/2, \qquad \int E^2\, d\nu_\phi(E) \le C_\phi = v^2\|\nabla\phi\|^2 + U_0^2/2.$$

Only the second-moment inequality passes to the limit. Negative response frequency denotes removal; the two charge-tagged excitation energies are nonnegative with respect to the original positive GNS generator L.

Suppose a proposed physical identification supplies an antiunitary Θ preserving the same ground and L, and mapping the sixteen smeared annihilators A_s into their adjoints by a unitary matrix Q:

$$\Theta\Omega = \Omega, \qquad \Theta L\Theta^{-1} = L, \qquad \Theta A_s\Theta^{-1} = \sum_t Q_{st}A_t^*.$$

The real even packet is preserved by the spatial inversion and conjugation in the usual CPT action. For every spectral set B, antiunitarity and unitarity of Q give

$$\sum_s \|P_L(B)A_s^*\Omega\|^2 = \sum_s \|P_L(B)A_s\Omega\|^2.$$

Thus the averaged addition and removal measures agree, ν_ϕ is reflection symmetric, and its first moment is zero. This precise same-field, same-ground, same-clock CPT identification is therefore impossible for $U_0 > 0$. No CPT theorem is applied without its hypotheses: this does not exclude Lorentz covariance alone or other field dictionaries.

With ν_ϕ^- the reflected measure and total variation defined as $\sup_B |\nu_\phi(B) - \nu_\phi^-(B)|$, Cauchy–Schwarz proves

$$\mathrm{TV}(\nu_\phi, \nu_\phi^-) \ge \frac{\mu^2}{C_\phi}, \qquad W_1(\nu_\phi, \nu_\phi^-) \ge 2\mu.$$

There is a bounded odd witness: for $R = C_\phi/(2\mu)$, clipping E to $[-R, R]$ gives f_R with $|E - f_R(E)| \le E^2/(4R)$ and $\int f_R\, d\nu_\phi \ge \mu/2$. Gaussian packets of width ℓ have $C_\phi = 3v^2/(2\ell^2) + U_0^2/2$. At $v = 1/8$, $U_0 = 1$ and $\ell = 1$, the total-variation lower bound is $32/67$; broad packets make it tend to $1/2$. These are rigorous bounds, not computed infinite-ground spectra or executed native measurements.

A scalar charged-field rephasing changes the generator to $L - \kappa Q_{\mathrm{charge}}$ and the signed first moment to $\mu - \kappa$. Consequently $\kappa = U_0/2$ is the unique necessary centering for this CPT condition. It is a change of charged dynamics, not an additive energy-origin change or a rest mass. Positivity on the old ground is not automatic: the empty two-mode ground of $H = n_0 + 3n_1$ has a centered response of zero mean at $\kappa = 2$, while $H - 2N$ has eigenvalues $0, -1, 1, 0$. The new reference and its ground property require proof.

The free $U(a) \to 0$ control has physical addition and removal energies $v|p|$, zero rest masses, and four Weyl chiralities of each sign. A massive Dirac control, $H_D = Z \otimes (\sigma \cdot p) + mX \otimes I$, has $H_D^2 = (|p|^2 + m^2)I$ and zero trace: its mass couples chiralities and does not create the scalar signed-energy shift. Exact full-CAR counterexamples and the conditional proof are retained in `model/continuum_cpt_constraint`; no physical species, interacting Lorentz completion or Einstein premise is thereby supplied.

A resolved canonical contact-force channel. The bounded-contact refinement also resolves a nonlinear operator in the original action. At each site set

$$b_\sigma = n_{\bar\sigma}c_\sigma, \qquad d_\sigma = (2n_{\bar\sigma} - 1)c_\sigma = 2b_\sigma - c_\sigma.$$

The onsite unitary $Z = I - 2n_\uparrow n_\downarrow$ obeys $Zc_\sigma Z = -d_\sigma$, so d is itself an exact CAR family, with $\|d(u)\| = \|u\|$. It is a composite of the original fields, not an added microscopic mode. The unchanged Hamiltonian satisfies

$$[H, c(u)] = -c((h + U/2)u) - (U/2)d(u).$$

Thus retaining the scalar term does not justify dropping the composite. For $B(u) = \sum_{x,\sigma} \overline{u_{x\sigma}} b_{x\sigma}$, exact local algebra gives

$$\{B(u), c(v)^*\} = \{B(u), B(v)^*\} = Q(u, v) = \sum_x \mathrm{d}\Gamma(q_x), \qquad q_x = (u_x^* v_x)I - v_x u_x^*.$$

Each q_x has rank at most one and $\|q_x\| = \|q_x\|_{\mathrm{HS}} = \|u_x\|\|v_x\|$. The anomalous mixed anticommutator is the pair sum

$$P(u, v) = \{B(u), c(v)\} = \sum_x \overline{u_{x\downarrow} v_{x\uparrow} - u_{x\uparrow} v_{x\downarrow}} c_{x\uparrow} c_{x\downarrow}.$$

Let D_a be the occupation defect above, $L = \|u\|\|v\|$, and $I_4 = \sum_x \|u_x\|^2 \|v_x\|^2$. The uniform four-field Wick comparison, $\omega_a(Q) = r_a \langle u, v\rangle$, and the free covariance identity $\mathrm{Var}_{\mathrm{free}}(\mathrm{d}\Gamma(q)) = \mathrm{Tr}[\Pi q^*(I - \Pi)q] \le I_4$ imply

$$\omega_a(Y_a^* Y_a) \le 4\big[I_4 + (8\sqrt{D_a} + D_a/2)L^2\big]$$

for either $Y_a = 2Q - \langle u, v\rangle I$ or $Y_a = 2P$. For $u = M_b J_a f$, $v = M_e J_a g$ with bounded smooth compactly supported smears, $I_4 \le a^3 \|f\|_\infty^2 \|g\|^2$ and the coefficient supremum is at most $a^3 \|f\|_\infty \|g\|_\infty$. These ground fluctuations therefore vanish.

The passage is an operator statement on the full field hierarchy. A degree-m CAR polynomial obeys $\|[\mathrm{d}\Gamma(q), A]\| \le m\|q\|\|A\|$ for $q = q^*$: its one-body unitary orbit has frequencies in $[-m\|q\|, m\|q\|]$, and the Bernstein interpolation bound applies. The same argument gives $\|[\sum_x n_{x\uparrow} n_{x\downarrow}, A]\| \le m\|A\|$, with degree at most $m + 2$ after the commutator. In the free interaction picture the order-N Dyson remainder is consequently bounded by

$$\|A\|(2U_{\max}|s|)^N \frac{(m/2)_N}{N!}.$$

This is uniform in spacing and volume. Divide $|t| \le T$ into $K = \max(1, \lceil 8U_{\max} T\rceil)$ intervals and choose $N = m + \ell$ at each step. The resulting approximation has degree at most

$$M = 3^K(m_{\mathrm{initial}} + \ell - 1) - (\ell - 1)$$

and relative norm error at most $(1 + 2^{-\ell})^K - 1$. The quadratic coefficient bound then proves $\|[Y_a, X_a]\| \to 0$ for every finite-time field polynomial X_a; choosing $2^{-\ell} \le a^3/(2K)$ gives an explicit $O(a^3 \log(1/a))$ bound at fixed time. Therefore

$$\|Y_a X_a \Omega_a\| \le \|[Y_a, X_a]\| + \|X_a\| \, \|Y_a \Omega_a\| \longrightarrow 0.$$

The same argument covers initial composite fields, whose degree is three.

Adjoining their equal-time smearings to the common subsequence constructs operators $D_b(f)$ alongside the original $C_b(f)$, with

$$\{D_b(f), D_e(g)^*\} = \delta_{be}\langle f, g\rangle I, \qquad \{D_b(f), C_e(g)^*\} = \{D_b(f), C_e(g)\} = 0.$$

Together with the original CAR and vanishing same-family anomalous anticommutators, these are two copies of the eight-valley CAR algebra. This is algebraic independence; the state is not asserted to factorize or obey Wick's rule. The original correlations and their cyclic positive-energy representation are preserved. Its continuous time action is not automatically extended to the entire enlarged GNS space: the composite derivative contains kinetic terms of order $1/a$. Thus the result resolves the contact-force channel but does not determine whether its bounded-frequency weight survives, escapes to infinite energy, or produces an interacting continuum. It supplies no physical species, Lorentz stress or geometric modular identification. A positive quadratic two-mode control makes this boundary explicit: for $h = \left(\begin{smallmatrix} E_0 & \kappa \\ \kappa & \Lambda \end{smallmatrix}\right)$, with $E_0\Lambda > \kappa^2$ and $\Lambda > E_0 > 0$, the memory Gram stays κ^2 but $\Sigma(z) = \kappa^2/(z-\Lambda) \to 0$. The field error from free energy E_0 is at most $(2\kappa+\kappa^2|t|)/(\Lambda - E_0)$. The full proof and exact controls are in `model/continuum_composite_channel`.

Regional modular data on the full particle ground. The invariant finite reference $\rho_G = P_G/5$ supplies regional modular data on the same ground used for the particle spectrum. In the hidden spin frame, its highest-weight vector is the Slater determinant of four negative spatial orbitals, with projector $P_- = (I - A/\sqrt{3})/2$. Averaging a common spin rotation yields ρ_G. For a region R, let λ_j be the eigenvalues of the spatial compression $(P_-)_R$. The Slater Schmidt decomposition and the spin average give, for every occupied natural-orbital subset I,

$$w_I = \prod_{j\in I} \lambda_j \prod_{j\notin I} (1-\lambda_j), \qquad \rho_{R,I} = \frac{w_I}{|I|+1} P_{\mathrm{sym},I}. \tag{1}$$

Here $P_{\mathrm{sym},I}$ projects onto the symmetric spin space of dimension $|I|+1$; all other states have zero weight. Indeed phase integration deletes unequal spin counts, and the remaining diagonal integral is $\binom{k}{l}\int_0^1 t^{k-l}(1-t)^l\,dt = 1/(k+1)$. Thus, retaining the complete kernel,

$$S(\rho_R) = \sum_j h_2(\lambda_j) + \sum_I w_I \log(|I|+1). \tag{2}$$

For $0 < \lambda_j < 1$ the modular generator on the positive support is

$$K_R = -\sum_j \log(1-\lambda_j) I + \sum_j \log\frac{1-\lambda_j}{\lambda_j} N_j + \log(N_R+1). \tag{3}$$

Endpoint occupations are handled as fixed blocks, without taking logarithms of zero. This is a support identity, not a full-CAR faithfulness claim.

Every adjacent two-site region has positive spectrum

$$\left\{ \frac{1}{6}\ [1],\quad \frac{2-\sqrt{3}}{12}\ [2],\quad \frac{2+\sqrt{3}}{12}\ [2],\quad \frac{1}{18}\ [3] \right\}, \tag{4}$$

where brackets denote multiplicities. Its eight other eigenvalues are zero. The support consists of the vacuum, the four one-particle states, and the three two-particle triplet states. With H_R the actual internal regional Hamiltonian, whose contact term vanishes on this support,

$$K_R = \log 6\, I + \beta_R H_R + \log(N_R+1), \qquad \beta_R = 16\log(2+\sqrt{3}). \tag{5}$$

Exact signed CAR traces of all five source ground vectors verify this identity on all twelve cube bonds; the sixteen nonadjacent pairs give $K_R = \log 4\, I + \log(N_R+1)$ instead. The same local

spin-frame change conjugates the density, support and Hamiltonian back to the original source coordinates.

For every Hermitian trace-zero variation δ within the positive support, including noncommuting variations,

$$DS_{\rho_R}(\delta) = \beta_R \operatorname{Tr}(\delta H_R) + \operatorname{Tr}\big(\delta \log(N_R + 1)\big). \tag{6}$$

The second term is central on the supported number-preserving algebra, not on the full CAR algebra. Since $\lambda^+_{\min} = (2-\sqrt{3})/12 > 1/48$, for $\delta = \sigma - \rho_R$ with σ supported there and $0 \le \epsilon < 1$,

$$0 \le D(\rho_R + \epsilon\delta \| \rho_R) \le \frac{24\epsilon^2 \|\delta\|^2_{\mathrm{HS}}}{1-\epsilon}. \tag{7}$$

In contrast, a density $\sigma_\perp$ in the orthogonal kernel obeys the full-state identity $S((1-\epsilon)\rho_R + \epsilon\sigma_\perp) = (1-\epsilon)S(\rho_R) + \epsilon S(\sigma_\perp) + h_2(\epsilon)$, whose derivative diverges at zero. The kernel is therefore retained in the operational domain; no supported first law is applied to such a disturbance. These are same-ground finite modular and entropy inputs. Identifying their central term with a geometric edge, their energy pairing with a diamond charge, or their coefficient with a physical clock requires the existing Einstein theorem's common geometric and scaling premises.

Source spectra and supplied mass channels. The centered clock $K = H - UN/2$ has a unique half-filled cube ground at $c = 1/8$, $U = 1$, with hidden spin and eta spin zero. Certificates resolve 228 charged lines, 135 spin/density lines each, and a full-state gap in $[0.015021254176, 0.015021254177]$. Actual cubic rotations give G1+G2+H on each visible charged line and A1+E on the first neutral triplet. The complete onsite anticommuting quadratic catalogue has four supplied coefficients and rest energies $|m_0 \pm |\mathbf{m}||$, with four states per generic signed level. Any two nonnegative energies can be selected this way; no mass ratio is predicted. The exact finite gaps, coefficient designs and zero-mass free limit are collected in the mass and energy table. None identifies a measured species or calibrated physical mass.

Local exchange-charge preparation and transport. An explicit two-label extension realizes the protected exchange charge on local ports. Complete native instruments prepare and transport a localized coherence with unchanged initial exterior state. In the one-A/one-B sector, a deterministic preparation confined to C has optimal charge magnitude $\binom{|C|}{2}/66$; every sharp charge state has exterior trace distance at least $\binom{|C^c|}{2}/132$ from the tracial reference. The transported contrast is a pair coherence; that reference has scalar modular generator and zero modular-energy excess. These results do not identify a localized physical particle. See `model/local_exchange_charge/`.

Positive energy in the actual continuum spin fields. The trajectory $c = J|E|^4$, $U = J|E|^8$ produces interacting lattice Heisenberg dynamics. Its selected ordered ground has $m_s > 1/1000$ and constructs one regular continuum Weyl (CCR) pair with continuous translations and finite propagation. Both actual Stone fields create nonzero positive-energy vectors, with explicit retained weight in energy bands proportional to each arbitrarily small momentum scale. The continuum dynamics is nonstatic and gapless. Exact continuity reduces its first-energy tail to a specified inverse-current response; fractional source-energy tails are uniformly controlled, but a sharp shell and surviving interaction are unproved. Complete finite filters retain every rejection and have success lower bounds proportional to p^4 at the stated input accuracy. Their cube energies are certified; an effective approximation rate for the infinite ground remains open. See `model/spin_current_spectrum/` and `model/spin_packet_readout/`.

Finite entropy response for the existing Einstein stack. The centered charged hierarchy is free for $U = o(c^{2/7})$, $c = v/a$. At $U = 0$, actual finite regional entropy changes approach the Weyl stress response on a common shrinking schedule with nine independent preparations. For the original nine normalized shells, the actual Araki relative entropy satisfies

$$D_R(\theta) = \frac{32 c_\gamma^2 \theta^2}{35\pi^2}(R/L)^6 + o((R/L)^6), \qquad c_\gamma = \frac{4249388}{150088295},$$

uniformly on compact amplitude intervals. The nonlinear correction is therefore $o(R^4)$ at fixed shell scale L, and explicit lattice meshes retain the finite-amplitude stress response. See `model/small_ball_entropy_asymptotic/`. Responsive geometry, area and edge identification, Newton normalization and entropy stationarity remain separate requirements. These are free-source rest-ball matter results.

Other finite interfaces. The full-CAR locality bounds used above control dynamics at fixed lattice spacing; a chiral spatial continuum remains a further limit. For partition expectations E_c and positive rates,

$$L = \sum_c \gamma_c (E_c - I), \qquad \operatorname{Tr}(X^* L X) = -\sum_c \gamma_c \|X - E_c X\|_{\mathrm{HS}}^2$$

identifies $\ker L$ with the intersection of the partition commutants, without assuming commuting pinchings or independent no-cancellation. Applying this theorem requires the stated generator. Equivariant rational chain maps and homotopies on the supplied finite support tower are homology maps, not quantum channels or a dynamical refinement limit.

What the negative controls exclude. Several finite benchmarks constrain possible identifications. A supplied Weyl/contact action has an isolated moving bound-pair band, complete contact residues and an optical identity, but its positive contact section makes the band topologically trivial. Its certified curvature fails the rest/kinetic mass relation with the same speed parameter. Contact interactions also couple proper Weyl momentum pockets to their complements. The lattice scattering theorem, finite thermal positivity and KMS identities do not remove those obstructions or supply physical asymptotic detectors, a chiral continuum or calibrated cross sections.

The centered repulsive dimer gives a sharper family-wide exclusion. For $t = c/2 > 0$, $U \geq 0$, $\mu = U/2$ and $d = \sqrt{U^2 + 16t^2}$, its two charge-one doublets have gaps $g_\pm = d/2 \pm t$, hence

$$1 < g_+/g_- \leq 3.$$

No common positive linear conversion yields the charged-lepton hierarchy or three distinct same-charge families there. Allowing another chemical potential can evade the ratio bound but does not create a third doublet. The fixed complete charged-frequency catalogue also fails its broader lepton-ratio comparison. The periodic four-site extension fails its declared $E(k)^2 = m^2 + c^2 \sin^2 k$ benchmark at $c = 1/8$; a two-parameter fit to two distinct momentum values does not rescue that test. These exclude stated finite dictionaries, not every larger action or a renormalized continuum limit. Electron/positron masses set by one common measured calibration, and retrospective parameter comparisons, provide no independent mass prediction.

Finite current response also has a precise scope. An open-bond phase is pure gauge and measures no magnetic moment; the contact kernel is not a Coulomb kernel. On the complete periodic ring, a genuine loop phase has gauge-invariant current and total curvature,

$$E_0' = \langle \partial_\phi H \rangle, \qquad E_0'' = \langle \partial_\phi^2 H \rangle - 2 \sum_{n \neq 0} \frac{|\langle n | \partial_\phi H | 0 \rangle|^2}{E_n - E_0},$$

under the isolated-ground hypothesis. Degenerate free and atomic controls lie outside that formula. The calculation retains all 256 Fock states but does not identify a photon or physical Lorentz spin.

Clock and memory exclusions are similarly conditional. A frozen real-operation grammar has a normalized trace-distance lower bound 1/2 for a specified nontrivial modular action of its nontracial reference. Finite independent copies of the original native reference also fail the required field modular-gap ratios. These do not exclude the explicitly enlarged clock and control laws above. The earlier pair/dark-reference encoding covers a fixed corner, not the whole spatial field. Finite stationary observer references concern source/buffer marginals at the stated repair rate and on their recurrent support; delayed variants need their own construction, and growing exported histories are not made stationary by these marginal references. Actively corrected coherent memories can have flat logical energy and a fixed core, so memory lifetime alone does not establish a moving massive particle. Properties proved on distinct carriers require an explicit intertwiner before they can be combined.

Physical target and remaining obligations. The target is an interacting quantum field theory with gapless photons and experimentally matched massive sectors, not a positive global mass gap. A gapless photon rules out such a global gap. An exact nontrivial positive-energy conformal theory in a flat-space vacuum representation with unbroken unitary dilations also cannot have a positive flat-space vacuum mass gap: for a spectral value $E > 0$ and proposed gap $g > 0$, scale covariance also puts $g/2$ in the spectrum. This does not exclude a gap in scaling dimensions or discrete finite-radius cylinder levels; the cylinder radius supplies an energy scale. Conformal behavior may describe a critical or short-distance regime, but a deformation or symmetry-breaking mechanism responsible for massive flat-space sectors must then be specified and implemented.

The finite results leave concrete physical joins. A localized defect must carry the protected charge and the same measured field observables. Spatial refinement must control the additional lattice modes and yield the required Lorentz propagation, while keeping the interactions and reference state. The electromagnetic field and gauge–Higgs interactions must be dynamical, with charge constraints, detector coupling and their infrared behavior treated on that same theory. Massive excitations need a common energy and clock normalization, including the relation $E''(0) = c^2/E_0$ for a relativistic branch with $E_0 > 0$, together with the appropriate stress and proper-time relations. The neutrino sector and other measured particle properties require their own matching tests. Neither a successful finite source compilation nor a calibrated spectrum alone discharges these obligations.

1 This paper's hypothesis and place in the tower

This paper is the executed part of layers two to four of the tower: the carrier and its repair law, the executable observer, and the fields and clocks read from one process. Its hypothesis is the executable-model hypothesis stated above, taken at the level a finite execution can test. The carrier, the native C/E repair law and its memory, the finite observer algebra, the declared services and the scheduler are fixed before testing, and the observers, records, local fields and event clocks of every result are read from a process run under that one specification. Fields, particles and geometry enter as descriptions of what observers record and jointly interpret; no fundamental field or spacetime is reconstructed first [48, 54, 4].

The paper proves finite theorems and certifies finite computations on the declared objects: the seam rule from four requirements, the classification of admissible quantum record laws, the exact gap, memory and recovery of the native channels, record capacity, finite observers and field

detectors with every history retained, the common interacting source and its occupation limits, overlapping-observer correspondence, local event clocks and matched screens. For one repair history on the refining spherical source it proves source limit theorems: a common continuum operator of spectral dimension two, Gaussian density paths, incident-clock parametrizations and their fluctuations. Conditional analytic theorems hold on supplied objects: a declared radial product with a chosen wave action, and the supplied interacting actions with their occupation limits.

None of these components is a physical clock, field theory, particle or gravitational response. The five conjunctive obligations of the executable hypothesis are open joins of this construction [31]: observer algebras with faithful references, executed modular clocks and a physical energy normalization; a nontrivial interacting chiral 3+1 field theory with controlled refinement and detector maps; native defects identified through charge, spin and statistics, propagation, interactions and energy, with the measured spectrum; the Einstein reconstruction premises realized on the same matter, state, modular flow, clocks and refining geometry; and joint validation of complete observation and intervention laws against independent measurements. Table 1 separates what this paper executes from what each obligation requires beyond it, and Section 11 orders the joins with their acceptance conditions and stopping tests.

A replayable failure of the specified law, state, access or readout contract rejects that candidate and its tested interpretation. A proof covering every realization of a frozen admissible class stops that class, and it rejects OPH only if the excluded choice is proved necessary. Section 11 records four such failures of the tested law: the native double-load defect decays, the static radial product cannot balance a scalar excitation, passive repair has no wave sector, and the tested mass laws miss the charged hierarchy.

Table 1: The five obligations of the executable hypothesis [31], the components this paper executes toward each, and the open join.

Obligation	**Executed here**	**Open join**
Clocks and energy normalization	Finite modular flow of actual observation states and the fixed-reference relation $t = -\beta s$ (Section 6.6); local, joint and active event clocks (Sections 3.8, 4.2); native incident clocks with source limits and fluctuation predictions, and faithful encoded-field states sharing one active clock (Section 7).	A reference selected by the process, regional compatibility, calibrated duration and the energy normalization. Passive repair is no nontrivial modular automorphism of an invariant observer algebra, and the uncapped native-return reference tends to the trace.
Interacting chiral 3+1 field theory	Free-scalar refinement transfer and two-site occupation limits (Section 5); an interacting feedback signal at infinite occupation (Section 6); the native continuum operator with density, clock and admitted-wave limits, a declared radial product with Lorentz-signature density, smoothing theorem and a quartic projection with an occupation limit at fixed band (Section 7).	The fields are free or two-mode; radial product, filters, preparations and inverse temperature are declared; no interacting refining family and no chiral matter. The native decay spectrum excludes a complete linear identification with a compact free scalar.
Native defects, identity and spectrum	A charged realization with branch charge covariance (Section 3.4); a flux-dressed fermion code with operator Gauss constraints (Section 6.1); the collision candidate tested (Section 11.2).	No localized protected native sector, maintenance-energy law or spectrum. The collision decays and the tested mass laws miss the charged hierarchy.

Table 1 continued

Obligation	Executed here	Open join
Realized Einstein premises	A rank-three response, native causal cones, the product's area and volume dictionary, one characteristic cone for scalar, spinor and Maxwell symbols, and a necessary spherical focusing equation (Sections 7, 8).	Responsive geometry, conserved stress of the native process, fixed-proper-volume generalized-entropy stationarity and a positive Newton normalization. The static radial product fails null balance.
Joint validation	A prospectively fixed count protocol for the record-law coherence and its sequential design; retrospective Bell and optical-clock replays; a retrospective coupling audit (Section 9).	One frozen physical dictionary compared with independent field, particle and gravity data.

The theorem library holds conditional theorems on consensus and repair [6], observers and records [7], gauge structure [8], spacetime and Einstein dynamics [9], and screen microphysics [10]; the flagship synthesis gives the informal account [3]. The labels A1, A2 and A3 used there name theorem premises: A1 is the carrier, federation and response layer of the specification, A2 the agreement condition with its separately adopted response-endogeneity clause, and A3 the adopted least-informative state-selection principle on its declared reference and constraints. Neither selection nor connected-response endogeneity follows from the closure equation alone. A theorem that uses them states an implication, and Section 10 records, route by route, which antecedents the construction supplies on its own finite objects.

The sections follow the tower. Section 2 states the specification (layer two). Sections 3 and 4 give the executable observer on one carrier and overlapping observers, event clocks and matched screens on one native process (layer three). Sections 5 to 7 give the fields and clocks (layer four): field detectors on supplied actions, the common interacting source, and the geometry, continuum operator, clocks and fields of one repair history, followed by the common matter metric with its gravitational control (the interface to layer six). Section 9 gives the prediction protocols, Section 10 the premise map and Section 11 the ordered joins. Every number is copied from a retained receipt or a model document of the construction [11], cited at a pinned revision of the public repository [2].

Results carry the claim classes of Table 2. Actions, preparations, couplings, instruments, services and schedules are inputs of the specification, and apart from the retrospective replays of Section 9.5 no result is a comparison with independent observations. Numbered theorems and propositions are finite theorems unless they state a limit; items that quote a certified enclosure, table or replayed count are certified finite computations; the occupation limits of Section 6 are conditional analytic theorems on their supplied actions; the native continuum statements of Section 7 are source limit theorems with their domain, normalization and order of limits; Sections 9.5 and 9.3 are diagnostics, and the count protocol of Section 9.1 is a prospectively fixed conditional test.

Seven claim records of the repository [2] state the scope of the executed components (Table 3). Further model records own the published-oscillator and anharmonic-transfer comparisons and the matter, common-field and static-product witnesses; they are cited where those results appear.

2 The specification

The specification is layer two of the tower. The carrier, the repair law and its record laws, the observer algebra, the services and the time parameters below are fixed before any test, and every later result runs on them.

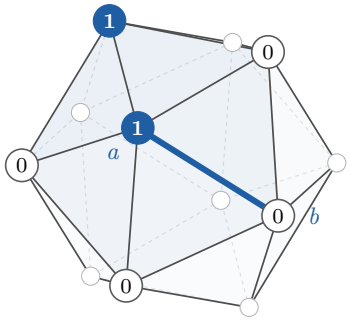
Table 2: Claim classes used in this paper, from the flagship synthesis [3].

Class	Meaning
Finite theorem	A deductive result proved from displayed premises on a stated finite domain. An exhaustive computation counts as part of the proof only with a specified domain and a soundness argument.
Certified finite computation	An executable witness whose inputs, controls, and output are fixed by a reproducible certificate; execution alone does not promote its interpretation to a theorem.
Conditional analytic theorem	A theorem whose hypotheses include a supplied continuum action, state space or limit; its conclusion holds on that supplied object.
Source limit theorem	A theorem about the limit of a declared family of finite source executions along a stated subsequence, with its normalizations and order of limits displayed; the limit is derived, and its physical identification is a separate claim.
Reconstruction implication	An exact implication whose antecedents include displayed geometric or physical premises.
Unsupplied realization map	A required map from a finite object to a physical event, current, field, clock, scale, or continuum structure for which no construction is asserted.
Diagnostic	A certified computation compared against measured values that it consumes; never counted as a prediction.
Prospectively fixed conditional test	A diagnostic with a custody-bound target definition, kill band, precision floor, and decision rule fixed before comparison.

Table 3: Executed components and the repository's claim records that state their scope.

Component	Sections	Claim record
Two-event native observer, classification of record laws, quantum mixing of the repair channels, recovery from complete records, phase protocol and its source-executed implementation	2.3, 2.4, 3.1, 3.5, 3.6, 9.1, 9.2	`OPH-MODEL-NATIVE-OBSERVER-EXECUTION`
Correctable record capacity and compact controller	3.3	`OPH-MODEL-OBSERVER-RECORD-CAPACITY`
P-dependent binary detector and retrospective coupling audit	3.4, 9.3	`OPH-MODEL-P-RESPONSE-EXECUTION`
Memory certificate at observer-local ticks for the reciprocal exchange law	3.8	`OPH-MODEL-N4-LOCAL-TICK-CERTIFICATE`
Scalar detector, two-read continuation and two-site source apparatus	5.1, 5.2, 5.3	`OPH-MODEL-SCALAR-RESPONSE-EXECUTION`
Common interacting source, occupation and readout certificates, smooth continuation, compiled native attachment and encoded memory, observer readouts, history repair, geometric and field readouts	6, 7.1	`OPH-MODEL-COMMON-SOURCE-EXECUTION`
Overlapping observers, local and active event clocks, record-law identification, federations and matched screens, the native continuum operator and its limits, the sequential protocol and public-record replays	7.2–7.6, 4, 9.4, 9.5	`OPH-MODEL-NATIVE-PROCESS-RECOVERY`

2.1 The carrier and its load configurations



twelve ports, thirty seams, twenty faces;
a binary load of total two (filled ports)

$d = x_a - x_b, \qquad j = \frac{1}{2}\big[d - (2c-1)(|d| \bmod 2)\big]$

d	$c=0$	$c=1$	effect of the attempt
2	1	1	balance
1	1	0	swap or wait
0	0	0	no move
-1	0	-1	wait or swap
-2	-1	-1	balance

The move j carries j units from a to b, and the fair coin c reads no load. At $|d| = 2$ the attempt lowers $V = \sum_i x_i^2$; at $|d| = 1$ it keeps V fixed, so the native process keeps moving. The seam drawn has $d = 1$.

Figure 1: The twelve-port carrier and its native repair. Left: the icosahedral incidence of the twelve ports, with thirty seams and twenty faces, in an orthographic view with back edges dashed; the filled ports carry a binary load of total two, and the drawn seam (a, b) has $d = 1$. Right: the seam rule for every mismatch the total-two carrier admits.

The specification has one carrier with twelve primitive ports. Its incidence is the oriented boundary of an icosahedron, with thirty seams and twenty faces; five seams meet at every port, the neighbours of each port form a five-cycle, and every seam lies on two oppositely oriented faces [10, 3]. Write A for the adjacency matrix of the seam graph and $L = 5I - A$ for its Laplacian. The spectrum of L is 0, $5 - \sqrt{5}$ with multiplicity three, 6 with multiplicity five and $5 + \sqrt{5}$ with multiplicity three, so its characteristic polynomial is $x(x-6)^5(x^2 - 10x + 20)^3$ [58, 99, 92].

A load configuration assigns a nonnegative integer x_p to each port p, with total load $\sum_p x_p = 2$. There are 78 configurations. The 66 with two distinct occupied ports form the binary shell; the other twelve are collisions $|2u\rangle$ with load two at one port u. We write $\mathcal{H}_{78} = \mathbb{C}^{78}$ for the quantum load space with basis $|x\rangle$, $\mathcal{H}_{66}$ for the span of the binary shell, and M_{78}, M_{66} for the corresponding full matrix algebras. For a binary configuration occupied at ports u and w we also write $|uw\rangle$ or $|\{u, w\}\rangle$.

Ports are numbered 0 to 11 by the frame vectors u_p: $(-1, \varphi, 0)$, $(1, \varphi, 0)$, $(-1, -\varphi, 0)$, $(1, -\varphi, 0)$, $(0, -1, \varphi)$, $(0, 1, \varphi)$, $(0, -1, -\varphi)$, $(0, 1, -\varphi)$, $(\varphi, 0, -1)$, $(\varphi, 0, 1)$, $(-\varphi, 0, -1)$, $(-\varphi, 0, 1)$, with $\varphi = (1 + \sqrt{5})/2$. The thirty seams are the pairs with $|u_a - u_b|^2 = 4$, each written (a, b) with $a < b$, and that order is the orientation used by the repair law below [92, 93].

Premise A1 of the theorem library is the layer of carriers, their federation through typed seams and coherent triple overlaps, the oriented spherical support and the complete reversible port response [5]. The construction realizes one carrier of that layer: its ports, seams and faces, its load configurations and the repair events on its seams. Coherent quantum operations run through a six-level processor whose twelve source generators, one per port, exponentiate to every executed gate on two- and four-dimensional proper codes through declared load and export services [79, 70]. Spherical-screen executions join many carriers under explicit access and refinement contracts [12]; they identify no physical current for the complete response algebra and do not discharge the full A1 attachment.

2.2 Native repair and its quantum record laws

Fix an oriented seam $e = (a, b)$ and a fair placement coin $c \in \{0, 1\}$. The native repair law is

$$d = x_a - x_b, \qquad j = \tfrac{1}{2}\big[d - (2c-1)(|d| \bmod 2)\big], \qquad y = x + j\,(\mathbf{e}_b - \mathbf{e}_a), \tag{8}$$

where $\mathbf{e}_p$ is the unit load vector of port p and y is the repaired configuration. The two endpoint loads become the floor and the ceiling of their mean; the coin assigns the ceiling to a when $c = 1$; all other loads are unchanged.

Proposition 2.1 (The classical seam rule)**.** *Let a stochastic rule on one seam (i) conserve $x_a + x_b$, (ii) end at a configuration minimizing $x_a^2 + x_b^2$, (iii) break any remaining tie by a random choice independent of the loads, and (iv) be invariant under exchange of the two endpoints. Then it is* (8) *with a fair coin.*

Proof. By (i) and (ii) the outputs are the floor and the ceiling of the mean. An odd total leaves two outputs, and by (iii) endpoint a receives the ceiling with a constant probability q. Exchange of the endpoints maps q to $1 - q$, so (iv) gives $q = \frac{1}{2}$. □

Without (iii), the rule that leaves the extra unit with the endpoint holding more satisfies the other three requirements. It fixes every binary configuration, so the binary shell carries no motion and the records carry no geometry. Requirement (iii) is an independent restriction on the admitted tie-breaker: it excludes retaining a preference for the endpoint holding more at the input. Internal realization alone does not impose that restriction.

The record is $r = (c, j)$, the coin and the signed current. Compatibility on a seam is specified independently of the update; it requires the two endpoint loads to differ by at most one. On the quantum load space the record defines the current instrument C with Kraus operators and record-discarded channel

$$L^e_{(c,j)} = \frac{1}{\sqrt{2}} \sum_{x:\, j(x,c)=j} |y(x,c)\rangle\langle x|, \qquad \mathsf{C}_e(\rho) = \sum_r L^e_r\, \rho\, L^{e\dagger}_r. \tag{9}$$

Both maps are written in Kraus form [120]. The contrast-record variant E keeps $r = (c, d)$ instead, with Kraus operators $M^e_{(c,d)} = 2^{-1/2} \sum_{x:\, x_a - x_b = d} |y(x,c)\rangle\langle x|$ and record-discarded channel E_e. Uniform independent seam attempts form the baseline schedule; on diagonal binary states the repair is then the lazy symmetric exclusion process with two particles, with generator $\frac{1}{60}\sum_e (S_e - I)$ per attempt, S_e exchanging the endpoint loads [51, 58].

Proposition 2.2 (Native repair instrument)**.** *Let $e = (a, b)$ be an oriented seam of the twelve-port carrier, let the placement coin be fair, and let C_e and E_e act on $\mathcal{H}_{78}$. Then:*

(i) For every record value the branch map $x \mapsto y(x, c)$ is an injective translation of its support. Consequently $\sum_r L^{e\dagger}_r L^e_r = I$, and C_e is completely positive and trace preserving (CPTP), also when tensored with an untouched reference system. The same holds for E_e.

(ii) Every branch conserves the total load, leaves spectator ports unchanged, never increases $\sum_p x_p^2$, and maps the binary shell into itself.

(iii) The current labels satisfy $c \in \{0, 1\}$ and $j \in \{-1, 0, 1\}$; all six labels occur on $\mathcal{H}_{78}$ and four occur on the binary shell.

(iv) C *and* E *induce the same classical transition on every configuration, and*

$$\mathsf{E}_e^2 = \mathsf{E}_e, \qquad \mathsf{E}_e\mathsf{C}_e = \mathsf{C}_e\mathsf{E}_e = \mathsf{E}_e, \qquad \mathsf{C}_e^n = \mathsf{E}_e + 2^{1-n}(\mathsf{C}_e - \mathsf{E}_e) \quad (n \geq 1).$$

On the binary shell E_e *is a trace-preserving conditional expectation.*

(v) With the seam-total parity $p = (x_a + x_b) \bmod 2$ *one has* $d = 2j + (2c-1)p$*, and* E *equals* C *followed by a binary record of* p *and this relabelling, as an identity of complete unnormalized branches.*

Proof sketch. The inverse of a fixed-current branch adds j at a and subtracts it at b; every basis vector has one branch for each coin, so each column of the joint record/source isometry has two entries $2^{-1/2}$ with disjoint supports [56]. For fixed contrast, the unchanged spectators and the endpoint sum reconstruct the input, which gives (i) for E. Items (ii) and (iv) are the single-seam relations of the candidate register [51]. Item (v) follows from $j = \lceil d/2 \rceil$ for $c = 0$ and $j = \lfloor d/2 \rfloor$ for $c = 1$; the parity-refined Kraus operator $\Pi_p L_{(c,j)}$ has exactly the contrast group as support [73]. The executable checks all 312 outputs of the observer's two seams against an independent floor and ceiling rule [56]. □

The current instrument is no classical coarse-graining of the contrast instrument: grouping the labels of E sums density operators and cannot restore the coherent Kraus sums of C. Both belong to the family of quantum record laws classified in Section 2.3. One further law of the candidate register, the reciprocal exchange N4 that couples the native current to private six-level response factors, enters only the local-tick certificate of Section 3.8 [51].

2.3 Admissible record laws

The classical rule fixes the diagonal of a record law and leaves its coherences free. For the seam $e = (a, b)$ write $\kappa(x) = (x_a, x_b)$ for the seam key and $D = \mathbf{e}_b - \mathbf{e}_a$. A channel Φ on $\mathcal{H}_{78}$ is an admissible record law for e if it acts as the identity on the ten spectator ports, reproduces the classical repair distribution on every $|x\rangle\langle x|$, commutes with $U_t = \exp(i \sum_p t_p N_p)$ for every $t \in \mathbb{R}^{12}$, and is invariant under the carrier rotation that exchanges a and b with the current reversed. The third condition makes every local load phase a symmetry of the unread channel. It is a premise of the classification, a local charge-superselection rule for the repair channel, and it is what excludes a recorder of a pre-repair endpoint load. It guarantees that a definite-current Kraus representation exists; it does not force every recorded implementation of the channel to commit a definite current, since a change of record basis before commitment mixes the current labels and leaves the channel unchanged [50]. The classification below concerns unread channels, and statements about current histories use the compatible definite-current instrument.

Theorem 2.3 (Coherence tables)**.** *A channel* Φ *is an admissible record law if and only if*

$$\Phi(|x\rangle\langle x'|) = \sum_j G_j(\kappa(x), \kappa(x'))\, |x + jD\rangle\langle x' + jD|$$

for endpoint-symmetric positive semidefinite matrices G_j *on the seam keys that can carry the current* j*, with* $G_j(\kappa, \kappa) = \Pr(j \mid \kappa)$*. On* $\mathcal{H}_{78}$ *the free entries are five multipliers:* k *between the empty and the full seam,* g *between the empty seam and a waiting single load,* h *between the full seam and a waiting single load, a real* m *between the two waiting orientations, and* n *between a moving single load and a moving collision. Positivity gives* $|g|, |h|, |n| \leq 2^{-1/2}$ *and* $|m| \leq \frac{1}{2}$*, and* $k = 1$ *forces* $g = h$*. Repairing one seam with* (k, g, h, m, n) *and then with* (k', g', h', m', n') *gives* $(kk', gg', hh', mm', n\bar{h}')$*.*

Exactly two admissible laws are idempotent, those with $g = h = m = n = 0$ and $k \in \{0, 1\}$, and every law with $k = 1$ converges under repetition to the idempotent law with $k = 1$.

Proof sketch. Covariance places $\Phi(|x\rangle\langle x'|)$ in the span of units $|y\rangle\langle y'|$ with $y - y' = x - x'$. Complete positivity confines nonzero Choi entries to pairs with nonzero diagonal, which the classical law makes $(x, x + jD)$, so both sides carry the same current and the Choi matrix is the direct sum of the G_j. Trace preservation and the classical law fix the diagonals, and spectator invariance makes each G_j a function of the seam keys. Under repetition every coherence keeps its seam key, so each multiplier applies again; idempotence requires $k, g, h, m \in \{0, 1\}$ and $n = n\bar{h}$, and the bounds leave $g = h = m = n = 0$. Exact arithmetic in $\mathbb{Q}(\sqrt{2})(i)$ checks the composition law on every pair of seam keys for 64 ordered pairs of tables and on all 6,084 matrix units for two pairs [49]. □

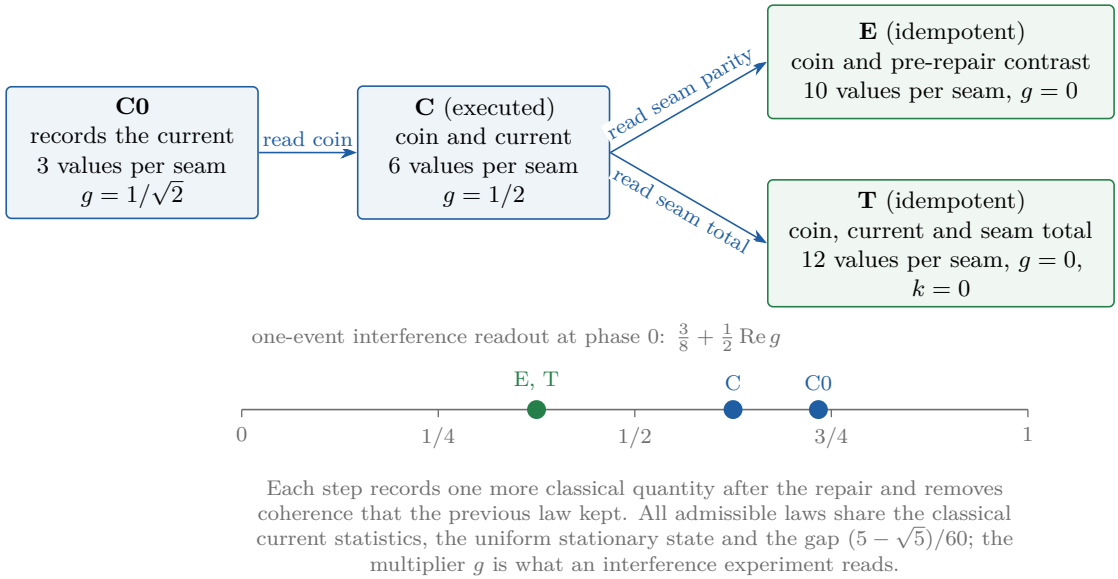


Figure 2: The record-law ladder of Theorem 2.3 and Table 4, and the one-event interference readout $\frac{3}{8} + \frac{1}{2}\,\mathrm{Re}\,g$ at phase zero that separates its members.

Four members form a ladder (Table 4 and Figure 2): C is C_0 followed by a diagonal readout of the coin, E is C followed by a readout of the seam parity as in Proposition 2.2(v), and T is C followed by a readout of the seam total. The completion of C in Proposition 2.2(iv) is the case $g = h = n = \frac{1}{2}$, $m = 0$, and E is the idempotent law with $k = 1$. A recorder of an endpoint's pre-repair load keeps the classical law and fails covariance, because one of its branches carries two currents. C_0 records only what conservation requires, C records the tie-break as an event, E is the coarsest law for which one repair completes the seam, and T records the full seam occupation; none of these principles follows from the theorem premises. The construction executes C as its convention and compares it with E. Under a common definite-current record interface all members assign the same probabilities to every classical current observation; Section 9 reads g in one coherent event, and Section 4.3 identifies all nine real coordinates. Every admissible table, including complex phases and different laws on different seams, has the uniform binary density as its unique stationary source state and the same spectral gap $(5 - \sqrt{5})/60$, so equilibrium selects none of the coordinates; a finite unread warm-up of 221 attempts on the binary shell, or 318 on the full carrier, brings an arbitrary initial density within $1/1000$ of that state, uniformly over untouched references [49].

Table 4: Four distinguished admissible record laws [49].

Law	Each event records	Values per seam	k	$g = h$	Idempotent
C_0	the current	3	1	$2^{-1/2}$	no
C	coin and current	6	1	1/2	no
E	coin and pre-repair contrast	10	1	0	yes
T	coin and both endpoint loads	12	0	0	yes

2.4 The finite observer algebra

An observer of length n on the carrier consists of a preparation on $\mathcal{H}_{78}$, a record tape $R^{\otimes n}$, seam policies s_k that map committed records and classical randomness to seams or to supplied detector outcomes, the coherent writes W_{s_k}, optional supplied instruments whose committed bits may enter s_k, and a halting rule after n writes; a policy reads only committed records. Definition 2.4 is the instance $n = 2$, and Sections 3.4, 5 and 6 add declared instruments to it.

Definition 2.4 (Two-event native observer)**.** Let $R = \mathbb{C}^6$ carry one basis vector per current label (c, j). The observer carrier is

$$\mathcal{H}_{\mathrm{cap}} = \mathcal{H}_{78} \oplus (R \otimes \mathcal{H}_{78}) \oplus (R \otimes R \otimes \mathcal{H}_{78}),$$

with sectors of dimensions 78, 468 and 2808, total 3354; the sector index is the phase, the number of filled record slots.

(a) *Preparation.* The scored preparation is uniform on the 66 binary configurations; the full input family uses all 78.

(b) *Measurement and record.* At phase zero the coherent write $W_e = \sum_r |r\rangle \otimes L_r^e$ acts on the seam $e = (0, 1)$.

(c) *Record-only feedback.* At phase one the seam is $s(r) = (0, 5)$ if the stored current is positive and $(0, 1)$ otherwise, and $V_1(|r\rangle \otimes \psi) = |r\rangle \otimes W_{s(r)}\psi$.

(d) *Forecast.* Before the second event the controller reads its stored letter and issues a supplied forecast $q(r)$ of a positive second current.

(e) *Halting.* At phase two the observer acts as the identity.

The controller reads only the committed first record; it has no access to source loads or future outcomes.

Proposition 2.5 (One CPTP map with classical-prefix preservation)**.** *Let $V_0 = W_{(0,1)}$ and V_1 be extended by zero to the other sectors, giving A_0 and A_1, and let P_2 project onto the full sector. Then:*

(i) V_0 and V_1 are isometries on every nonfull sector, including unreachable combinations of letters and loads, and $T(\rho) = A_0\rho A_0^\dagger + A_1\rho A_1^\dagger + P_2\rho P_2$ is CPTP;

(ii) on the inhabited subspace $K = \mathcal{H}_1 \oplus \mathcal{H}_2$ every first-record projector Q_r obeys $T^*(Q_r) = Q_r$, so a committed first record is reread with probability one;

(iii) for the blank input with occupations $(1, 1, 0, \ldots, 0)$*, the first-record projector of the letter* $(0, 0)$*, extended by zero to phase zero, has expectation* 0 *before the tick and* $1/2$ *after it, so writing a record is a transition distinct from preserving it;*

(iv) for the equal superposition of $\{0, 2\}$ *and* $\{1, 2\}$*, the reduced-density entry of the first-record pair* $((0, 1), (0, 0))$ *equals* $1/4$ *after the first write and* $1/8$ *after the second while the classical diagonals are unchanged, so coherence between different record values need not survive feedback.*

Proof sketch. Orthogonal retained labels and $W_s^\dagger W_s = I$ give $V_0^\dagger V_0 = V_1^\dagger V_1 = I$, hence $A_0^\dagger A_0 + A_1^\dagger A_1 + P_2 = I$. Every subsequent map intertwines Q_r on K. In (iv) the working state attached to the entry has occupations $\{1, 2\}$, on which the isometries of $(0, 5)$ and $(0, 1)$ overlap by $1/2$ [56]. The checkpoint adapter verifies all $78 + 468 + 2808$ columns and both the preservation identity on the 3276-dimensional checkpoint space and the blank-sector control [79, 105]. □

Lemma 2.6 (Feedback requires classical record preservation)**.** *Let* Λ *be a CPTP map from* $R \otimes W$ *to* $R \otimes W'$ *with* $\Lambda^*(B \otimes I) = B \otimes I$ *for every* B *in the full matrix algebra of* R*. Then* $\Lambda^*(I \otimes O) = I \otimes \Phi^*(O)$ *for a unital completely positive map* Φ^**, so changing only the record preparation of a product input changes no output statistic on* W'*.*

Proof sketch. The operators $B \otimes I$ lie in the multiplicative domain of the unital completely positive map Λ^*, because the Schwarz inequality is an equality for them [112]. Then $\Lambda^*(I \otimes O)$ commutes with every $B \otimes I$ and has the stated form [56]. □

Lemma 2.6 fixes the shape of the observer contract. A record that steers the working state is preserved as a classical record, with its projectors fixed, while coherence between different record values may change.

2.5 Declared services

Each component names the services it assumes; they belong to the specification and enter the resource counts where a component counts resources. The services are coherent proper-code load and export, with complementary failure outcomes retained by a complete channel definition; blank initialization, reset, storage and routing of quantum registers and classical records; Born readout of pointers and fair coins; and analogue precision of source pulses. Particular components add a calibrated external flux and probe (Section 3.4), incident-event recognition and bare-occupation readout (Section 3.8), coherent phase preparation and readout (Section 9), and, for the native process results, independent uniform seam scheduling, uniform fixed-number preparation, full initial and current readback, coherent wave processing, the return-length sampler and the response controller (Sections 7, 4 and 9). The coherent services are not products of the native grammar: committed C/E repair with diagonal accessible inputs and fresh independent diagonal ancillas cannot prepare the declared coherent reference to arbitrary accuracy, even with adaptive heralding. This coherence floor of the native interface makes the coherent services a necessary resource of the specification. The sharp joint instrument of Section 6.1 uses native integer repair as a declared primitive, and Section 6.5 attaches the compiled 78-dimensional C/E channels, routing and encoded recovery to the continuing work history. The processor algebra is M_6; its twelve generators are the exact source generators of the compiler, one per port [70, 79]; a proper code is a two- or four-dimensional subspace of $\mathbb{C}^6$ with its load and export channels and the complementary failure outcome; and the service error model is the compositional contract of Section 9.2. Exact certificates exclude numerical execution error and physical service error unless a component states otherwise.

2.6 The closure inputs P and N

The pixel ratio P is the area of one screen cell in units of $\ell_\star^2$, and N is a record capacity in nats [7, 64]. The construction accepts both as closure-derived inputs and fits neither to an output. P enters through a declared coupling map, the source-only structured-running branch of the fine-structure closure (one-loop running and matching conventions with the tree-level m_Z closure), defined in the P-derivation code of the repository [2] and consumed unchanged; the map is a declared selection, not a result of this construction. At the pinned closure value $P = 1.630972095858897\ldots$, the midpoint of the pinned forward-closure interval, the map gives $\alpha^{-1}(P) \approx 136.9948351774$ and $g(P) = \sqrt{4\pi\alpha(P)} \approx 0.302867613117$ in Heaviside–Lorentz natural units, numerical evaluations of the executed map [62, 101]. N enters only as a bound, and no component consumes its closure value. For a declared capacity carrier of dimension D, the record capacity $N_{\text{record}} = \ln D$ bounds every correctable classical record code by $\ln M_0 \le N_{\text{record}}$ [48, 57]. The cosmological relation $N = 3\pi/(\Lambda\ell_\star^2)$ additionally requires a physical identification of capacity with horizon area, which the construction does not make.

2.7 Time parameters

The construction keeps the time parameters of Table 5 distinct [54]. Conversions between them need explicit maps. The construction uses three: the fixed-reference relation $t = -\beta s$ of Section 6.6 between a model coordinate and a modular parameter, the instrument-valued conversion between two local event clocks of Section 4.2, which keeps their whole count law, and the rescaling of native attempt count along the refining screens of Section 7. A calibration of native calls against a monotonic host counter is an engineered software correspondence. No preferred universal physical clock is required.

Table 5: Time parameters and where they occur.

Parameter	**Meaning**	**Occurrence**
Host time	Runtime of a software execution; a resource cost	A software calibration of native calls; never a physical quantity
Solver counter	Position in a search or repair schedule; the attempt count of a declared schedule	History repair (Section 6.7); attempt counts in the observer theorems
Causal order	Precedence of recorded events; counts of native operations	Observers, detectors, phase protocol, local ticks
Local event count	Attempted repairs incident to an observer region, or occupation changes at a port	Overlapping observers, local and active event clocks, matched screens (Section 4)
Scaled attempt count	Native attempt count n at level l rescaled to $\tau = n/(1260 \cdot 16^l)$ on the refining screens	Native density limits and the controlled repair response (Sections 7.3, 9)
Coherent wave parameter	Parameter of the admitted coherent continuation $\cos(t\sqrt{A})$	Native waves and the return reference (Sections 7.2, 7.7)
Model coordinate	Parameter of a declared Hamiltonian or action	Scalar detectors, two-site apparatus, common source
Modular parameter	Parameter of the modular automorphisms of an algebra-state pair	Modular clock and fixed-reference relation (Section 6.6)
Operational clock	Observer-accessible reading with calibrated duration	Supplied by no component

3 The finite observer

This section is layer three on one carrier: an observer that prepares, records, retains and acts on native records, with its capacity, memory and recovery limits.

3.1 The native observer

The native observer executes Definition 2.4 on the scored preparation [55, 56]. A trial is one fresh preparation followed by two repair events and a halt; running many trials repeats the preparation and does not extend one observer's memory. Event order is the time parameter. The tape, prior, forecast table, seam selector and halting rule are designed apparatus, and the forecast is a fixed table without learning.

Proposition 3.1 (Statistics of the native observer)**.** *Prepare the 66 binary configurations uniformly, use fair coins and the controller of Definition 2.4, and let Y indicate a positive second current. Then:*

(i) the 264 histories, two coins for each of the 66 configurations at each of the two events, each have weight $1/264$*;*

(ii) the first record has masses $14/33$*,* $14/33$*,* $5/66$*,* $5/66$ *for* $(c, j) = (0, 0), (1, 0), (0, 1), (1, -1)$*, the conditional probabilities* $q(r) = \Pr(Y = 1 \mid r)$ *are* 0*,* $5/56$*,* 0*,* $1/2$ *in the same order, and* $\Pr(Y = 1) = 5/66$*;*

(iii) the Brier risk of forecasting Y is $395/7392$ *with* $q(r)$*,* $140/2013$ *with the best forecast based on the chosen seam alone, and* $12875/150304$ *when q is evaluated on an independent record resampled within the same seam;*

(iv) the mean final load at port 0 *is* $35/264$ *under record feedback and* $1/6$ *when the selector is disconnected or replaced by an independent selector with the same seam frequencies; the feedback effect is* $-3/88$*.*

Proof sketch. For (iii), $\mathbb{E}[(Y - q(R))^2] = \mathbb{E}[\mathrm{Var}(Y \mid R)]$. The chosen seam S is a function of R; the best S-only forecast has excess risk $\mathbb{E}[\mathrm{Var}(q(R) \mid S)]$, and an independent resample within the seam adds twice that amount. For (iv), a positive first current occurs on ten binary inputs with coin zero, mass $5/66$. On that branch $x_0 = 0$, $x_1 = 1$ and $\mathbb{E}[x_5] = 1/10$; a second repair on $(0, 1)$ gives port-0 mean $1/2$ and one on $(0, 5)$ gives $1/20$. Hence the change is $(5/66)(1/20 - 1/2) = -3/88$ and the mean is $1/6 - 3/88 = 35/264$. An independent selector leaves the uniform shell uniform under either seam [56]. □

The other two current letters occur only on the full carrier, where the forecast abstains. The exact receipt retains all 264 histories [96], and a seeded execution of 20,000 fresh trials matches every trial against its exact path reference, with sample final port-0 loads 0.1327 with feedback, 0.16645 disconnected and 0.1668 with the independent selector [97]. Since the selector is a design choice and the forecast table is the exact conditional law, (iii) and (iv) are consistency checks: the Bayes forecast beats coarser forecasts and the record-wired selector moves the occupation.

3.2 Continuing memory and the actual stationary reference

The finite campaign [2] extends the two-event observer with an explicitly declared memory law. It keeps the same source and complete C/E instruments. A classical tag chooses C with probability λ

and E otherwise; a repair occurs with probability r, with an identity hold otherwise. A controller chooses a preferred seam from its retained records. The actual seam distribution is

$$p(e \mid m) = \frac{\epsilon}{30} + (1-\epsilon)\pi(e \mid m), \qquad r, \epsilon \geq \frac{1}{16}, \quad 0 \leq \lambda \leq 1.$$

The rational parameter bound and permitted controller tables are part of the campaign, not measured physical constants. The executed controller proposes $(0, 5)$ after a positive current and $(0, 1)$ otherwise. It does not read the hidden source load. A proposal intervention preserves the uniform exploration floor. Holds are recorded with current zero.

For each classical memory input, every source branch is $K_b = \sqrt{w_b} M_b$, with M_b a native partial injection or identity and $\sum_b K_b^* K_b = I_{78}$ exactly. Tensoring with the identity preserves an arbitrary untouched reference. The tests independently reconstruct the branch supports by endpoint floor/ceiling arithmetic, including their off-diagonal action. Composition therefore defines every finite history with its unnormalized quantum output and actual probability. Exact prefix replay and retained six-attempt source/reference/last-record marginals verify C, E and the delayed tagged mixture; the complete history tree remains implicit in that channel kernel.

One to eight complete records occupy the declared buffer. Under the executed one-slot export policy, n attempts leave one record and export $n-1$ records for $n \geq 1$; expected repair and hold counts are nr and $n(1-r)$. Concatenating exports and the remaining buffer reconstructs the history. The halt-on-full control instead preserves its source and records with an explicit timeout outcome. These are different resource contracts. Bounded working memory with growing exports is not closed finite-resource operation for unlimited duration. No new native source-word implementation of the supplied initialization, selection, write, export or delivery services is claimed.

The stationary reference must include this controller. For $r = 1$ and $\epsilon = 1/2$, a uniform binary source with blank memory becomes nonuniform after two attempts: 30 source probabilities change, with maximum deviation $1/960$ from $1/66$. Exact rational solution of the actual source/current-sign kernel instead gives one stationary law on a 129-state recurrent subset of the 132 source/bit states. Applying the complete C/E observer instrument to that law constructs the full-record reference. Its projection recovers the bit law, and one further actual observer step preserves the entire full-record density exactly. The resulting 128 C or 192 E record blocks have 4,218 positive diagonal entries overall. Positivity establishes faithfulness only on the explicitly compressed support of each block. The growing exported tape is not a stationary object, and the delayed member's hold updates require a separate stationary solve.

This reference also makes a clock obstruction precise. In an actual nontracial block choose positive weights with ratio $q = p_i/p_j > 1$ and the dimensionless modular parameter $s_* = \pi/(2 \log q)$. The state-dual modular action takes the real pair probe $(|i\rangle + |j\rangle)/\sqrt{2}$ to a density with off-diagonal entry $-i/2$. Every admitted history, with the declared real joint ancillas and classical memory, preserves real density matrices. A norm-one imaginary pair observable separates the target from every such normalized output, giving trace-distance error at least $1/2$. This witness is not a native detector, and the probe is an all-input map test, not an asserted native preparation. On the accessible commutative record algebra the modular action is trivial. Other algebras, encodings or coherent services have different contracts; no general physical-clock exclusion follows. The exact reference, implementation bound and full qualifications are retained with the executable model [2].

3.3 Record capacity

The retained records are evaluated with the correctable-public-record construction of the theorem library [7, 57]. There M_0 is the largest correctable set of reachable public records, the independence

number of the compound checkpoint confusability graph; the record capacity is $N_{\mathrm{record}} = \ln M_0$ nats; and $M_0 \le D$ for a declared carrier of dimension D. The counts below describe the reachable record interface; program, source, probe and service memory are additional.

Proposition 3.2 (Record capacity of the native observer)**.** *Take halt and exact classical reread as the declared terminal continuations of the observer of Definition 2.4. Then the compound graph has no edges, and:*

(i) $M_0 = 11$ *on the binary preparation (four reachable first records, 264 histories) and* $M_0 = 17$ *on the full family of 78 inputs (six first records, 312 histories);*

(ii) each terminal word projector has rank 78 in the 3354-dimensional carrier, distinct projectors are orthogonal, and the eleven reachable ones have total rank 858, so the bound is unsaturated with slack $\ln(3354/11)$*;*

(iii) the binary preparation reaches $1{+}4{+}11 = 16$ *classical prefixes, so four bits encode the complete controller history; a compact controller that reads only this code reproduces the phase, action, forecast, source loads, record and halt of all 264 histories; the full family reaches 24 prefixes and needs five bits;*

(iv) coarser retained readouts give correctable codes of size 4 for the first letter alone, 4 for the second letter alone, 3 for the forecast alone, 2 for the chosen seam alone, 4 for forecast and seam, and 1 after complete erasure.

Proof sketch. Halt and reread leave the tape unchanged, so the joint kernel is the identity and the existing evaluator returns the number of reachable terminal words. The terminal projector of word w occupies the basis coordinates $546 + 78w$ to $546 + 78w + 77$ of the carrier [57, 98]. □

The terminal capacity is $\ln 11$ nats, about 3.459 bits, on the binary preparation and $\ln 17$ nats on the full family; the Shannon entropy of the benchmark record law, about 2.023 nats, is a different quantity. The library's global proposal adds closure across the complete terminal fiber, $N_\star = \ln M_0(U_{N_\star})$, and a de Sitter horizon identification giving $\Lambda \ell_\star^2 = 3\pi/N_\star$ [7, 57]; the native records supply an actual finite input to that evaluator, and the two further steps are not taken here.

3.4 The P-dependent detector

The detector carries the closure input P through the electromagnetic coupling into a recorded binary observation that controls the next repair [62, 63]. The charge assignment, probe, calibrated flux, readout and controller are specified apparatus (Section 9.3).

Choose the charged realization $Q_v = g(P)N_v$ of the native occupation N_v. A current record j on the seam (a, b) then transfers the charge $\Delta Q = g(P)\, j\, (\mathbf{e}_b - \mathbf{e}_a)$, and total charge is conserved. For port gauge parameters λ_v and $U_\lambda|x\rangle = e^{ig\sum_v \lambda_v x_v}|x\rangle$, every current-grouped Kraus operator obeys $U_\lambda L_{(c,j)} U_\lambda^\dagger = e^{igj(\lambda_b - \lambda_a)} L_{(c,j)}$. The probe is a unit-charge path qubit prepared in $|{+}X\rangle$. The first native current routes it around a calibrated external loop with signed exposure $j\chi$, where $\chi = \oint A{\cdot}dx$ is prepared independently of g, so the relative phase is $\varphi_j = g(P)\, j\, \chi$, the Aharonov-Bohm phase [108]. A Y measurement commits $y = \pm 1$ with amplitudes $m_y(j) = \frac{1}{2}[e^{-i\varphi_j/2} - iy\, e^{i\varphi_j/2}]$ and effects $E_y = \sum_r \frac{1}{2}[1 + y \sin\varphi_{j(r)}]|r\rangle\langle r|$ on the old current register. The actuator receives only y. It repairs $(0, 5)$ for $y = +1$ and $(0, 1)$ for $y = -1$, keeps y unchanged, and the protocol halts.

Proposition 3.3 (P-dependent response)**.** *Prepare the 66 binary configurations uniformly, run the protocol above, and put* $s = \sin(g(P)\chi)$*. Then:*

(i) $\{E_y\}$ *is a binary instrument on every input, including coherent records; each* $M_y = \sum_r m_y(j(r))|r\rangle\langle r|$ *commutes with the old record projectors, and the composed experiment is one CPTP map that preserves the committed bit;*

(ii) $\Pr(y \mid j_1, P, \chi) = \frac{1}{2}[1 + y \sin(g(P) j_1 \chi)]$*;*

(iii) at $\chi = 1$ *all 528 state, coin, detector and coin histories have positive probability, and*

$$\mathbb{E}[y] = 0, \quad \mathbb{E}[y j_1] = \tfrac{5}{33} s, \quad \mathbb{E}[j_2] = \tfrac{5}{264} s, \quad \mathbb{E}[g(P) j_2] = \tfrac{5 g(P)}{264} s;$$

(iv) for weak exposure, $\mathbb{E}[g\, j_2] = \frac{5\pi}{66}\alpha(P)\chi + O(\chi^3)$*;*

(v) disconnecting the probe, or replacing y *by an independent fair bit, gives* $\mathbb{E}[j_2] = 0$ *with the same detector and seam frequencies.*

Proof sketch. The eigenvalues of E_y lie in $[0, 1]$ and sum to one, which gives (i). For (iii), after the first repair let the source be z; averaging the second coin gives $\mathbb{E}[j_2 \mid z, j] = \frac{1}{4}(2z_0 - z_1 - z_5) + \frac{sj}{4}(z_1 - z_5)$. The first term averages to zero because the first repair preserves the uniform shell. Currents ± 1 each have mass $5/66$, and on either branch the other load is uniform over ten ports, so $\mathbb{E}[j(z_1 - z_5)] = 5/66$. An independent floor and ceiling enumerator reproduces every coefficient over all $66 \cdot 2 \cdot 2 \cdot 2 = 528$ paths [63, 65]. □

At the computed P and $\chi = 1$ the plus probability is 0.649129261647 for $j = +1$, 0.350870738353 for $j = -1$ and $1/2$ for $j = 0$; the mean second current is 0.00564883566843, the mean transferred charge 0.00171084937579 in natural units, and the final port-0 load 0.161017830998235 against $1/6$ for the controls. Off-closure controls at $0.999P$ and $1.001P$ change the detector and feedback probabilities at fixed χ, and a seeded run of 200,000 trials gives mean second current 0.006390 with standard error 0.00085715 [62, 101]. The audit record (r_1, y, r_2) has 24 terminal words and $M_0 = 24$ [64]. If a cut enclosing a retained record obeys the area-capacity law with area K chart cells, then $N_{\mathrm{cut}} = KP/4$ and $\ln M_0 \le N_{\mathrm{cut}}$ give $K \ge 4 \ln M_0 / P$; with the pinned P one detector bit needs $K \ge 2$ ($4 \ln 2 / P \approx 1.6999608573$) and the full audit $K \ge 8$ ($4 \ln 24 / P \approx 7.7942567832$).

3.5 Quantum mixing and memory

Discarding every seam, coin and repair record after each event turns the repairs into channels Φ_{C} and Φ_{E} on the source. Under independent seam attempts these channels have an exact spectral theory [58]. Theorems 3.4 and 3.5 concern this record-discarded channel under independent uniform attempts; they do not describe the retained tape or the feedback controller.

Theorem 3.4 (Signed-difference reduction and quantum gap)**.** *Let* G *be a finite connected graph with* $n \ge 2$ *vertices, positive edge weights* w_{ab} *and* $W = \sum w_{ab}$*. Let each event choose the edge* ab *with probability* w_{ab}/W *and a fresh fair coin, and discard its record. On the binary space* $\mathcal{H}_k$ *of load* k*,* $1 \le k \le n - 1$*, with* $D = \binom{n}{k}$*:*

(i) for a matrix unit $|x\rangle\langle y|$ *the signed difference* $z = x - y$ *is invariant; if* z *has* r *positive entries and* $Z = \{v : z_v = 0\}$*, then on the sector of* z

$$\Phi_{\mathsf{L}}\big|_z = I - \frac{L_{\mathrm{excl}}(G[Z], k - r) + K_{\mathsf{L}}(z)\, I}{2W}, \qquad \mathsf{L} \in \{\mathsf{C}, \mathsf{E}\},$$

with the exclusion Laplacian L_{excl} of the induced graph, $K_{\mathsf{C}}(z) = \sum w_{ab}|z_a - z_b|$ and $K_{\mathsf{E}}(z) = 2\sum w_{ab}\mathbf{1}[z_a \neq z_b]$;

(ii) both channels are unital and Hilbert-Schmidt self-adjoint with spectrum in $[0,1]$, their fixed algebra consists of the scalars, and both have gap $\gamma = \lambda_1(G)/(2W)$, where $\lambda_1(G)$ is the first positive Laplacian eigenvalue; and $\|\Phi_{\mathsf{L}}^t - \mathrm{Dep}_D\|_{2\to 2} = (1-\gamma)^t$;

(iii) for every input and every untouched reference R, $\mathrm{dist}((\Phi_{\mathsf{L}}^t \otimes \mathrm{id}_R)(\rho), \tau \otimes \rho_R) \le \min\{1, \frac{D}{2}(1-\gamma)^t\}$, with $\tau = I_D/D$ and $\mathrm{dist}(\rho,\sigma) = \frac{1}{2}\|\rho - \sigma\|_1$.

For the twelve-port carrier at load two ($n = 12$, $W = 30$, $\lambda_1 = 5 - \sqrt{5}$, $D = 66$) the gap is $\gamma = (5-\sqrt{5})/60 \approx 0.046066$, and the slowest off-diagonal sectors contract by exactly $r_{\mathsf{C}} = 5/6$ and $r_{\mathsf{E}} = 7/10$ per attempt. On the full 78-state carrier the gap is the same, the full channel is block triangular with a diagonal collision block whose entries lie in $[0, 5/6]$, and the unique stationary density is the uniform binary density, of rank 66 on dimension 78.

Proof sketch. A matrix unit survives a coin branch exactly when its two records agree, and then both configurations translate by the same current; comparing the coin branches gives (i). The diagonal sector is the symmetric k-particle exclusion chain, whose gap equals $\lambda_1(G)$ by the theorem of Caputo, Liggett and Richthammer [111]. For a nonzero sector, $K_{\mathsf{E}} \ge K_{\mathsf{C}} \ge \frac{1}{2}\sum w_{ab}(z_a - z_b)^2 \ge r\lambda_1 \ge \lambda_1$, so no off-diagonal mode is slower. Item (iii) uses Schmidt rank at most D and $\|Y\|_1 \le D\|Y\|_2$. On the carrier, $r = 1$ gives $K_{\mathsf{C}} = 10$ and $K_{\mathsf{E}} = 2(10 - \mathbf{1}[uv \text{ edge}])$, while $r = 2$ gives $K_{\mathsf{C}} \ge 16$ and $K_{\mathsf{E}} \ge 24$. The collision block has largest entry $1 - d_{\min}/W = 5/6 \le 1 - \gamma$ [58]. The executable checks the reduction against the native primitive on all 4,356 binary matrix units and the transient structure on all 6,084 full matrix units [99]. □

Choose adjacent ports u, v and the two mixtures $\rho_\pm = \frac{1}{20}\sum_{w\notin\{u,v\}}(|uw\rangle \pm |vw\rangle)(\langle uw| \pm \langle vw|)$. They have identical load populations and orthogonal supports, and

$$\mathrm{dist}(\Phi_{\mathsf{L}}^t(\rho_+), \Phi_{\mathsf{L}}^t(\rho_-)) = r_{\mathsf{L}}^t, \qquad \mathrm{dist}(\Phi_{\mathsf{C}}^t(\rho_+), \Phi_{\mathsf{E}}^t(\rho_+)) = \tfrac{1}{2}(r_{\mathsf{C}}^t - r_{\mathsf{E}}^t).$$

The second distance is $1/15$ after one attempt and peaks at four attempts at $12259/101250$; the binary effect $\frac{1}{2}(I + A_{uv} + A_{uv}^\dagger)$ with $A_{uv} = \sum_{w\notin\{u,v\}}|uw\rangle\langle vw|$ attains it. Load populations, classical load histories and diagonal geometric observables obey the same exclusion process under both laws, so only coherent preparations and readouts distinguish them under this schedule.

Theorem 3.5 (Finite quantum memory)**.** *Under the schedule of Theorem 3.4 on the twelve-port carrier:*

(i) on the binary shell both record-discarded channels are entanglement breaking after at most 134 attempts;

(ii) on the full carrier let u, v be adjacent and $|\Psi\rangle = (|2u\rangle|0\rangle + |uv\rangle|1\rangle)/\sqrt{2}$ with a reference qubit; after t attempts the coefficient of $|2u\rangle\langle uv|$ is a_{L}^t, with $a_{\mathsf{C}} = 23/30$ and $a_{\mathsf{E}} = 7/10$, and the partial transpose on the reference has a principal 2×2 minor equal to $-a_{\mathsf{L}}^{2t}/4 < 0$; no finite power of either full channel is entanglement breaking;

(iii) every input converges to $\tau_P \otimes \rho_R$, where τ_P is the uniform binary density, with $\mathrm{dist} \le \min\{1, \mu^{s/2} + \mu^s + \frac{D}{2}(1-\gamma)^{t-s}\}$ for all $0 \le s \le t$, $\mu = 5/6$ and $D = 66$.

Proof sketch. For the normalized Choi state J_t, $\|J_t - I/D^2\|_2^2 = D^{-2}\sum_{i>0}\lambda_i^{2t}$. The separable ball of Gurvits and Barnum [116] and the equivalence of separable Choi states with entanglement-breaking channels [118] reduce (i) to $65\,r_{\mathrm{mix}}^{2t} + 4290\,r_{\mathsf{L}}^{2t} \le 1/4355$, where $r_{\mathrm{mix}} = 1-\gamma$. The rational enclosure $559/250 < \sqrt{5} < 2237/1000$ certifies this at $t = 134$ and its failure at $t = 133$; treating all modes with r_{mix} would certify only 178. For (ii), 25 edges avoid u and four of them touch v; C keeps one coin on each of those four and E keeps neither, and the triangular structure prevents any return from the binary space. Separable states have positive partial transpose [122]; the complete argument is in [58, 99]. □

Replacing the full carrier by its attracting binary shell therefore changes an exact finite-time quantum property, although both converge to the same stationary density. The number 134 is a sufficient certificate time; the exact index of entanglement breaking on the shell is not determined.

3.6 Recovery from complete records

Preserving committed records and recovering the unknown input from them are different questions [59]. Here a seam policy may depend on committed records and classical randomness and may not inspect hidden loads or insert quantum corrections; a final decoder may act on the full load register and tape, keep the tape, use blank ancillas and return a state on the original carrier. That decoder access is stronger than the record-only access of the observer's controller.

Theorem 3.6 (Optimal recovery from complete native records)**.** *Let X be the 78 configurations or the 66-state shell and $D = |X|$. For a complete admitted history h let X_h be its compatible inputs, P_h the projector onto them, $d_h = \operatorname{rank} P_h$, and w_h the product of its seam and coin factors. Then:*

(i) the history Kraus operator is $K_h = \sqrt{w_h}\,U_h P_h$ with $U_h^\dagger U_h = P_h$ and $\sum_h w_h P_h = I$; a tape-controlled inverse outputs $\sum_h w_h |h\rangle\langle h| \otimes P_h \rho P_h$, so every initial load population is recovered;

(ii) among all final CPTP decoders the optimal entanglement fidelity for the maximally mixed input is $F_{\mathrm{opt}} = D^{-2}\sum_h w_h d_h^2 = D^{-2}\sum_{x,y} H_{xy}$, with $H_{xy} = \sum_{h:\,x,y\in X_h} w_h$;

(iii) for the two-event observer, F_{opt} equals 5543/8712 (binary) and 1301/2028 (full) under C, and 769/1452 and 3329/6084 under E with the same current-sign selector applied to its decoded current;

(iv) the maximal exactly correctable algebra on the whole input is the commutant of $\{P_h\}$, and one event with full-support random seam choice reduces it to the diagonal load algebra; nevertheless the qubit code $|0_L\rangle = (|\{0,2\}\rangle + |\{1,3\}\rangle)/\sqrt{2}$, $|1_L\rangle = (|\{0,3\}\rangle + |\{1,2\}\rangle)/\sqrt{2}$ satisfies the Knill-Laflamme condition $QP_hQ = \alpha_h Q$ for every seam, coin and record of either law and is recovered exactly after one event;

(v) under independent seam attempts $H_{xy}(t) = q_{\mathsf{L}}(x-y)^t$, with largest off-diagonal rates 5/6 for C and 7/10 for E on both carriers; $F_{\mathrm{opt}}(t) \to 1/D$, and neither retained instrument is entanglement breaking at any finite horizon;

(vi) a fixed E seam word identifies every full-carrier input exactly when at most one visited component has size one or two, and every binary input exactly when at most one component is a singleton and at most one has size two; the minimal such words have eight and seven distinct edges;

(vii) no bounded record-only adaptive policy makes C *entanglement breaking; a* C *word that doubles a minimal forest, 16 or 14 events, excludes every nontrivial exact code; the policy that repeats each forest edge until its coin first differs identifies the input almost surely, with stopping time* $T = m + \sum_{i=1}^{m} G_i$ *for independent geometric* G_i *of success probability one half,* $\Pr(T = t) = \binom{t-m-1}{m-1} 2^{-(t-m)}$ *for* $t \geq 2m$*, and* $\mathbb{E}[T] = 3m$*, namely 24 on the full carrier and 21 on the binary shell.*

Proof sketch. Every retained current determines a translation, so reversing them gives the input uniquely from the terminal configuration and h. Optimality follows from Cauchy-Schwarz on each classical branch: $\sum_a |\mathrm{Tr}(R_{ha} U_h P_h)|^2 \leq d_h^2$. For (iv), commutation is sufficient by (i), and necessity follows from equality in the Schwarz inequalities in a Stinespring dilation; the code has nonadjacent occupied pairs, equal one-port means and vanishing products $N_a N_b$ on every edge, which gives the Knill-Laflamme identity [119]. For (vi), matching E histories are those for which z is constant on every visited component. For (vii), the inputs $\{u, w\}$ and $\{v, w\}$ with u, v nonadjacent keep equal currents for at least one coin on every edge, so a history of length at most T shares a rank-two fiber with probability at least 2^{-T}; repeating an edge with the opposite coin reveals the initial parity through $d = 2j_1 + j_*$ [59, 100]. □

The minimal fixed E words are the forests $(0,1),(0,5),(2,3),(2,4),(6,8),(8,9),(7,10),(10,11)$ on the full carrier and $(1,5),(2,3),(2,4),(6,8),(8,9),(7,10),(10,11)$ on the binary shell, with component sizes $(3,3,3,3)$ and $(1,2,3,3,3)$. The full-input correctable blocks of the two-event observer have sizes $46, 10, 9, 1$ on the binary shell and $56, 10, 9, 1, 1, 1$ on the full carrier, for both laws. Existing source inverse words implement the single-seam branch reversal of (i) while keeping the classical archive [59, 100]. Native current records thus suffice to reverse classical load transport, without restoring arbitrary coherence lost across record groups. The source encoder and active recovery of Section 6.5 instead preserve the stated logical qubit between events; they do not invert the whole bare-load channel.

3.7 A coherent continuation and its selection boundary

An exact finite experiment attaches a reflection walk to the binary native branches, with coherent event access in place of a committed tape [60]. For each binary configuration x, seam e and coin c, let (y, j) be the native output and current. On the $66 \cdot 60 = 3960$ arc basis vectors define

$$S|x,e,c\rangle = |y,e,c'\rangle, \qquad c' = \begin{cases} c, & j = 0, \\ 1-c, & j \neq 0, \end{cases} \qquad A|x\rangle = \frac{1}{\sqrt{60}} \sum_{e,c} |x,e,c\rangle.$$

Every arc is checked: $S^2 = I$, with 300 exchanged pairs and 3360 fixed arcs, and the outgoing native record (y, e, c, j) is recovered bijectively from the reversed arc. Thus $A^\dagger A = I$ and $A^\dagger S A = K$, the actual symmetric native transition kernel, and $W = S(2AA^\dagger - I)$ satisfies the hypotheses of the quantum-walk spectral mapping theorem with discriminant K [151, 193]. The identity $A^\dagger W^n A = T_n(K)$, with T_n the Chebyshev polynomial, gives for the centered loads $X_{x,p} = x_p - 1/6$

$$KX = X(I - L/60), \qquad U_{n+1} - 2U_n + U_{n-1} = -U_n L/30, \qquad U_n = A^\dagger W^n A X.$$

These are compressed amplitudes, and n counts the supplied coherent continuation with no identification as duration. The binary domain is essential: on the full carrier $(2, 0, \ldots) \mapsto (1, 1, \ldots)$ has no native reverse event, so its asymmetric kernel equals $A^\dagger S A$ for no involutive unitary on the same domain. Twelve port addresses and sixty seam/coin events give a 720-arc source-compiled walk

with discriminant $K_{12} = I - L/60$; the map $F_{x,p} = (x_p - 1/6)/\sqrt{10}$ satisfies $F^*F = I - \mathbf{1}\mathbf{1}^T/12$ and $K_{66}F = FK_{12}$, so it realizes the shared mean-zero density sector. Among the sixty proper carrier rotations a unique inverse-symmetric order-five class $\mathcal{C}$ of size twelve gives $L = 6I - \frac{1}{2}\sum_{g\in\mathcal{C}} U_g$, with multiplicity-free density representation $1 \oplus 3 \oplus 5 \oplus 3$. The other order-five class is isospectral but assigns probability zero to a native seam whose actual one-attempt probability is $1/60$, so the seam records select the class.

The emitted walk word uses ten address qubits, five blank scratch qubits and 114,910 source pulses per walk with 229,820 load/export events; uniform event preparation costs a separate 36,288 pulses. The reversible circuit is checked on all 32,768 basis inputs, every clean-scratch address returns clean scratch, including 304 padding addresses, and the fifteen-qubit execution reuses raw pulse factors without target-matrix insertion or output normalization. Floating routine guards sum to $s = 829/(5 \cdot 10^9)$, giving complete output trace-norm error at most $2\delta + \delta^2$ with $\delta \le s/(1-s)$; these are numerical guards, not physical precision certificates.

A finite reference and its modular clock. Biased native event amplitudes, the compiled reversible seam shift, inverse event preparation and a herald realize $F_p = I - pL/60$ on the protected site code. From the supplied $I_{12}/12$, n successive successes give $F_p^{2n}/\operatorname{Tr} F_p^{2n}$ with success probability at least $1/12$, every failure recorded; for $p = 30t/n \le 1$ this tends to $e^{-tL}/\operatorname{Tr} e^{-tL}$ with relative-entropy and total-variation errors at most $16t^2/[n(1-4t/n)]$. The execution at $p = 1/2$, $n = 2$, $t = 1/30$ tests all twelve input columns with success $4382551/5184000$ and bound $1/105$, and a source interference readout has contrast $260343/43825510$ against zero after dephasing. This state is central on the 44-dimensional represented group algebra, where its modular flow is trivial; nontrivial modular gaps require the full protected matrix algebra [21].

Positive subordination gives square-root weights on the same register. With $F = I - hL/2$, $h = p/30$, a Poisson number of Sibuya jumps gives positive weights with

$$\sum_n w_n F^{2n} = e^{-R\sqrt{I-F^2}} = e^{-b\sqrt{L-hL^2/4}}, \qquad b = R\sqrt{h}.$$

Finite jump and count caps and dyadic rounding retain missing mass as sampler rejection. The source instance $p = R = 1/2$ uses 64 native coin records and at most four raw filters, retains each first failure and all sixteen successful interference outcomes, and has Gibbs trace-distance bound below 0.217 and spectral diagnostic 0.01344. Its finite polynomial $M = \sum_n w_n F^{2n}$ fixes $H = -\log(M/m_0)$, $m_0 = \sum_n w_n$. Raw source words execute that clock, a current-controlled intersector kick and complete spectral-pair detectors on the same register, retaining all 3,600 two-attempt native tapes, sampler and herald failures, sixteen detector addresses and twelve reference columns. The modular identity $\sigma_s = \alpha_{-s}$ holds for the ideal successful marginal $M/\operatorname{Tr} M$ on $\mathcal{B}(\mathbb{C}^{12})$; it does not hold for every retained request block.

This density supplies eleven pairs $Q_j = X_{0j}/\sqrt{2\Delta_j}$, $P_j = Y_{0j}/\sqrt{2\Delta_j}$, $\Delta_j = p_0 - p_j > 0$, with $\langle [Q_j, P_k] \rangle = i\delta_{jk}$, symmetric covariance $\delta_{jk} \coth(g_j/2)/2$, $g_j = \log(p_0/p_j)$, and frequency g_j under H. For chronological factors $U_N = e^{iA_m/\sqrt{N}} \cdots e^{iA_1/\sqrt{N}}$ built from these pair observables on independent identical copies, a parity commuting with the state and anticommuting with every A_k gives

$$\left| z_N^N - e^{-C} \right| \le \frac{D^4/24 + |C|^2/2}{N}, \qquad z_N = \operatorname{Tr}(\rho U_N),\ D = \sum_k \|A_k\|,$$

with $C = \frac{1}{2}\sum_k \langle A_k^2 \rangle + \sum_{k>j} \langle A_k A_j \rangle$. Source pointer reads execute the one-copy factors for three band representatives with all 32 work/pointer addresses retained; a growing copy register is not executed. With physical normalization $q_j = Q_j/\sqrt{\omega_j}$, $\pi_j = \sqrt{\omega_j} P_j$ and $g_j = b\omega_j$, the clock H/b

gives thermal covariance $\coth(b\omega_j/2)\cos(\omega_j t)/(2\omega_j) - i\sin(\omega_j t)/(2\omega_j)$. For m smeared factors with $R = \sum_k R_k$, $R_k^2 = \sum_j (f_{kj}^2/\omega_j + \omega_j h_{kj}^2)$, gaps $g_j, b\omega_j \geq \gamma > 0$, $\vartheta = \gamma/(1+\gamma)$ and $Z = 1/p_0$,

$$|z_N^N - e^{-C}| \leq \frac{R^4}{\vartheta^2 N}\left(\frac{(m+1)Z}{96} + \frac{1}{8}\right),$$

so $Z/N \to 0$ suffices on a fixed positive band; the family $\rho = \mathrm{diag}(2,1,\ldots,1)/(d{+}1)$ with $N = d$ has instead the limit $e^{3(\cos(1/\sqrt{2})-1)} \neq e^{-3/4}$. On the native family $D_l = 4^l L_l$ one has $\lambda_n(D_l) \geq n/81940$, hence $\mathrm{Tr}\, e^{-b\sqrt{D_l}} \leq 1 + 163880/b^2$, and with the lazy filter $p = 630h16^l \leq 1$,

$$Z \leq \frac{1 + 327760/b^2}{m_0}, \qquad P_{\mathrm{success}} \geq \frac{m_0}{240\,4^l};$$

both prepared and target gaps exceed $b/182$ when $(1-m_0)/m_0 \leq b/182$. Preparation error E with success floor $p > 2E$ gives conditional trace error at most $4E/p$, and with one-copy word error ϵ the N-copy amplification is at most $N\zeta/(1-N\zeta)$, $\zeta = 4E/p + \epsilon$. On a fixed band $0 \leq D_0 \leq \Lambda$ with $h\Lambda < 2$, retained positive weights with missing mass $\delta < a \leq e^{-b\sqrt{\Lambda}}$ give

$$\|\log(m_0)I - \log M - b\sqrt{D_0}\| \leq \frac{\delta}{a-\delta} + \frac{bh\Lambda^{3/2}}{4(1+\sqrt{1-h\Lambda/4})}.$$

A constructive growing word family sets $k = 2l + c$ with $4^c \geq b^2 \max(630, \Lambda)$, $R = 2^k$, $h = b^2 4^{-k}$, $a = 4^{-k}$, jump cap 2^{6k}, count cap $6R + 2k$ and Taylor order $4k + 3$; the zero-request branch keeps weight at least $f = 2^{-2R}$. For $d = 240\,4^l$, request cap K and word length P, primitive and load/export error at most $fa/(64dKP)$ gives complete stopped preparation error $E \leq fa/(32d)$ and

$$\|dS_{\mathrm{raw}} - M\| \leq 2dE, \qquad \|\log(dS_{\mathrm{raw}}) - \log M\| \leq \frac{a}{16-a}$$

for the actual successful marginal S_{raw}, without a reducing-band premise. The derived partition bound permits $N = 2^l$ copies with copy, preparation and detector errors tending to zero on a fixed positive band. Growing execution, spectral tails, accurate source services and a zero-energy anchor for the radial product remain obligations, and fixed caps with $N_c h \to 0$ force the bounded-band generator to zero; the uncapped sampler has infinite expected work because the constant sector survives every filter [21].

Selection controls. Two exact controls bound what the committed records select. For $|z| = 1$, $W_z = S[AA^\dagger + z(I - AA^\dagger)]$ has the same first event $W_z A = SA$ but $A^\dagger W_z^2 A = K^2 + z(I - K^2)$; a complete added Ramsey read gives two-step plus probability 1 for $z = 1$ and $113/150$ for $z = -1$. Multiplying each coherent record branch on seam $(0,1)$ with current $+1$ by -1 preserves every committed C and E branch on arbitrary references, yet the twisted embedding gives $\mathrm{Tr}[(A_\eta^\dagger S A_\eta)^3] - \mathrm{Tr}(K^3) = -1/900$. Both real continuations conserve load, respect all sixty proper rotations, fix the uniform coherent state and admit the time reversal $\Theta = S$ conjugation; under full event-permutation covariance, real orthogonal coins fixing the uniform vector are identity or reflection. A sign change on every moving arc preserves proper covariance and the committed instruments but gives $\mathrm{Tr}(D^3) - \mathrm{Tr}(K^3) = -1/90$ for $D = 2\,\mathrm{diag}(K) - K$. Scalar record phases cancel throughout committed-record histories. A source-executed interface test that reads the low unread record bit in the X basis gives complete X-plus probabilities $3/4$ and $1/4$ for one preparation, against $1/2$ for both after commitment, using 5380 C or 5383 E source pulses per attempt [61]. These are restrictions on selection: coherent access is a supplied interface, and the test selects no phase.

3.8 The local event clock

N4 is a reciprocal extension of the binary repair law in the construction's candidate register [51]. Each port carries a private six-level response factor, and one coherent exchange $W_e = \exp[\theta(Y \otimes \Gamma_e + X \otimes \Delta_e)]$, with $X = S_e(N_a - N_b)^2$, $Y = iS_e(N_a - N_b)$ and response generators Γ_e, Δ_e built from the endpoint response matrices, conjugates the binary repair, $K_r = W_e(L_r \otimes I)W_e^\dagger$, with supplied coupling θ. Private preparations change bare load probabilities: at fixed bare load and $\theta = 1/8$ an endpoint occupation probability moves from 0.482020673 to 0.517979327. An interval certificate proves an edge-Gram margin $\mathcal{G} \geq mI$ at $\theta = 1/16$ and $\theta = 1/8$ [51, 76, 2].

Fix an observer port v. A tick is any attempted seam incident to v, including attempts that leave its load unchanged. Under independent uniform seam selection a tick has probability $p = 1/6$ per attempt, and remote attempts continue between ticks. In independent trials the full tracial state is prepared, the binary occupation N_v is measured, the unmodified source runs until one or two ticks have occurred, and N_v is measured again, with no intermediate measurement, reset or postselection. With $f = N_v - 1/6$ and $\|f\|_\tau^2 = 5/36$, write $c_k = \mathbb{E}[(X_0 - \frac{1}{6})(X_k - \frac{1}{6})]/(5/36)$ for the endpoint correlation after k ticks [76, 77]. The numbers of Theorem 3.7(iii) are certified lower bounds; the full N4 two-tick correlation is not evaluated, and preparation, incident-event detection and bare-occupation readout are supplied operations that the detector of Section 3.1 does not implement.

Theorem 3.7 (Memory at observer-local ticks)**.** *Let $\mathcal{A}$ and $\mathcal{B}$ be the averages over the thirty seams of the remote and incident seam channels, so that the first-hit channel is $\mathcal{K} = \mathcal{B}(I - \mathcal{A})^{-1}$. Then:*

(i) $c_k = \langle f, (\mathcal{K}^*)^k f\rangle / \|f\|^2$, *and* $c_1 = 1 - 6\alpha_{\mathrm{occ}}/11$ *for the occupation compression factor* α_{occ};

(ii) the statistic

$$b_{\mathrm{tick}} = c_2 - c_1^2 - \frac{r^2(209r + 180)}{1331r^2 + 3300r + 1800}, \qquad r = 1 - c_1,$$

vanishes for the native law and for the one-term exchange controls, which have $c_1 = 5/11$ *and* $c_2 = 93/407$;

(iii) for both N4 completions, built on C *and on* E, $b_{\mathrm{tick}} \geq 3m/50$, *and the certified lower bounds are*

$$b_{\mathrm{tick}} \geq \frac{139996257}{12500000000000} \approx 0.00001119970056 \quad (\theta = 1/16),$$
$$b_{\mathrm{tick}} \geq \frac{11335290321}{50000000000000} \approx 0.00022670580642 \quad (\theta = 1/8);$$

(iv) $c_{2,\mathsf{C}} \geq c_{2,\mathsf{E}}$ *and* $b_{\mathrm{tick},\mathsf{C}} \geq b_{\mathrm{tick},\mathsf{E}}$.

Proof sketch. Each seam channel is a positive semidefinite contraction of Hilbert-Schmidt space, with spectrum in $\{0, 1/2, 1\}$ for C and $\{0, 1\}$ for E; N4 conjugation preserves this. Hence $0 \leq \mathcal{A} \leq (1-p)I$, the first-hit series converges, and the stopped maps are unital and trace preserving. Remote seams fix f, which gives (i). The closed comparator with seam mean maps $I - \frac{\alpha_{\mathrm{occ}}}{2} w_e w_e^\top$ yields the correction term in (ii) after substituting $\alpha_{\mathrm{occ}} = 11(1-c_1)/6$. With the Schur complement S of $I - \mathcal{A}$ on the centered occupations, $S \leq I$ and $b_{\mathrm{tick}} = \langle z, S^{-1}z\rangle/(p\|f\|^2) \geq \|z\|^2/(p\|f\|^2)$. The hidden defect is $z = -\frac{1}{60}\sum_e t_e S_e$ with five incident coefficients ± 1, so $\|z\|^2 \geq 5m/3600$ and $b_{\mathrm{tick}} \geq 3m/50$. On each seam $\mathsf{C} - \mathsf{E}$ is positive semidefinite, which gives (iv) [76]. □

The correction used for global attempts, $(1 - c_1)^2/10$, assigns $-16/2035$ to the native null at local ticks; the correction of Theorem 3.7(ii) is the one adapted to this stopping instrument. The

receipt and eight independent checks bind the result, including a comparison with the full 66-state native stopping kernel and a complete dimension-six quantum model of the resolvent and Schur identities [76, 102].

4 Observers, clocks and screens on one process

This section completes layer three on one native process. Its components run the native C or E instrument itself, with the uniform independent seam schedule and fair placement coin as declared scheduling inputs. Their time parameters are attempt counts and local event counts. Each result states its preparation, observer access and services; none identifies a count with laboratory duration [15].

4.1 Two overlapping observers

On the 66 binary total-two configurations, prepare the uniform state and disclose the bits at ports $(0, 1, 5)$; all eight labels stay in the output menu. Observer A accesses $(0, 1)$ and observer B accesses $(0, 5)$, so the regions overlap at port 0. Each receives its initial bits and every incident local current, including zero, and ticks when an attempted seam meets its region. The protocol requests two incidents on the union with at most two native attempts per interval; after the first it applies one native repair on $(0, 1)$ if the reconstructed n_0 is one and on $(0, 5)$ otherwise, and that repair ticks both observers.

Proposition 4.1 (Exact overlapping-observer correspondence)**.** *Normalize the native remote and incident branch counts by* 60*, write $P = A + B$ with $B = \sum_m B_m$ over incident marks m, and put $D = I - A$, $p = 2/5$ and $G = D/p$. With $K_m = B_m D^{-1}$ and $F_m = D^{-1} B_m$,*

(i) $K_m = G F_m G^{-1}$ and $\mathbf{1}^{\mathsf{T}} G = \mathbf{1}^{\mathsf{T}}$; remote projectors, the selective feedback maps and the disclosed uniform preparation commute with or are fixed by G;

(ii) for attempt cap h the marked and timeout maps are $K_{m,h} = B_m(I + A + \cdots + A^{h-1})$ and $Q_h = A^h$, their effective conjugates are $F_{m,h} = G^{-1} K_{m,h} G$ and A^h, and the terminal decoder G returns the complete unnormalized source state on every output word of every finite adaptive protocol in this commuting intervention family;

(iii) the direct endpoint execution and the effective recursion agree exactly on all 5,768 *output words,* 689 *of positive probability, including all terminal populations, and the timeout probability is* $369/625$*; and*

(iv) the unread completed transfer $F = D^{-1} B$ is self-adjoint in the metric G with $1/20 \le F \le 1$, so $H = -\log F$ is an effective Euclidean generator per completed union incident tick.

Proof sketch. Remote repairs leave the union fields fixed and every remote seam is disjoint from the feedback ports, so the conjugation by G cancels between successive reads and allowed feedback maps; induction over the protocol gives (ii). Every native seam count equals $I + \mathrm{SWAP}$, which makes $GF = B/p$ symmetric positive semidefinite, and the native triangle $(0, 1, 5)$ contributes at least $3I/60$ to B [16]. □

At every positive-weight prefix each observer's decoder, which sees only its local initial pair and incident increments, returns fields that obey native continuity, and the two port-zero values agree. The transformed finite-cap maps preserve the transported cone $G^{-1}\mathbb{R}^{66}_+$; they are not entrywise-positive stochastic maps, and the comparison is of decoded full states. The quantum extension

checks the native remote resolvent on all 4,356 matrix units with coherent preparations, untouched references and complete local clock records, and an ordinary completely positive effective instrument for local outputs is a positive mixture of native repairs; its one-particle coherent experiment retains all 81 outputs [17]. The compiled clock history replaces the ideal preparation, intervention and pointer of that experiment by words of the six-level source compiler. Exact rational propagation of the binary source arrays keeps all 512 code addresses and 1,656 clock, timeout and complement-abort outputs within 10^{-11} of the ideal history, at most 51 pulses (54 with the compiled Bell preparation), two code services per pulse and five native repair attempts [18]. The native instrument, the scheduler and the code services are specified inputs.

4.2 Joint and active event clocks

A local event clock ticks on actual occupation changes at an observer port, or on attempted repairs incident to a region, while remote repairs continue; its stopped instrument sums the unresolved wait and retains finite timeouts and conditional quantum outputs. Two such clocks on ports 0 and 72 of joined carriers read one continuing history with an observer-chosen field algebra. Exact elimination of the hidden factor gives a record-conditioned quantum memory recurrence valid for arbitrary untouched references. At completed events the experiment admits a four-mark measure-and-prepare description, while unresolved waits retain coherence that the completed instrument erases. C and E share the completed clock law and predict field-plus probabilities 35/54 and 14/27 after three native attempts. Without feedback each two-tick cycle of clock A carries B-count mean two and variance 16; instrument-valued clock conversion preserves that count law, and direct native elimination and elimination through the joint event instrument agree, including B-triggered field feedback [19].

An active event clock lets each native event incident to port 0, including a zero-current one, trigger one fixed compiled modular word $U(1) = \exp(iH)$ generated by the native three-state attempt transfer; the carrier pauses during probe services, which are charged separately. For caps $c \in \{1, 2\}$ the execution retains all 3,661 joint output labels, the first timeout has probability $(5/6)^c$ and both ticks occur with probability $[1 - (5/6)^c]^2$. Closed native histories derive a faithful thermal reference for that generator, and passive repair itself cannot act as a nontrivial modular automorphism on an invariant observer algebra [20]. On the complete movement-bit tape $B_t = |j_t|$, including zero currents, four consecutive covariances $c_n = \mathbb{E}[B_t B_{t+n+1}] - p^2$ invert to the two transfer roots $\lambda_\pm = (25 \pm \sqrt{3})/30$ with $p = 5/33$, and the same roots predict the coherent quadratures $q_X = (1 + \cos(sE_j))/2$ and $q_Y = (1 + \sin(sE_j))/2$, $E_\pm = -\log \lambda_\pm$, with no frequency fit. The one-exponential approximation matching c_0, c_1 misses c_2 by exactly $1/2673000$. The inference does not distinguish C from E, whose classical tapes agree [21].

4.3 Identification of the record law

Nine source-executed scalar measurements identify all nine real coordinates (k, g, h, m, n) of the record-law table of Theorem 2.3, and with them the coarse definite-current instrument on all 78 total-two states with arbitrary spectator and reference coherences. The execution comprises 36 compiled probes and ten complete two-event histories, five with untouched quantum references, run through the canonical observer runtime with serialized checkpoint validation and continuation. An admissible collision law agrees with E on the whole binary shell, and this experiment separates it. A reference-uniform bound carries measurement uncertainty through the finite feedback histories, so narrowing the admissible probability tubes narrows the complete-history bound [22]. The probes are synthetic; they identify an instrument within the classified family and select no law for nature.

4.4 Federations and matched screens

Federations join 20, 80 or 320 identical twelve-port carriers through the declared seam wiring, and native Kraus maps preserve signed-difference operator blocks with individual outcomes and reference correlations. The endpoint instruments fix the local Liouville action $H_{\mathrm{native}} = \sum_e (I - \Phi_e)$, with $h_e = P_1 + \frac{1}{2}P_{1/2}$ for C and $h_e = P_1$ for E on orthogonal projectors; the discrete uniform transfer is $I - H_{\mathrm{native}}/M$ for M seams. For an observed region R, maximum degree Δ and retained radius r, truncating distant ports changes the complete local record, quantum and reference output by at most $\min(1, |R|(\Delta s)^{r+1}/(r+1)!)$ at auxiliary per-seam Poisson parameter s [23]. Actual source neighborhoods are identified at levels 1/3, 2/4 and 3/5. On the 15,360- and 245,760-port screens matched radius-nine initial restrictions give complete root-observer output distance below 0.000382 at $s = 1/4$; after four local incident ticks, with no auxiliary time coordinate, the pair allowance is below 6.656×10^{-9}, and a density-1/3 preparation keeps a certified two-time field covariance above 0.15332581621. Full-screen executions retain the actual seam scheduler, current records, feedback repair costs, timeouts and serialized continuation. Conditional on a conserved total, native connected exchange derives the unique diagonal equilibrium, so the screens at totals 5,120 and 81,920 keep density 1/3, the feedback reversal of the connected four-field statistic and their finite-number correlations without product preparation [24].

A coherent observer runs on the same matched screens. A logical qubit on ports 0 and 1, possibly entangled with an untouched reference, passes through two native incident ticks and a local phase Z_a after first mark 1, while every other port is occupied with probability 1/3. A degree-two operator dual evaluates the common 222-port ball with 561 seams and all 21 capped record labels, and each full-screen completed output lies within 7.779×10^{-10} of it, uniformly over references. C and E give identical classical records in this protocol and different reference coherences [25]. The native continuum operator of the refining screens and its density, wave and response limits are stated in Section 7.

5 Field detectors on supplied actions

Layer four begins with detectors that couple native records to fields whose actions are supplied. They provide complete-output and refinement machinery that a source-derived field description must reuse; they derive no field.

5.1 The scalar detector and its refinement transfer

The scalar detector joins the native observer to the free-scalar continuum family of the theorem library [66, 67]. The scalar action, quantization, vacuum, preparation, Born readout and model clock are supplied. The field is the massive scalar on the golden tensor grids $q \in \{13, 34, 89, 233\}$, with Dirichlet boundary and mass one; after mass whitening its one-particle operator is $\mathsf{A} = I + B \otimes I \otimes I + I \otimes B \otimes I + I \otimes I \otimes B$, with $\Omega = \sqrt{\mathsf{A}}$ and $H = d\Gamma(\Omega)$. A preparation profile f and a detector profile g lie in compact left and right slabs.

One trial runs as follows. (1) Prepare one of the 66 binary configurations uniformly, repair $(0, 1)$ and retain (c_1, j). (2) Prepare the field vacuum and apply the momentum displacement of amplitude $\gamma j f$, $\gamma = 4$. (3) Evolve to model time t and read the centered binary Weyl instrument

$$K_y = \tfrac{1}{2}[e^{-iX/2} - iy\, e^{iX/2}], \qquad X = a^\dagger(v) + a(v), \qquad v = \Omega^{-1/2} g/\sqrt{2},$$

with $K_y^\dagger K_y = \frac{1}{2}[I + y \sin X]$. (4) Commit y, repair $(0, 5)$ for $y = +1$ and $(0, 1)$ for $y = -1$, and halt. Model times lie in $[19/20, 1]$, with primary setting 39/40, and carry no laboratory calibration.

Proposition 5.1 (Scalar feedback and refinement transfer)**.** *For the process above at model time t, let $r(t) = \Pr(y = +1 \mid j = 1) - \frac{1}{2}$. Then:*

(i) the branch operators $M_{(r,y,s)} = (L_s L_r) \otimes (K_y e^{-iHt} V_j)$, with V_j the controlled displacement, form one CPTP instrument on arbitrary joint source and field inputs, and each branch retains its conditional field state as a superposition of two coherent components with their relative phase;

(ii) $p(y \mid j) = \frac{1}{2} + yj\, r(t)$ with $r(t) = \frac{1}{2} e^{-\|v\|^2/2} \sin \mu(t)$ and $\mu(t) = \gamma \langle g, \Omega^{-1} \sin(\Omega t) f \rangle$;

(iii) over the 528 histories, $\mathbb{E}[j_2] = 5r/132$, $\mathbb{E}[N_{0,\mathrm{final}}] = \frac{1}{6} - 5r/132$, $\mathbb{E}[yj] = 10r/33$ and $\Pr(y = \pm 1) = \frac{1}{2}$; disconnecting the preparation or replacing y by an independent fair bit gives $\mathbb{E}[j_2] = 0$ with the same seam frequencies, and reversing the kick reverses the sign;

(iv) if $|r_q - r_\infty| \le \varepsilon_q$ on the window, then the complete labelled histories satisfy the total-variation distance $\mathrm{TV} = \frac{5}{33}|r_q - r_\infty| \le 5\varepsilon_q/33$, and $|\Delta \mathbb{E}[j_2]| \le 5\varepsilon_q/132$.

Proof sketch. The Weyl operators $e^{\pm iX/2}$ are displacements by $\pm iv/2$, which gives the branch states and (ii). Conditioned on the initial configuration and first coin, the two detector laws differ in total variation by $|j|\,|r_q - r_\infty|$, and the subsequent native update is common [67]. □

At $q = 233$ the parent certifies, uniformly for $19/20 \le t \le 1$,

$$\varepsilon \le \frac{12513633763}{250000000000}, \qquad -\frac{111322580247}{500000000000} \le r \le -\frac{34438518047}{250000000000},$$

so the finite-grid mean second current satisfies

$$-\frac{37107526749}{4400000000000} \le \mathbb{E}[j_2] \le -\frac{34438518047}{6600000000000},$$

negative throughout the window, with the continuum mean within $\frac{5}{132}\varepsilon \approx 0.001896$ [103, 68]. At $t = 39/40$ the plus probabilities for $j = +1$ are about 0.334515, 0.319421, 0.317093 and 0.316771 on $q = 13, 34, 89, 233$, and the finest-grid prediction is $\mathbb{E}[j_2] \approx -0.00694050$. A seeded execution with 200,000 fresh trials per grid tests Born outcomes against predictions computed before sampling, with reversal, donor and fixed-seam controls; at $q = 233$ all 12,487,168 field modes contribute. An independent diagonalization gives largest relative frequency errors 2.137%, 0.2875% and 0.04116% at $q = 13$, 34 and 89 against the supplied dispersion $\omega^2 = 1 + \pi^2(n_x^2 + n_y^2 + n_z^2)$ [95]. A local kick-drift-kick extension through the six-neighbour stencil runs at $q = 13$ and 34 and transfers the combined spatial and temporal bound through the same factors 5/33 and 5/132; at $q = 233$ the temporal term is at most 0.000008071512 and the combined probability error at most 0.050062606564 [66].

5.2 The retained two-read continuation

The continued observer keeps the disturbed field after its first read [69]. The field is read at model times 19/20 and 1; the first bit y_1 selects the second native seam, the conditional field evolves to time 1, the second bit y_2 selects the third seam, and the observer halts with five records readable. This continuation uses exact continuous spectral evolution and carries its own error bounds.

Proposition 5.2 (Two reads of one field)**.** *In the continued protocol:*

(i) all 2,112 labelled histories are retained; the record-prefix counts at the six phases are $1, 4, 8, 24, 48, 124$, so the 209 reached prefixes have an eight-bit encoding and the 124 terminal words need seven bits;

(ii) with the Gram kernel G_{ab} of the two detector vectors, $V = G_{11} = G_{22}$, $C = \operatorname{Re} G_{12}$, $c = 2 \operatorname{Im} G_{12}$ (so that $[X_1, X_2] = ic$), $S_a(j) = e^{-V/2} \sin(j\mu_a)$, $D = \cos(c/2)$ and $R_j = \frac{1}{2}[e^{-(V-C)} \cos(j(\mu_1 - \mu_2)) - e^{-(V+C)} \cos(j(\mu_1 + \mu_2))]$, the joint law is

$$p_j(y_1, y_2) = \tfrac{1}{4}[1 + y_1 S_1(j) + y_2 D\, S_2(j) + y_1 y_2 R_j];$$

(iii) on the positive-current preparation, performing the first read changes the second plus probability by $(-r(t_2))[1 - \cos(c/2)]$, certified greater than 0.000022281162 at $q = 233$ and greater than 0.0000252366 in the declared compact continuum; a commuting Gaussian control with the same means and real covariance has $D = 1$ and no such disturbance; the zero-current vacuum correlation exceeds 0.045619388905 at $q = 233$;

(iv) the total variation distance of the complete labelled history from the declared continuum is at most 1, 0.451806, 0.052189606637 and 0.025709913643 on $q = 13, 34, 89, 233$, including the zero-current histories.

Proof sketch. A state is stored as $\sum_n c_n D(i \sum_a h_{na} v_a)|\alpha\rangle$ with exact half-integer shifts, its Gram matrix is explicit in G, and both terms of each Kraus operator are applied before the branch norm is taken. For four atoms with Walsh-moment differences a, b, d, $\mathrm{TV} = \frac{1}{2} \max(|a|, |b|, |d|, \frac{1}{2}(|a| + |b| + |d|))$, bounded in directed rational arithmetic [69]. □

At $q = 233$ the execution gives $\mathbb{E}[j_3] \approx -0.00130126$ and $\mathbb{E}[y_1 y_2 \mid j = 0] \approx 0.0514838$; deleting the first field record from the forecast raises its Brier risk by about 0.00064677. The 50,000-trial runs per grid do not resolve the small disturbance statistically.

5.3 The source-compiled two-site apparatus

The two-site apparatus executes preparation, evolution and pointer as emitted twelve-port control words [70, 71]. Its action lies outside the golden refinement family, so its occupation limit carries no spatial refinement. The Hamiltonian is

$$H = \tfrac{1}{2}(P_0^2 + P_1^2 + Q_0^2 + Q_1^2) + \tfrac{\kappa}{2}(Q_0 - Q_1)^2, \qquad \kappa = \tfrac{9}{32},$$

with normal-mode frequencies 1 and 5/4 and $d \in \{4, 8, 16\}$ number states per site. The executed Hamiltonian uses the compressed squares $Q_d^2 + \frac{d}{2}|d-1\rangle\langle d-1|$ and $P_d^2 + \frac{d}{2}|d-1\rangle\langle d-1|$, so that $H_d = \Pi_d H \Pi_d$. The prepared state is the normalized projection of the coupled vacuum onto total occupation at most $d-2$; at $d = 16$ its binary amplitude tree costs 5,379 native pulses. The current impulse is $e^{ijQ_{0,d}}$, 64 or 128 symmetric split steps reach $t = 1$, and the pointer coupled by $e^{-iZQ_{1,d}/2}$ and read in the Y basis gives $\Pr(y = +1 \mid j) = \frac{1}{2} + j r_d$ and $\mathbb{E}[j_2] = 5 r_d / 132$.

Theorem 5.3 (Complete ideal error budget of the two-site instrument)**.** *For the declared two-site action, model time $0 \le t \le 1$ and native current $j \in \{-1, 0, 1\}$:*

(i) with A the onsite and B the bond part of H_d and $S(h) = e^{-ihA/2} e^{-ihB} e^{-ihA/2}$,

$$\|S(t/n)^n - e^{-it(A+B)}\| \le \frac{t^3}{n^2}\Big(\frac{\|[A,[A,B]]\|}{24} + \frac{\|[B,[A,B]]\|}{12}\Big),$$

uniformly over inputs and references;

(ii) the prepared state differs from the infinite coupled vacuum in trace distance by less than 0.007576, 0.000079899 and 1.0324×10^{-8} at $d = 4, 8, 16$;

(iii) the trace distance between the complete ideal finite instrument, including its classical outcome and unnormalized terminal field branches, and the infinite two-oscillator instrument is below 0.799447, 0.077171 *and* 0.000196 *at* $d = 4, 8, 16$, *and the bound tends to zero as* d *grows;*

(iv) at $d = 16$ *and* $n = 128$ *the combined occupation and splitting error is below* 0.001464436, *and the ideal source-word instrument satisfies* $0.514957066204 \leq \Pr(y = +1 \mid j = 1) \leq 0.517885937293$, *an interval that excludes* $1/2$.

Proof sketch. Item (i) is a defect bound for the symmetric product formula [125] with nested commutators bounded by exact rational Frobenius norms. The infinite state is a displaced squeezed vacuum throughout; the Gaussian generating function bounds the number tails along impulse, evolution and both pointer arms, and projected Duhamel comparison with the orthogonal leakage gives (iii) without dividing by a Born probability. Item (iv) adds a rational enclosure of the infinite reference probability [71, 104]. □

The executed plus probabilities for a positive unit current, with their rational splitting bounds, are 0.516680958 (0.000103957) and 0.516679563 (0.000025990) at $d = 4$ with 64 and 128 steps, 0.516424174 (0.000693013) and 0.516422803 (0.000173254) at $d = 8$, and 0.516423330 (0.005076875) and 0.516421959 (0.001269219) at $d = 16$. The infinite reference has detector variance $9/20$ and mean $\mu(t) = \frac{1}{2}(\sin t - \frac{4}{5}\sin(5t/4))$, so $\Pr(y = +1 \mid j) = \frac{1}{2}(1 + e^{-9/40}\sin(j\mu(t)))$, equal to 0.516421501748 at $t = 1$ and $j = 1$ [70]. The $d = 16$, 128-step protocol uses nine work qubits, 18,612,125 native pulses and 37,224,250 proper-code load and export events, and retained 200,000-trial runs record the reversal, disconnected and donor controls.

6 The common interacting source

6.1 Construction

The common source places quantum electric flux, a charged fermion pair and a neutral scalar on one evolving work register. The sharp protocol measures it twice and feeds its records into the register and the declared native integer primitive [78, 79]; its four-read smooth extension joins compiled native quantum channels and encoded recovery (Section 6.5). The action, preparation, coupling and model-coordinate schedule are construction inputs.

Order four fermion modes as L_0, R_0, L_1, R_1. The two channels carry charges $q_0 = +1$ and $q_1 = -1$ and one fermion each; $b_j = 0$ means left and $b_j = 1$ right occupation of channel j. The electric link has basis $|\ell\rangle$, $\ell \in \mathbb{Z}$, with $\mathcal{E}|\ell\rangle = \ell|\ell\rangle$ and $U|\ell\rangle = |\ell + 1\rangle$. The isometry

$$W|b_0, b_1\rangle = |\text{modes } b_0 \text{ and } 2 + b_1 \text{ occupied}\rangle \otimes |-b_0 + b_1\rangle$$

maps onto four orthonormal occupation and flux vectors.

Proposition 6.1 (Flux-dressed matter code)**.** *On the image of* W*:*

(i) both operator Gauss constraints hold, $G_L W = G_R W = 0$ *with* $G_L = \mathcal{E} - Q_L$ *and* $G_R = -\mathcal{E} - Q_R$, *the supported flux set is* $\{-1, 0, 1\}$ *inside the infinite link, and fermion parity is even;*

(ii) the dressed hopping terms $h_j = c_{Lj}^\dagger U^{q_j} c_{Rj} + c_{Rj}^\dagger U^{-q_j} c_{Lj}$ *preserve both ambient Gauss values and the code, with* $W^*(h_0 + h_1)W = X_0 + X_1 = H_{\text{hop}}$ *and* $W^*\mathcal{E}^2 W = \frac{1}{2}(I - Z_0 Z_1)$, *and* $\|[H_{\text{hop}}, \mathcal{E}^2]\| = 2$*;*

(iii) with $\theta = \operatorname{atan2}(4,3)$, $u = (3I - 4iX)/5$, $U_\theta = u \otimes u = e^{-i\theta H_{\mathrm{hop}}}$ *and* $V_\theta = e^{-i\theta\mathcal{E}^2} = \operatorname{diag}(1, z, z, 1)$, $z = (3 - 4i)/5$, *the echo* $U_\theta^* V_\theta U_\theta$ *applied to* $|00\rangle$ *gives right occupation of channel* 0 *with probability exactly* 576/3125, *and without* V_θ *the probability is zero.*

Proof sketch. On every image column $Q_L = -b_0 + b_1 = \mathcal{E}$ and $Q_R = -\mathcal{E}$. The four-mode algebra of canonical anticommutation relations (CAR) is checked on all 16 occupation states with independent Jordan-Wigner signs, and all entries of the echo unitaries are Gaussian rationals [79]. □

The echo runs on the source controls with 22 native pulses and 44 coherent service events. Two neutral scalar sites join the same register, and with onsite quadratics compressed from the infinite oscillator the declared dimensionless action is

$$\begin{aligned} H &= H_{\mathrm{hop}} + \mathcal{E}^2 + H_{\mathrm{scalar}} + H_{\mathrm{int}}, \\ H_{\mathrm{scalar}} &= \sum_{j=0,1} \tfrac{1}{2}[P_j^2 + (1+\kappa)Q_j^2] - \kappa Q_0 Q_1, \qquad H_{\mathrm{int}} = g\sqrt{2}\, Q_0(n_{L0} - n_{R0}), \end{aligned} \tag{10}$$

with $\kappa = 9/32$ and $g = 1/4$. The interaction commutes with both Gauss generators and with fermion parity and couples the sectors, $[H_{\mathrm{int}}, H_{\mathrm{hop}}] = 2ig\sqrt{2}\, Q_0 Y_0$. The experiment proceeds in four steps. (1) Native repair on $(0,1)$ records a current j, which controls the gauge pulse $V_j = e^{-ij\theta X_0}$. (2) The register evolves for interval 1 (U_1), and a source pointer commits the right occupation a of the first charged species (P_a). (3) Only a selects the hopping pulse $F_1 = (3I - 4iX_0)/5$, with $F_0 = I$, and the register evolves for interval 1/2 (U_2). (4) A second pointer commits the sign b of Q_0 (R_b), and only b selects the second native seam, $(0,5)$ for $b = 1$ and $(0,1)$ otherwise.

Proposition 6.2 (One joint instrument with retained histories)**.** *The operators* $K_{(a,b)} = R_b U_2 F_a P_a U_1$ *satisfy* $\sum_{a,b} K_{(a,b)}^\dagger K_{(a,b)} = I$, *and* $M_{(r,a,b,s)} = (L_s^{e(b)} L_r^{(0,1)}) \otimes (K_{(a,b)} V_j)$ *form one CPTP instrument on arbitrary joint native and quantum inputs. Under the uniform binary preparation and the declared quantum vector* ψ, *the* $66 \cdot 2 \cdot 4 \cdot 2 = 1{,}056$ *histories have probabilities* $p(a,b \mid j)/264$ *with* $p(a,b \mid j) = \|K_{(a,b)} V_j \psi\|^2$, *each bound to its terminal density* $|v_{j,a,b}\rangle\langle v_{j,a,b}|/264$, *and the first native record keeps the masses* 14/33, 14/33, 5/66, 5/66. *For a complete record-controlled instrument* $B_y = \sum_r |r\rangle\langle r| \otimes K_{(r,y)}$ *with channel* $\Lambda(X) = \sum_y B_y X B_y^\dagger$, *every old record projector* Q_r *satisfies* $\Lambda^*(Q_r) = Q_r$, *and the block readback* $E_{\mathrm{block}}(X) = \sum_r Q_r X Q_r$ *and the public readback* $E_{\mathrm{public}}(X) = \sum_r \operatorname{Tr}(Q_r X Q_r) Q_r / \dim W$ *are distinct conditional expectations.*

Proof sketch. Summing $M^\dagger M$ over s, then (a,b), then r gives the identity. Each B_y commutes with Q_r, so $\Lambda^*(Q_r) = Q_r \sum_y B_y^\dagger B_y = Q_r$; every reported probability is checked against the norm of the retained unnormalized vector [79, 105]. □

At 512 steps per interval the emitted words use 4 working qubits, 176,182 native pulses and 352,364 coherent load and export events at $d = 2$, and 6 qubits, 1,879,628 pulses and 3,759,256 events at $d = 4$. Disconnected, feedback-disabled, fixed-seam and independent-bit controls isolate the causal effect of the records [78].

6.2 The feedback contrast at finite and infinite occupation

Theorem 6.3 (Finite feedback contrast)**.** *For the common source with intervals* 1 *and* 1/2*:*

(i) every palindromic product $F(h)$ *of the component exponentials whose generators sum to* H *satisfies* $\|F(t/n)^n - e^{-itH}\| \le S^3|t|^3/(3n^2)$ *with* $S \le 3 + (1+2\kappa)(2d-1) + |g|\sqrt{4d-2}$, *on all inputs and references; at* $n = 512$ *and* $t = 1$ *the half-diamond bounds are below* 0.000728 *for* $d = 2$ *and* 0.004184 *for* $d = 4$*;*

(ii) at $d = 2$ and zero native preparation current, exact rational Taylor arithmetic of degree 64 with $\|H\| \le 109/16$ encloses the unconditional probability of a positive scalar coordinate and the feedback contrast for the continuous finite Hamiltonian:

$$\begin{aligned} &\textit{feedback enabled:} && [0.370697726608684,\ 0.370697726608685],\\ &\textit{feedback disabled:} && [0.402359206973491,\ 0.402359206973492],\\ &\textit{contrast:} && [-0.031661480364808,\ -0.031661480364807]; \end{aligned}$$

(iii) for the ideal source word at 512 steps per interval the contrast lies in

$$[-0.033297300667839,\ -0.030025660061775],$$

which excludes zero.

Proof sketch. The multinomial Leibniz expansion bounds the third derivative of a product of unitaries by S^3, palindromic symmetry gives second-order agreement, and unitary telescoping gives (i). For (ii) the integral Taylor remainder gives $\|T_m(t) - e^{-itH}\| \le (St)^{m+1}/(m+1)!$, and at $m = 64$ the probability radius is below 7.091×10^{-37}; measurements and feedback contract trace distance, which gives (iii) [79]. □

With $g = 0$ the gauge-only feedback cannot change the scalar marginal, and a coarse splitting or inadequate Taylor degree yields an interval containing zero, which the evaluator reports as no resolved signal. The finite pointer measures the sign projector of the truncated coordinate, which differs from the infinite half-line effect; at $d = 2$ the two differ in norm by $\frac{1}{2} - 1/\sqrt{2\pi} \approx 0.10105772$ [83], so the infinite-occupation statement needs the infinite detector.

Theorem 6.4 (Infinite-occupation feedback contrast)**.** *Take the action (10) on two infinite oscillator Fock spaces, the infinite free coupled vacuum, proportional to* $\exp[-(a_0^\dagger - a_1^\dagger)^2/36]|0,0\rangle$ *with squared normalization* $4\sqrt{5}/9$*, the blank gauge code, zero native preparation current, the matter record and feedback of Section 6.1, and the half-line readout of* Q_0*. Then:*

(i) the unconditional positive-coordinate probability with feedback minus that without lies in $[-0.024045670143,\ -0.024043450955]$*, and without the interaction the contrast is exactly zero;*

(ii) both matter records and both half-line outcomes retain their quantum states, including the post-read infinite occupation tail;

(iii) along $d = 2^k$*, with exact palindromic factors and* $n = d^2$ *steps per interval, the complete finite sign instruments converge in trace distance to this infinite instrument, terminal states included, and the finite-family contrast is strictly negative for all sufficiently large* d*.*

Proof sketch. The linear force is relatively bounded by the coupled oscillator Hamiltonian, so H is self-adjoint and finite Hermite polynomials lie in every $\mathcal{D}(H^k)$. Each of 24 steps of length $1/16$ applies a degree-16 Taylor polynomial of H, charges its remainder exactly, truncates above total occupation 16 and rounds to a 2^{-112} grid, charging both differences. Half-line probabilities are $r + s/\sqrt{2\pi}$ with exact rationals, and π is enclosed by Machin's formula. For (iii), $P_d Q_0 P_d \to Q_0$ in the strong resolvent sense and the spectral measure of Q_0 has no atom at zero [80, 126, 123]. □

The $d = 2$ finite experiment has contrast about -0.03166148; replacing only its detector by the half-line effect gives about -0.02526221, and removing the cutoff gives the interval of Theorem 6.4. The signal survives both limits, which change its size; the interval is specific to zero native preparation current and the declared two-site action [80].

6.3 Occupation refinement and complete-output comparisons

Let P_d project both oscillators onto occupations below d and let $N = n_0 + n_1$ be the total oscillator occupation, a different object from the closure input N.

Theorem 6.5 (Uniform pre-readout occupation bound)**.** *Let the matter start in an arbitrary state entangled with an untouched reference, let the scalar start in the fixed coupled vacuum, and let the finite preparation retain total occupations* $0, 2, \dots, d-2$*. Then:*

(i) $M(s,t) = \langle e^{sN} \rangle$ *satisfies* $\partial_t M \le \frac{11}{16} \sinh(s)\, \partial_s M + \frac{25}{32} \sinh(s)\, M$ *through the matter measurement and feedback;*

(ii) at total model time $T = 3/2$ *the resulting moment bound holds with base* $\varrho = 113/64 \ge 7/4$ *and* $M < 13$*, where* $\varrho \le e^{s(T)}$*;*

(iii) $\|P_d H (1 - P_d)\| \le C_d = \frac{\kappa}{2}\sqrt{d(d+1)} + \frac{3\kappa}{2} d + \frac{1}{4}\sqrt{d}$*, and the retained-history trace distance up to the final scalar read is at most* $\left[(\sqrt{2 v_d} + T C_d \sqrt{\tau_d})^2 + \tau_d \right]^{1/2}$ *with* $\tau_d = M/\varrho^d$ *and* v_d *the omitted preparation probability, an* $O(d\, \varrho^{-d/2})$ *rate;*

(iv) this bound is below 0.011408*,* 0.000002498 *and* 6.175×10^{-14} *at* $d = 32, 64, 128$*.*

Proof sketch. In common and difference modes only the squeezing term and the force change the number; weighted commutator, Cauchy and Young inequalities that hold for matter-entangled vectors give (i), and integration along $\tanh(s(t)/2) = e^{-11t/16} \tanh(s_0/2)$ from the initial value $\sqrt{80/17}$ at $\varrho_0 = 8$ gives (ii). The record dilation and feedback commute with N, and projected Duhamel comparison gives (iii) [81]. □

Theorem 6.6 (Fourth moment and strong convergence of the sharp instrument)**.** *Let* $w(N) = (N+2)^4$ *and let* $M(t) = \langle w(N) \rangle$ *be summed over the unnormalized record branches. Then:*

(i) $|\dot{M}| \le C M$ *with* $C = 8\kappa + 4|g| = 13/4$*, and every finite preparation has* $M(0) \le M_0 = 143748821/8192000$*;*

(ii) the record-dilation vectors immediately before the final read satisfy

$$\|\Psi_d - \Psi\| \le \left[(e_0 + I_T L_d)^2 + \frac{M_0 e^{CT}}{(d+2)^4} \right]^{1/2} = O(1/d),$$

with $I_T = \sqrt{M_0}\, (e^{CT/2} - 1)/(C/2)$*,* $L_d = \kappa/(d+2) + |g|/(d+2)^{3/2}$ *and* e_0 *the preparation error;*

(iii) the sign projectors of $Q_{0,d}$ *converge strongly to those of* Q_0*, and the complete unnormalized classical and quantum history, together with its native continuation, converges in trace norm for all three native currents and both feedback settings.*

Proof sketch. On occupation at most m, $\|H\psi\| \le 4(m+2)\|\psi\|$ and H raises occupation by at most two, so finite-particle vectors are analytic and core convergence gives strong resolvent convergence [124, 126]. The moment coefficients are bounded by 8κ and, through a positive polynomial certificate, $4|g|$; the gauge measurement and rotations preserve $w(N)$ summed over branches [82]. □

Strong convergence supplies no finite-d rate for the sharp readout. The complete-output certificate supplies fixed-state enclosures at $d \in \{2, 32, 64, 128\}$ for the zero-current preparation and the named rectangular occupation codes [83, 106].

Theorem 6.7 (Complete outputs of the sharp detector)**.** *For zero native preparation current, with both matter records and both sign outcomes retained:*

(i) for two complete post-read outputs with rank-one branches, the half trace norm of their difference is $\frac{1}{2}\sum_b \sqrt{(p_b+q_b)^2-4|z_b|^2}$*, summed over earlier records, where* p_b*,* q_b *are the branch weights and* z_b *their overlap;*

(ii) the complete continuous-output distances between the finite and the infinite sharp histories lie in the intervals of Table 6; at $d=2$ *the ideal 512-step source budget widens them to* $[0.454336, 0.455973]$ *with feedback and* $[0.456824, 0.458461]$ *without;*

(iii) every output state supported in the rectangular occupation code of $d=256$ *lies at distance above* 0.122 *from the true output of either prepared history;*

(iv) for the vacuum control, whose two sign probabilities are both $1/2$*, the best approximation by a state supported in the code with* $d=2m$ *has distance between* $\sqrt{3/(8\pi)}\,m^{-1/4}$ *and* $\sqrt{4/(3\pi\sqrt{3})}\,m^{-1/4}$*, hence of order* $d^{-1/4}$*, and distance at most* ε *requires* $d \geq 9/(32\pi^2\varepsilon^4)$*.*

Proof sketch. Item (i) is the trace norm of a difference of rank-one operators. For (iii), the operator exchanging the normalized inside and outside directions of each branch is a norm-one witness, giving distance at least $\sum_{a,b}\sqrt{a_{ab}t_{ab}}-\epsilon_{\mathrm{poly}}$; half-line Hermite matrix elements are rational up to $1/\sqrt{2\pi}$ [115]. For (iv) the omitted vacuum mass per sign branch obeys $1/(4\pi\sqrt{m}) \leq T_{2m} \leq 2/(3\pi\sqrt{3m})$ [83]. □

Table 6: Certified complete continuous-output distances between the finite sharp history and the infinite sharp history, zero native preparation current, rounded outward [78, 106].

Onsite dimension d	Feedback enabled	Feedback disabled
2	[0.455154, 0.455155]	[0.457642, 0.457643]
32	[0.203713, 0.226530]	[0.203935, 0.226752]
64	[0.180917, 0.180924]	[0.181090, 0.181097]
128	[0.152201, 0.152204]	[0.152341, 0.152344]

At $d=64$ the pre-readout error is below 0.000002498, so the sharp detector dominates the final error: accurate sign probabilities do not certify an accurate post-measurement state.

6.4 The smooth Ramsey instrument

A detector variant keeps the preparation, evolution, first record and feedback and replaces the final sharp read by a source Ramsey pointer with work-space Kraus maps

$$M_0 = e^{-i\pi/4}\cos\left(\tfrac{Q_0}{2}-\tfrac{\pi}{4}\right), \qquad M_1 = e^{i\pi/4}\cos\left(\tfrac{Q_0}{2}+\tfrac{\pi}{4}\right),$$

and effects $\frac{1}{2}(I \pm \sin Q_0)$ [84]. Bit zero denotes the plus effect, a detector meaning different from the sharp sign record, so the sharp-sign certificates do not carry over.

Theorem 6.8 (Smooth instrument)**.** *For the declared initial vacuum family, uniformly over the three native current kicks and both feedback settings, with* $T=3/2$ *and the constants of Theorem 6.6:*

(i) the complete recorded instrument, including the pointer interaction and both unnormalized outcomes, lies within trace distance $\min(1, E_d)$ *of its infinite counterpart, where* $A_T = \sqrt{M_0}e^{CT/2}$ *and*

$$E_d = \Big[\Big(e_0 + I_T L_d + \frac{A_T(e-1)}{2(d+2)^{3/2}}\Big)^2 + \frac{M_0 e^{CT+2}}{(d+2)^4}\Big]^{1/2} = O(1/d);$$

(ii) with $n(d) = d^2$ *symmetric steps per interval the combined ideal temporal and occupation error is* $O(1/d)$*;*

(iii) on normalized inputs in the form domain of the infinite Hamiltonian the unconditional increment of the full energy is exactly $1/8$ *in the declared dimensionless convention.*

Proof sketch. Along the pointer parameter the generator $\frac{1}{2}Z_{\mathrm{anc}}Q_0$ obeys $dM/ds \le 2M$, which gives the pointer term. For (iii), $f_0(q) = \cos(q/2 - \pi/4)$ and $f_1(q) = \cos(q/2 + \pi/4)$ satisfy $\sum_b f_b^2 = 1$, $\sum_b f_b f_b' = 0$ and $\sum_b (f_b')^2 = \frac{1}{4}$, so the kinetic term gains exactly $\frac{1}{8}\|\psi\|^2$ [84]. □

The bound is about 0.2063 at $d = 64$, 0.04096 at $d = 256$ and 0.008840 at $d = 1024$, and vacuous at the executed $d = 2$ and $d = 4$ [78]; the energy identity belongs to the infinite oscillator and fails at a finite cutoff [84, 95].

6.5 Continued instruments, compiled native repair and encoded memory

One execution engine carries the sharp and smooth protocols and the extension

$$U(1),\ P \to r_0,\ F(r_0),\ U(1/2),\ M \to r_1,\ U(1/2),\ M \to r_2,\ F(r_2),\ U(1/2),\ P \to r_3,$$

with P the complete gauge instrument, M the smooth instrument and F the gauge feedback, model time $5/2$ and all sixteen unnormalized work/reference branches retained, with no intermediate work preparation [85]. Complete Kraus families preserve every old-prefix effect, and imported checkpoints are validated by prefix reexecution; this supplies neither physical quantum serialization nor cloning. Carrying a moment bound A and an error e from stage to stage, with $L_H = 9/[32(d+2)] + 1/[4(d+2)^{3/2}]$ and $L_S = 1/[2(d+2)^{3/2}]$, each unitary interval of length t adds $A(e^{at} - 1)L_H/a$ $(a = 13/8)$, each smooth read $A(\mathrm{e} - 1)L_S$, and the final tail gives $E_d = [e^2 + A^2/(d+2)^4]^{1/2}$; with $n(d) = d^2$ the complete output distance is $O(1/d)$ for the scalar vacuum family and vacuous at the executed $d = 2, 4$; the classical native attachment enumerates 4,224 joint outcomes at each executed cutoff. A separate state-specific certificate retains all sixteen outputs for the blank-gauge and coupled vacuum preparation and each $j \in \{-1, 0, 1\}$ [86]; the rank-one trace-norm identity, complete-channel contractivity and directed rational evaluation at 512 splits per interval give

$$\max_{j\in\{-1,0,1\}} D_{d,j}^{\text{ideal product},\infty} < \begin{cases} 0.264092, & d = 2, \\ 0.053956, & d = 4. \end{cases}$$

This bound transfers through the native and protected-memory attachments for arbitrary admitted native and reference inputs with this independently prepared work state; it is not an arbitrary work-input bound and excludes floating execution and physical service errors.

Coherent native attachment. Let $L_h^{A,e}$ be the compiled native Kraus operator for law $A \in \{\mathsf{C}, \mathsf{E}\}$, including its fair coin, and $K_r^{j(h)}$ the work branch including the first record's current-controlled kick. Source routing implements seam $(0, 5)$ by exchanging port labels 1 and 5. With

$e(0) = (0, 5)$ and $e(1) = (0, 1)$,

$$M_{h,r,s} = (L_s^{A,e(r_2)} L_h^{A,(0,1)}) \otimes K_r^{j(h)}, \qquad \sum_{h,r,s} M_{h,r,s}^\dagger M_{h,r,s} = I_{78} \otimes I_W,$$

complete on arbitrary entangled native, work and reference inputs [87]. The execution keeps all 512 native amplitudes, padding and abort outcomes, with 295,106 terminal basis outcomes for C and 820,166 for E. For input $(|\{2,3\}\rangle + e^{i\theta}|\{1,2\}\rangle)/\sqrt{2}$ and independent work, the native interference effect gives $p_C(\theta) = B + \frac{1+q}{8}\cos\theta$ and $p_E(\theta) = B$, with $q = \Pr(r_2 = 0 \mid j = 0)$; the C phase contrast is about 0.349153278 at $d = 2$ and 0.351351028 at $d = 4$, against zero for E. Cyclic Gray ordering of the shared nine-qubit controlled-flip macro reduces its source pulse count from 11,782 to 2,044 while preserving the full 512-dimensional operator, including global phase, inverse and padding [201, 87, 88].

Active encoded memory. The code $V|0\rangle = (|\{0,2\}\rangle + |\{1,3\}\rangle)/\sqrt{2}$, $V|1\rangle = (|\{0,3\}\rangle + |\{1,2\}\rangle)/\sqrt{2}$ has the flagged Knill–Laflamme identity $V^\dagger L_h^\dagger L_h V = p_h I_2$, $p_h = k_h/4$, $k_h \in \{0,1,2\}$. A source encoder and record-controlled full-register unitary R_h realize $R_h L_h V = \sqrt{p_h} V$ [88]. Correcting after each native event, with final seam $(2,3)$ for $r_2 = 0$ and $(0,1)$ otherwise,

$$M_{h,r,s}^{\text{corrected}}(V \otimes I_W) = \sqrt{p_h p_s^{e(r_2)}}(V \otimes K_r^{j(h)}),$$

which preserves the logical and reference marginal after summing all work and record outputs, including initially entangled inputs. The passive control without recovery has complete plus/minus output distance $1 - \bar{q}$, with $q_j = \Pr(r_2 = 0 \mid j)$ and $\bar{q} = (q_{-1} + 2q_0 + q_1)/4$, so every later decoder errs by at least $\bar{q}/2$ on one input, while the corrected distance is one; for fixed seams $(0,1), (2,3)$ passive Bell fidelity is $1/2$ and corrected fidelity one. Encoding, blank registers, source precision, storage and record services are supplied. This is active protection of the stated qubit before readout, not arbitrary-carrier recovery, fault tolerance or an autonomous observer.

6.6 Observer readouts and the fixed-reference clock

The readouts below extract observer representations, modular flow and regional structure from the same executed history [89]. The quantum algebra, state pairing and access choices are premises; the Born rule is not derived from native repair.

Proposition 6.9 (Observer representations of the executed history)**.** *Let v_{ab} be the four unnormalized terminal vectors of the history at $d \in \{2,4\}$ and $\rho_{\text{hist}} = \bigoplus_{a,b} |v_{ab}\rangle\langle v_{ab}|$ on $\bigoplus_{a,b} M_{4d^2}$.*

(i) For a positive functional ω the Gelfand-Naimark-Segal (GNS) construction $\langle [A], [B] \rangle = \omega(A^*B)$, $\pi(X)[A] = [XA]$, $\Omega = [I]$ has dimension dr on a block M_d whose density has rank r. The record and work algebra has GNS dimension $4(4d^2)$, that is 64 and 256; the four committed record projectors give a four-dimensional classical GNS space; the site-0 observer that forgets both records has GNS dimension d^2, that is 4 and 16; and the regional maps $J_R[A] = [\text{embed}_R(A)]$ are isometric and intertwine the regional actions.

(ii) For faithful ρ_0 the modular group $\sigma_s(A) = \rho_0^{is} A \rho_0^{-is}$ moves number observables; the commutative record algebra has identity flow; states conditioned on the sharp terminal bit are nonfaithful on the full scalar algebra, with rank at most $d/2$.

(iii) *At $d = 2$, $\rho_0 = \frac{1}{2}(I - \delta X)$ with $\delta = P(b = 0) - P(b = 1)$ and modular frequency $\nu = \log(P(b = 0)/P(b = 1))$. A source-compiled continuation of every terminal branch by $Z(\pi/4)$, $X(s\nu/2)$ and a fresh number pointer preserves the old two-bit record marginal and has readout law $P(1 \mid s) = \frac{1}{2}(m + \delta \sin(\nu s))$, where m is the total branch mass, equal to one for the normalized history.*

(iv) *Scalar regional restrictions are completely positive conditional expectations, normalized partial trace followed by identity embedding; they are unital and idempotent and in general not multiplicative. The centered components of the actual Hamiltonian give an interaction graph that is a path through scalar site 1, scalar site 0 and the shared charged-matter and link sector.*

Proof sketch. For $\rho = V \operatorname{diag}(\lambda) V^*$ of rank r the map $[A] \mapsto AV\sqrt{\operatorname{diag}(\lambda)}$ is the GNS isometry [109]; the modular identities, including the finite Kubo-Martin-Schwinger (KMS) condition, are checked with this sign convention [110], and at $d = 2$ the probe implements the Schrödinger dual of σ_s [89]. □

In the executed history $\nu \approx 0.5292248981159262$ and $\delta \approx 0.2586046272446501$ [107]; ensemble frequencies read a modular interval on a declared monotone window, and full-cycle aliases are indistinguishable. The post-measurement site-0 state taken as its own modular reference fails to reproduce the interacting dynamics: its modular flow preserves it while its density changes under H with derivative norms about 0.2086 at $d = 2$ and 0.6603 at $d = 4$ [95]. A fixed reference on the full work algebra removes this obstruction [90].

Proposition 6.10 (Fixed-reference modular relation)**.** *Let H be the finite interacting Hamiltonian on the work space of dimension $4d^2$, $\beta > 0$ and $\sigma_\beta = e^{-\beta H} / \operatorname{Tr} e^{-\beta H}$. Then*

$$\sigma_\beta^{is} A \, \sigma_\beta^{-is} = e^{-i\beta s H} A e^{i\beta s H} = \alpha_{-\beta s}(A), \qquad \alpha_t(A) = e^{itH} A e^{-itH},$$

so continuous Hamiltonian segments correspond to modular flow at $t = -\beta s$. For the full matrix algebra this identification of the two automorphism groups forces the Gibbs form of the reference. With record blocks $\Sigma_\beta = \bigoplus_r q_r \sigma_\beta$, $q_r > 0$, the weights cancel and every record projector is fixed. The actual state $\omega(t) = e^{-itH} \omega e^{itH}$ can change while σ_β is stationary, and $\operatorname{Tr}(\omega(t)E) = \operatorname{Tr}(\omega \, \sigma_{-t/\beta}(E))$ for every effect E.

Proof sketch. $\log \sigma_\beta = -\beta H - \log Z_\beta \, I$, and equality of the generators requires $[\log \sigma + \beta H, A] = 0$ for every matrix A, whose commutant is scalar [90, 114]. □

The reference, β and full-work access are declared, and measurements and feedback act outside this one-parameter group. The receipt uses $\beta = 1$ and reference weights $(1/10, 2/10, 3/10, 4/10)$, with source continuations at model times 0, 1/2 and 1 appending a gauge-number pointer to every post-sharp branch and retaining all eight outcomes [90].

6.7 History repair of the completed record

Theorem 6.11 (Local repair of the completed history)**.** *Fix $d \in \{2, 4\}$, the native current j and the feedback setting. Let twelve event nodes carry unnormalized work vectors subject to the compatibility equations $x_{\text{root}} = \text{prepared}(j)$, $x_1 = U_1 x_{\text{root}}$, $x_{(a)} = P_a x_1$, $y_{(a)} = F_a x_{(a)}$, $z_{(a)} = U_2 y_{(a)}$ and $w_{(a,b)} = B_b z_{(a)}$, of depths 0 to 5. Then:*

(i) *exactly one assignment satisfies all equations;*

(ii) *the common fixed set of the local updates $x_v \leftarrow A_v x_{\text{parent}(v)}$ equals that compatibility set;*

(iii) *every node-fair schedule reaches the compatible assignment from arbitrary initial vectors after finitely many updates and stays there; five covering rounds suffice, a forward topological pass needs eleven updates, and a reverse order may need 55;*

(iv) *with subtree weights w_v, equal to* 11, 5, 4, 3 *and* 1 *at depths one to five, the potential* $\Phi = \sum_{v \neq \mathrm{root}} w_v \|x_v - A_v x_{\mathrm{parent}(v)}\|$ *decreases by at least the repaired residual at every active ideal repair;*

(v) *every deterministic extraction from the completed assignment, including branch probabilities, terminal densities, GNS representations and modular parameters, is independent of the schedule.*

Proof sketch. Agreement at depth zero forces agreement at each further depth. A node whose ancestors are correct becomes correct permanently at its next selection, and after the kth covering round every node of depth at most k is correct. Updating v zeroes its residual and changes each child's residual by at most $\|r_v\|$, since all maps are contractions [91]. □

The solved object is the complete branching instrument law for given apparatus inputs; its uniqueness selects no realized record pair, and the auxiliary solver is a host computation distinct from a native hardware process.

6.8 A same-action particle corner and regional interfaces

The current finite attachment uses the supplied Weyl/contact CAR action $H = d\Gamma[-(1/8)\sigma \cdot \sin k] + \sum_x n_{x0} n_{x1}$ on its complete two-fermion sector. An isolated bound-pair band has an independently checked position-space wavepacket, number and energy currents, and complete contact/scattering spectral records. Its topology is trivial; it supplies neither a nontrivial native defect invariant nor a physical mass assignment. Its exact rational dispersion enclosure gives a kinetic coordinate mass different from the value inferred from its energy difference and the same speed parameter. All of these tests use this one action.

At total momentum (π, π, π), an on-site singlet d and a zero-contact singlet r have energies one and zero, respectively. They have the same particle number and total momentum. A supplied isometry maps the native configurations $\{1, 2\}$ and $\{7, 2\}$ to (r, d). The native product reference restricts to $(264/7505)\,\mathrm{diag}(25/33, 8/33)$, furnishing an exactly faithful Gibbs reference on this invariant M_2 corner, with $\beta = \log(25/8)$ and Heisenberg time $t = -\beta s$.

The executed native protocol retains an initial coarse corner herald, a balanced probe, every uniformly chosen C/E seam outcome, a second herald, the record-derived clock, an analyzer and the final current/coin energy label. Rejection, leakage, zero slots and export aborts remain outputs. Each coarse herald is a 2,068-pulse source word checked on all 512 original register inputs. Preparation error is charged before the first herald, and the complete average half-trace bound is below 0.018762 without success conditioning. The selected two levels do not prepare or encode the full spatial field. Clock units, the many-site isometry and a full-field native reference remain further obligations.

The same local field action also has signed CAR restrictions retaining all regional number sectors zero, one and two. Their Kraus maps prove complete positivity, trace preservation, untouched-reference compatibility and composition. Number-preserving local observables have true algebra inclusions; compressed odd operators do not inherit a CAR homomorphism. The corresponding ambient conditional expectations satisfy tower laws but are not multiplicative. This instantiates the CP restriction interface, not the older impossible scalar-bottom homomorphic retraction.

Increasing open boxes admit exterior-square inclusions J satisfying $J^* H_{\mathrm{large}} J = H_{\mathrm{small}}$. Boundary hopping makes $H_{\mathrm{large}} J - J H_{\mathrm{small}} \neq 0$, so this equality alone does not intertwine dynamics. The complete two-read feedback histories instead use the existing finite-boundary and product-formula bounds, followed by contractive regional restriction. Every quantum record and discarded local particle sector remains in that comparison. The growing boxes hold spacing fixed and do not reconstruct Lorentzian geometry.

On the same full finite action, local energy continuity and quantum Gibbs/entropy identities supply further finite premises. The regional modular generator fails the proposed affine identification with bare regional energy even after number and number-squared terms are allowed. The full interacting Weyl action also couples every proper momentum pocket to its complement. Consequently neither a regional clock nor a chiral sector follows by merely deleting terms or modes. The source-bound construction is recorded in `model/bound_pair`, `model/particle_clock`, `model/field_regions`, `model/field_stress` and `model/field_tastes`.

The separate continuing native memory. The endpoint-biased 60-code process also continues through every forbidden antipodal flag by a supplied coherent decode/re-encode service. It preserves the unknown logical state and keeps boundary, zero and abort outcomes. Positive endpoint coins leave the exact flagged-code conditions unchanged. Every all-seam correctable sharp-$q = 1$ source qubit, without persistent ancillary storage of its logical state, has both core occupation compressions equal to $I_2/2$, so its core kernel is independent of the history-selected code. For this entire class, the exact absorption potential $h(c) = 1 + \sum_d Q(c,d)h(d)$ has a unique maximum $H \simeq 203.025378$ attempted repairs at core $(0,1)$. Every counted reset raises h by at most H, so telescoping gives $\mathbb{E}N_n \geq \max\{0, (n - h(c_0))/H\}$. Resetting only at the boundary to $(0,1)$ attains asymptotic rate $1/H$. Exact Bellman checks through horizon 64 are recorded separately from this all-horizon asymptotic argument. The compiled family attains the bound; arbitrary exact code choices cannot improve it under the same law and reset contract. Active recovery is supplied after every repair attempt; neither reset counts nor source-pulse counts are physical energy. The complementary fair-coin mobile-memory construction has a different law and a mean edge excursion of 150 attempts. Its finite-horizon theorem is separate from the biased asymptotic result. Neither controlled memory is the interacting field pair or a native topological particle. The complete service and its bounds are retained in `model/biased_reset` and `model/biased_code_universality`. A separately certified variant omits 3,120 identity recoveries among 11,580 positive flags, saving about 48,391 supplied pulses per stationary native attempt. It preserves the admitted logical/reference channel and does not change the reset rate.

Smaller exact controls and a separate clock. At side four and zero total momentum, the contact-generated cyclic subspace has seven dimensions. Its free spectral weights are $(1,3,3,2,3,3,1)/16$ at energies $2c(-\sqrt{3}, -\sqrt{2}, -1, 0, 1, \sqrt{2}, \sqrt{3})$. The resulting rank-one resolvent gives exact rational pole enclosures and intertwines complete contact/complement histories with arbitrary references. Other spatial observations require the full field sector. This control is implemented in `model/pair_cyclic_control`.

Complete Fock dynamics and the thermodynamic limit. The same local action acts on all 2^{2V} occupations of a box with V sites. For E spatial bonds, each of norm at most c, its full-Fock stability bound is $-cEI \leq H \leq (cE + UV)I$, with $E \leq 3V$. The completed two-mode hopping, contact and fermionic-route words apply to all number sectors and arbitrary references. With $B = cE + UV$, a symmetric n-step split has ideal operator error at most $B^3|t|^3/(3n^2)$. The

corresponding complete-history error is summed over segments without conditioning on outcomes. This bound depends on volume; a bound restricted to two particles is not substituted for it.

Removing the onsite contact terms by a local interaction picture leaves even, time-dependent two-site bonds of norm at most c. The commutator-chain estimate for disjoint, initially commuting observables is

$$\|[\tau_t(A_X), B_Y]\| \le \|A_X\|\|B_Y\|\,|X|\,2^{-d(X,Y)}(e^{48c|t|}-1).$$

At least one of the observables is even; odd–odd locality uses the graded CAR relation. For a normalized observable in a radius-r cube and strictly padded radius $R > r$, the comparison to any larger box is bounded by

$$\min\{2,\, 6c|X|(2R+1)^2|t|\,2^{-(R-r)}(e^{48c|t|}-1)\}.$$

The limit gives a strongly continuous automorphism group on the quasi-local CAR algebra at fixed spacing. The contact contribution $[H_U - H_0, a_0^* a_2] = U a_0^* a_2 (n_1 - n_3)$ remains nonzero in that limit. Thus the construction is an interacting lattice field with three spatial coordinates and continuous Hamiltonian model time. Neither an infinite-volume physical state nor a Lorentzian chiral continuum is selected. The executable full-Fock control and rational locality bounds are in `model/full_fock_field`.

A corrected memory branch and a clock instrument on a separate factor compose exactly, including correlated logical/clock/reference inputs. The clock requires its preparation, control, readout and event association; the tensor identity does not establish a stationary product reference, physical duration, dispersion relation or memory mass.

Extension and obstruction for the full band. The same native record-rate estimator drives counted local source words on the full 8,128-state field sector, with all 64 pair momenta and all four terminal occupation-pointer outputs. Its uniform complete average half-diamond bound is below 0.037991 for $|s| \le 1$. This requires a supplied globally normalized field Gibbs reference and a rate-transfer actuator; it is not an inherited native encoding. Every finite independent product of the canonical native reference has integer modular gaps in units $\log(25/8)$, whereas the same zero-momentum pair has $13/12 < E/U < 3/2$ relative to an eta dark state. The inherited intertwiner therefore fails, with generator error greater than $1/24$ in the stated units. A separate native coherent state has an exactly reference-matched exterior through every native record history and a computed finite contrast lifetime, but its coherent core is immobile and its modular-energy excess is zero. These distinctions are implemented in `model/band_clock` and `model/native_localization`.

7 One repair history: geometry, continuum, clocks and fields

This section derives what one native repair history supplies on the carrier and on the refining spherical source. Its statements are dimensionless carrier displacements, attempt counts and event counts; physical position, length scale and propagation speed need their own operational identification.

7.1 Native geometry, causal cones and field relations

Proposition 7.1 (Native geometric readout)**.** *(i) Exact enumeration of all 4,680 state, seam and coin branches gives the one-attempt mean $T = I - L/60$ under uniform seam selection.*

(ii) With Q removing the constant mode and $C_n = QT^{2n}Q$, the normalized response $12C_n/\operatorname{Tr} C_n$ tends to $G = 4P_{\mathrm{slow}}$, where P_{slow} projects onto the rank-three band $5-\sqrt{5}$; all eleven nonconstant modes are present at every finite attempt count.

(iii) *For the readout $x(n) = \sum_p n_p u_p$ with frame directions u_p, a committed current j on the seam (a, b) gives the relative displacement $x(n') - x(n) = j(u_b - u_a)$, computable from the record without hidden loads.*

(iv) *The two-event controller spans two displacement directions. A declared continuation that sends the final scalar bit to $(0, 5)$ or $(0, 7)$ after the first event re-executes all 1,056 second events, and its three displacement directions $(2, 0, 0)$, $(1, 1-\varphi, \varphi)$, $(1, 1-\varphi, -\varphi)$, with $\varphi = (1+\sqrt{5})/2$, have determinant 4.*

Proposition 7.2 (Native causal cones)**.** *For a fixed seam schedule and shared coins, the load at a port v after the schedule is a function of the initial loads and coins in its backward dependency set. An initial difference supported on D influences v only along a time-ordered path, so $\mathrm{dist}(D, v) \leq$ edge crossings on that path $\leq$ elapsed seam events, and at most one edge is crossed per causal layer of disjoint simultaneous events.*

Exhaustive checks cover all 78 states for 27 schedule and coin pairs, and two executed examples show influence at graph distance two arriving only after two properly ordered seams [93]. One edge per causal layer is a native signal speed of at most one seam per ordered repair event; remote repairs can carry a difference to a receiver before its first local tick [15]. Units are observer conventions [95]: if a common operational propagation law fixed a dimensionless speed c_{model}, choosing the SI speed of light would impose only $L/T = c_{\mathrm{SI}}/c_{\mathrm{model}}$ on the model's length and time units, an operational action constant would add $ET = \hbar_{\mathrm{SI}}/\hbar_{\mathrm{model}}$, and one overall scale would stay free. The construction supplies no c_{model} relating its native distance readout to its modular clock.

Proposition 7.3 (Native field relations)**.** *(i) Every event satisfies the discrete continuity law $x' - x = -b_e j$ with $b_e = \mathbf{e}_a - \mathbf{e}_b$.*

(ii) *Conditional on the past and the selected seam, $\mathbb{E}[j] = d/2$ and $\mathrm{Var}(j) = (|d| \bmod 2)/4$.*

(iii) *Under the uniform calibration the mean load obeys $\mu_{k+1} - \mu_k = -L\mu_k/60$, exactly, by $\mathcal{K}X = XT$ for the 78-state kernel $\mathcal{K}$ and load matrix X; the deviation $V = \frac{1}{2}\|\mu - \frac{1}{6}\mathbf{1}\|^2$ decreases strictly unless the mean is constant, while every binary trajectory keeps $\|x - \frac{1}{6}\mathbf{1}\|^2 = 5/3$.*

(iv) *The slow band has mean multiplier $q = (55+\sqrt{5})/60 \approx 0.953934466291663$, every nonconstant multiplier lies in $(0, 1)$, and the second-difference coefficient $q + 1/q - 2 > 0$ has the opposite sign to the coefficient $-4\sin^2(\omega\Delta t/2) \leq 0$ of an undamped oscillator; no positive rescaling of the attempt count turns this closed mean law into a wave law.*

Proof sketch of Propositions 7.1 to 7.3. The fair coin averages a seam to its endpoint mean, which gives the mean operator and the conditional current moments; the other two positive response eigenvalues have strictly smaller magnitude than $1 - (5 - \sqrt{5})/60$, which gives the limit in Proposition 7.1(ii). Conservation gives the displacement identity. For the causal cones, scanning the schedule backwards and adding both endpoints of every seam that touches the dependency set proves the dependency claim by induction [92, 93, 94]. □

The recovered frame also supplies an unbounded three-component *axial circulation*. With oriented incidence matrix B, seam current J_n and columns $W_{(a,b)} = \hat{u}_a \times \hat{u}_b$ of the unit frame vectors, exact source identities give $WB^T = 0$, $WW^T = 8I_3$, $\mathbb{E}[J_n \mid x_n] = -B^T x_n/60$ and $\mathbb{E}[J_n J_n^T] = I_{30}/198$ in the uniform binary equilibrium. Hence $Z_N = \sum_{n<N} W J_n$ is a record-readable martingale with $\mathrm{Cov}(Z_N) = (4N/99)I_3$, and $N^{-1/2} Z_{\lfloor Nt \rfloor}$ converges to three-dimensional Brownian motion with covariance $(4/99)\min(t, s)I_3$ and Cameron–Martin form $(99/8)\int |\dot{z}(t)|^2\, dt$ in a rescaled attempt

count. A three-event face loop restores the carrier state while adding a nonzero counter increment. The counter's memory grows with the observation horizon; it supplies neither a radial field nor a physical trajectory [12].

7.2 The common continuum operator

On the declared spherical source family, the native cell Green operators converge in norm along one common subsequence to a positive compact injective operator G on the mean-zero observation space. Its inverse $A = G^{-1}$ is densely defined and self-adjoint, and restoring the constant zero mode gives a conservative strongly local regular Dirichlet form with domain H^1 on the supplied flat-polyhedron comparison surface, energy bounded above and below by positive multiples of the gradient energy, and heat-trace spectral dimension two. A boundary-action test excludes a boundedly invertible L^2 energy identification of A with the static boundary action of a smooth compact three-dimensional Dirichlet bulk with smooth boundary, whose Steklov eigenvalues grow like $\sqrt{n}$; bounded-energy internal rank-three modes vanish at this scaling [12]. An exact terminal reduction gives local dual-edge conductances in $[5/8, 15/19]$ with Green-operator error at most $182\,2^{-l}$ relative to native cell means, so this weighted graph has the same continuum subsequences.

A positive native lag readout realizes the Poisson extension. For $K_l = I - L_l/(2M_l)$, Sibuya jump probabilities $p_n = \binom{2n}{n}/[4^n(2n-1)]$ and a Poisson number of jumps with mean R give $\mathbb{E}K_l^N = e^{-R\sqrt{I-K_l}}$. Restricting jumps to b and their count to j, with overflow retained, has maximum native lag bj and operator error at most $\min(1, R\binom{2b}{b}4^{-b} + R^{j+1}/(j+1)!)$; at $R = 1/2$, $b = 64$, $j = 8$ the full 66-state source gives 0.034581 within the bound 0.036. Taking $R_l = r\sqrt{2M_l 4^l}$ yields the compressed limit $e^{-r\sqrt{A}}$; here r labels a positive mixture of native lags, not a physical radius.

Full native density correlations converge to $e^{-\tau A}$, and the admitted coherent continuation converges strongly to $\cos(t\sqrt{A})$. The full-space extension $\bar{A} = A \oplus 0$ satisfies $\|\mathbf{1}_F e^{-\tau\bar{A}}\mathbf{1}_E\| \le e^{-d(E,F)^2/(2\tau)}$, and its cosine propagator vanishes between these supports when $|t| < \sqrt{2}\,d(E,F)$ in the declared flat comparison distance. Two consecutive long observation blocks of one history supply a continuous classical Gaussian wave readout with covariance $G\cos((t-u)\sqrt{A})$ in $C([0,T], H^{-2}(S^2))$, with observation count taken to infinity before spatial refinement [12]. A declared static filter $A_l^{1/4}$ changes it to $A_l^{-1/2}\cos((t-u)\sqrt{A_l})$, twice the real part of the free-field vacuum covariance; the stationary increments obey $2A_l^{-1/2}(I - \cos(\Delta t\sqrt{A_l})) \le (\Delta t)^2\sqrt{A_l}$ and, with $\sum_m \langle h_{jm,l}, \sqrt{A_l}h_{jm,l}\rangle \le 12j^2$ for centered cell averages of spherical harmonics, the geometric H^{-s} increment moment is at most $12[1 + 1/(2s-3)](\Delta t)^2$ for $s > 3/2$, giving convergence in $C([0,T], H^{-s}(S^2))$. The wave readout is declared classical processing, distinct from passive density waiting and from a quantum vacuum.

7.3 Native density histories and their limits

At level l, let $V_l = 240\,4^l$, $q_l = V_l/3$ and $\chi_l = q_l(V_l - q_l)/[V_l(V_l-1)]$. In the uniform canonical equilibrium, with independent uniform seams and fair repair, the actual snapshot readout of a mean-zero test function repeated on its twelve native ports,

$$Y_l(\tau; f) = \frac{\sum_p f_l(p)[n_{\lfloor 1260\,16^l\tau\rfloor}(p) - 1/3]}{\sqrt{V_l\chi_l}},$$

converges in joint distributions at finitely many times as space and the native attempt count are refined together. The limiting family is stationary and Gaussian, with covariance $\langle f, e^{-|\tau-u|A}g\rangle$.

Representing each unread repair as a fair identity-or-swap, the conditional profile overlaps concentrate with variance at most $4\|g\|_\infty^2\|f\|_{\ell^2}^2/V_l^2$ uniformly in attempt count, and a conditional finite-population central limit theorem applies. The declared filter $G_l^{1/2}$ gives covariance $Ge^{-|\tau-u|A}$. The law is reflection positive with one-particle semigroup $e^{-\tau A}$, generator A and not $\sqrt{A}$. The interpolated density converges in $C([0,T],H^{-3}(S^2))$ and its Green-filtered readout in $C([0,T],H^{-2}(S^2))$, by a forward–backward martingale decomposition with $g=2(I+P_l)^{-1}f$ and harmonic tail bounds; the step paths have the same Skorohod limits [12]. For $\phi=\Phi/\sqrt{2}$ a mode of eigenvalue λ has covariance $e^{-\lambda|\tau-u|}/(2\lambda)$, so the Gaussian precision is the Euclidean form

$$S_E(\phi)=\frac{1}{2}\int_{\mathbb{R}}\left(\|\partial_\tau\phi\|^2+\|A\phi\|^2\right)d\tau,$$

with reconstructed Hamiltonian $d\Gamma(A)$ and spatial precision A^2. The action follows from the native readout law and has diffusive dispersion.

Two restrictions bar identifying this sector with a relativistic field. On the shorter attempt scale $b_l=1260\,8^l$ both the density and Green-filtered paths freeze to their initial field, with expected squared supremum increments at most $50(T2^{-l}+a_l^{-1})$, $a_l=1260\,16^l$. For a complete linear readout with bounded injective dense-range composite map $B=TC^{1/2}$, $C=I$ or G, the left polar decomposition $B=D_0^{1/2}U$ gives

$$D_0^{-1/2}D_\tau D_0^{-1/2}=Ue^{-\tau A}U^*,$$

so normalization preserves the ordered decay rates $\lambda_n(A)\asymp n$. No common positive clock factor turns these into the $n^{1/3}$ Euclidean decay rates of a complete stationary free scalar on a smooth compact three-dimensional manifold without boundary, and the same argument excludes every spatial dimension $d\geq 2$, whose rates grow like $n^{1/d}$; a continuous nonconstant time change preserving the semigroup is a scalar factor. The excluded object is the complete linear Gaussian identification; restricted boundary observations, nonlinear dictionaries and interacting fields need their own analysis, and a causal microscopic theory may have this diffusive sector as an effective description [12].

7.4 Incident clocks and active waves

Let $a_l=1260\,16^l$, $M_l=630\,4^l$ and let a fixed event set contain $d_l>0$ seams. Its incident counter $N_l(n)$ includes zero-current repairs, with $p_l=d_l/M_l$; write $T_l(k)$ for the attempt of its kth tick. For $n_*=\lceil a_l(T+\delta)\rceil$,

$$\Pr\left[\max_{n\leq n_*}|N_l(n)-np_l|>a_lp_l\delta\right]\leq\frac{n_*(1-p_l)}{a_l^2p_l\delta^2},$$

and on the complementary event the normalized count $C_l(t)=N_l(\lfloor a_lt\rfloor)/(a_lp_l)$ and the inverse clock $U_l(t)=T_l(\lfloor a_lp_lt\rfloor)/a_l$ differ from t by at most $\delta+1/a_l$ and $\delta+1/(a_lp_l)$ on $[0,T]$. A primitive port has $d_l=5$ or 6 and a whole carrier $d_l=33$. The density path theorem transfers to these local count parametrizations without independence between clock marks and fields [14]. Passive port access chosen before each seam draw from earlier records also works: integer credits $R_n=\sum_{j\leq n}(30/d_j)I_j$ have conditional mean $30/M_l$ per attempt, deviation probability at most $n_*(M_l-5)/(5a_l^2\delta^2)$, and inverse clock error at most $\delta+M_l/(5a_l)$.

For fixed incidence counts d_i and shared-seam counts c_{ij}, centered clock counts divided by $\sqrt{a_l/M_l}$ converge to Gaussian increments with covariance $\min(t,u)c_{ij}$, asymptotically independent of the density snapshots. Adjacent degree-five port clocks have limiting correlation $1/5$, the endpoints of an external seam $1/6$, and nonadjacent clocks zero. These are native model predictions under the stated schedule and preparation, without an atomic-clock or proper-time dictionary [14].

The admitted coherent walk can use the same tick. A protected wave register receives one fixed step W_l per incident event at a degree-d port, including zero current, while the timing source pauses during coherent services. With $Q_l = 630\,16^l$, $D_l = 4^l L_l$, $\beta = 2d/\sqrt{630}$, $\eta = \delta + 1/a_l$ and $e_l = \Lambda/[6(4Q_l - \Lambda)]$, on the band $0 \le D_l \le \Lambda < 4Q_l$ and the count good event,

$$\sup_{\tau \le T} \left\| \cos(N_l(\lfloor a_l \tau \rfloor) \arccos K_l) - \cos(\beta\tau\sqrt{D_l}) \right\| \le \beta\sqrt{\Lambda}\,[\eta + (T+\eta)e_l],$$

so the compressed wave converges to $\cos(\beta\tau\sqrt{A})$. Step error ϵ_l accumulates at most $N_{\max}\epsilon_l$ with $N_{\max} = O(\sqrt{Q_l})$, so $\sqrt{Q_l}\epsilon_l \to 0$ suffices. An exact control retains 237,600 two-attempt clock histories; the continuing Ramsey-plus probability is 1297/1320 against 7901/7920 when zero-current triggers are dropped. The wave step is compiled apparatus; passive repair does not supply it [14].

7.5 An encoded quantum field and its modular clock

The classical covariance does not determine commutators. N independent proper-code registers in $|0\rangle$ give $Q_N = \sum_j X_j/\sqrt{2N}$ and $P_N = \sum_j Y_j/\sqrt{2N}$, with commutator $i\sum_j Z_j/N$, $\mathbb{E}Q_N P_N = i/2$ and $\mathbb{E}Q_N^4 = 3/4 - 1/(2N)$. For Weyl factors $W_N(v) = e^{i(sQ_N + tP_N)}$ the quantum central limit theorem [1] gives

$$\lim_{N\to\infty} \mathbb{E} \prod_j W_N(v_j) = \exp\left[-\frac{1}{4} \left| \sum_j v_j \right|^2 - \frac{i}{2} \sum_{j<k} \sigma(v_j, v_k) \right], \qquad \sigma((s,t),(s',t')) = st' - ts',$$

the CCR vacuum with its ordered-product phase. Source pulses execute four codes and one Ramsey pointer with all 32 outcomes retained; the pointer quadratures give $0.7452257133 - 0.3013609949i$ for the declared three-factor word, reversal conjugates the phase and the tracial preparation removes it. The native density bands fix the mode rotation $e^{i\theta_l Z/2}$ per incident trigger, $\theta_l = \arccos(1 - \lambda/(2Q_l))$, with encoded frequency $\Omega_l = \sqrt{Q_l}\theta_l \to \sqrt{\lambda}$, ordered covariance $e^{-i\Omega_l(t-u)}/(2\Omega_l)$ and kick response $\sin(\Omega_l(t-u))/\Omega_l$, executed for both kick signs at three native bands [12]. Polarized preparation, mode encoding, storage, pulses and readout are specified services. The sine kernel $A^{-1/2}\sin(t\sqrt{A})$ vanishes outside the comparison cone for mean-zero tests; unrestricted smearings add tP_{const}, and restoring the compact massless zero mode gives a free particle with no normalizable ground state, so the mean-zero field is no complete local net.

For $\rho_r = (I + rZ)/2$, $0 < r < 1$, and $H = \Omega(I - Z)/2$, the finite Gibbs and Kubo–Martin–Schwinger identity gives $\sigma_s = \alpha_{-bs}$ for every real s exactly when $\log\frac{1+r}{1-r} = b\Omega$, that is $r = \tanh(b\Omega/2)$; equal polarization at unequal frequencies fails a common b, the pure endpoint is not faithful and the tracial endpoint has trivial flow. With $Q_N = \sum X_j/\sqrt{2Nr}$ and $P_N = \sum Y_j/\sqrt{2Nr}$ the ordered copy limit has thermal covariance $C(\delta) = [\cos(\Omega\delta) - ir\sin(\Omega\delta)]/(2r\Omega)$ with $C(t - ib) = C(-t)$, executed on two codes with retained purifying references. The native bound $\lambda_n(A) \ge n/49920$ gives $\operatorname{Tr} e^{-b\sqrt{A}} \le 99840/b^2$, so the mean-zero Fock space has the faithful normal state $Z_b^{-1} e^{-b\,d\Gamma(\sqrt{A})}$ with the same modular flow, and under $t = \kappa\tau$, $\kappa = 2d/\sqrt{630}$, its modular parameter is $s = -\kappa\tau/b$. The common b is a measurable preparation condition, not a repair-selected temperature.

7.6 The product field, its clock and its Lorentzian density

Independent long blocks of one native history supply radial innovations. Their mean-zero covariance is $B_l = G_l - P/(2520\,16^l)$, and a second block adds a normalized constant-direction readout, giving

$B_l + P_{\mathrm{const}}$. On a declared interval $(0, L)$ with Dirichlet ends, a mode with $h = \sqrt{\lambda}$ has covariance

$$C_\lambda(r,s) = \frac{\sinh(h\min(r,s))\sinh(h(L-\max(r,s)))}{h\sinh(hL)}, \qquad C_0(r,s) = \frac{\min(r,s)(L-\max(r,s))}{L},$$

realized by applying $Q_{rs}^{1/2}(B_l + P_{\mathrm{const}})^{-1/2}$ to successive blocks, with Q_{rs} the conditional innovation. The whitening means the radial direction carries no native information beyond the spectrum of A. Complete 96-event current tapes check the finite decoder. The spatial subsequence gives the product operator $\mathcal{B} = -\partial_r^2 + \bar{A}$ with action $\frac{1}{2}\int_0^L (\|\partial_r\phi\|^2 + \mathcal{E}(\phi))\,dr$; for $L = \pi$ its count satisfies $\Lambda^{3/2}/(12\sqrt{2}) \le N_{\mathcal{B}}(\Lambda) \le 49921\Lambda^{3/2}$ for $\Lambda \ge 3$, and its tensor form gives comparison wave speed one in $d^2_{\mathrm{product}} = |r-s|^2 + 2d^2_{\mathrm{screen}}$. On a radial grid the sampled bridge has an explicit sine-mode covariance, and a spectral filter realizes the thermal covariance $\coth(b\sqrt{\mathcal{B}}/2)/(2\sqrt{\mathcal{B}})$ from the same records; the full-port generator and the cell Schur generator have compressed resolvents within $1/(8\,4^l)$ at each positive shift, so both give the same product target. Radial labels, filters, boundaries and block access are specified operations.

For a full-port angular mode set $a_l = \sqrt{Q_l}\arccos(1 - \lambda/(2Q_l))$ and $c = \sqrt{\rho_{k,h}}$. Each incident trigger applies the source Z pulse of angle $a_l/(2\sqrt{Q_l})$ and the radial X pulse of angle $c/(2\sqrt{Q_l})$; anticommutation gives $\omega_{l,k} = \sqrt{a_l^2 + c^2}$ with no fitted product frequency, and the split word differs from the joint Hamiltonian by at most $na_lc/(4Q_l)$ after n triggers. The faithful reference is polarized along $(cX + a_lZ)/\omega_{l,k}$ with $r = \tanh(b\omega_{l,k}/2)$; ordered two-time measurements retain all 32 outputs. A deterministic packet $v \propto (12I - L)^{1024}(\mathbf{1}_{\mathrm{cell}\,0} - \mathbf{1}_{\mathrm{cell}\,1})$ on the 240-port carrier has Rayleigh quotient $\lambda \simeq 0.044442251696$, residual $\|(L-\lambda)v\| < 0.000304$ and squared cell compression above 0.996458773594; Householder encoders and a Bell reference read it after incident-triggered phase, with 25-trigger Chebyshev discrepancy about 0.000149026 at $M = 630$, at a cost of 1,747,068 source pulses. Copy and spectral limits followed by radial refinement give

$$C_b(t) = \frac{\coth(b\sqrt{\mathcal{B}}/2)\cos(t\sqrt{\mathcal{B}}) - i\sin(t\sqrt{\mathcal{B}})}{2\sqrt{\mathcal{B}}}, \qquad D(t) = \frac{\sin(t\sqrt{\mathcal{B}})}{\sqrt{\mathcal{B}}}, \qquad \sigma_s = \alpha_{-bs},$$

and with $C_L = L(49920 + L^2/9)/3$, $\operatorname{Tr} e^{-b\sqrt{\mathcal{B}}} \le 6C_L/b^3$, so the encoded Fock algebra has the faithful normal state $Z_b^{-1}e^{-b\,d\Gamma(\sqrt{\mathcal{B}})}$ with $s = -\kappa\tau/b$. Vanishing count tolerances require $4^l\eta_l^2 \to \infty$ and complete-step errors $\sqrt{Q_l}\epsilon_l \to 0$. Radial design, common b, preparation, mode access and source precision are declared services [12].

Lorentzian density. The regular strongly local angular form with its H^1 domain admits a measurable uniformly elliptic tensor in each chart [137, Theorem 7.2 and Corollary 7.3], recoverable from complete native records: with the disclosed total-one kernel $K_l = I - L_l/(2M_l)$, including every zero-current hold, the ordered limits

$$\lim_{t\downarrow 0}\lim_{l}\int \phi\,\frac{\bar{P}_l(t)(fg) - f\bar{P}_l(t)g - g\bar{P}_l(t)f + fg}{2t}\,d\mu = \int \phi\,a^{ij}\partial_i f\partial_j g\,d\mu$$

recover weighted tensor averages. In barycentric face coordinates the one-attempt increment tensor obeys $\frac{3}{7}a_{\mathrm{bare}} \le a_{\mathrm{screen}} \le \frac{15}{19}a_{\mathrm{bare}}$ with $a_{\mathrm{bare}} = \begin{pmatrix} 1/36 & -1/72 \\ -1/72 & 1/36 \end{pmatrix}$, so the one-attempt-first limit reads a different tensor. The radial product adds $a^{rr} = 1$. With coordinate measure $\mu(x)\,d^3x$ the chosen wave action has coefficient density $C = \mu\,\mathrm{diag}(-1, a)$, and

$$g = \sqrt{-\det C}\,C^{-1} = \mu\sqrt{\det a}\;\mathrm{diag}(-1, a^{-1}), \qquad \sqrt{-\det g}\,g^{-1} = C$$

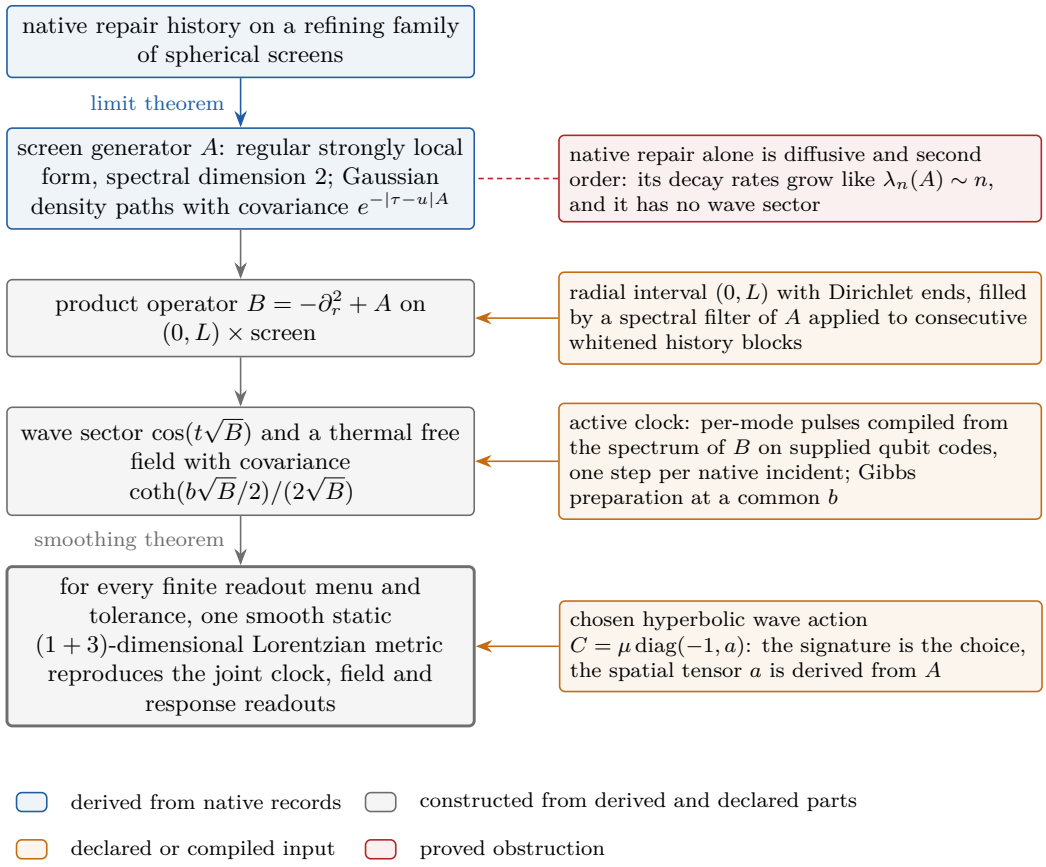


Figure 3: What the effective $(1 + 3)$ scalar description derives from native records and what it declares. Blue: derived from one native repair history by a limit theorem. Grey: constructed from derived and declared parts. Orange: declared or compiled inputs. Red: the proved obstruction that rules out a wave sector from passive native repair. The smoothing theorem is an existence result for fixed finite readout menus.

has signature $(-,+,+,+)$ almost everywhere, with weak wave equation $\partial_\alpha(C^{\alpha\beta}\partial_\beta\phi) = 0$ [198, Equations (23)–(26)]. The signature is that of the chosen action; the derived content is the measurable tensor a. The native measure is the radial pushforward of flat face area, with density $\rho_F(u) = c/(n_F \cdot u)^3$, $c = \pi r^2/(5a_0)$, on a face of unit normal n_F and plane distance r. Adjacent faces agree on their common edge but their spherical gradients differ by $-3c(n_F - n_G)/(n_F \cdot u)^4 \neq 0$, so the density is not C^1, and since $C^{tt} = -\mu$ the exact dictionary in this atlas fails the smooth-density premise. An alternative smooth dictionary would need a separate argument.

The density is globally Lipschitz relative to smooth sphere volume. Smoothing only the angular variables and the tensor, with the unitaries $U_\varepsilon f = \sqrt{\rho_\varepsilon/\rho}\, f$ between the changing L^2 spaces, gives transported Mosco convergence and then norm-resolvent convergence with gap $(\pi/L)^2$. Thermal covariance and sine response converge in operator norm, and ordered Weyl laws, finite adaptive Ramsey histories and finite detector outputs converge. Hence for any fixed finite admissible menu on bounded wave times and any positive tolerance one smooth static $1 + 3$ metric approximates the joint readouts. Both sides are free fields on the same supplied codes; the theorem removes the roughness of the native coefficients, not the declared radial product, clock or state, and gives no computable rate. In the smooth comparison $g = N^2\,\mathrm{diag}(-1, a^{-1})$, $N^2 = \mu\sqrt{\det a}$, the canonical

momentum is $\mu\partial_t\phi$ and the thermal covariance and response have the standard static CCR/KMS normalization with a reflecting radial boundary.

Angular observations do not select the radial law. For the 240-port L, three 720-port products $B_R = I_3 \otimes L + R \otimes I$ with the unit path, fourfold path and unit triangle Laplacians satisfy $J^*F(B_R)J = F(L)$ for the radial average J and every spectral function, while a radial-contrast probe has cosine second derivatives $-5/2, -10, -3$.

For positive finite-measure regions E, F of the product, let $d(E,F) = \sup_u(\operatorname{ess\,inf}_F u - \operatorname{ess\,sup}_E u)$ over bounded local-form functions with $\Gamma_{\mathcal{B}}(u) \le 1$. Davies–Gaffney estimates and the short-time theorem of Ariyoshi and Hino [130, Theorem 2.7], $\lim_{\tau\downarrow 0} -4\tau \log\langle \mathbf{1}_F, e^{-\tau\mathcal{B}}\mathbf{1}_E\rangle = d(E,F)^2$, combined with heat-cosine transmutation give

$$\inf\{t>0 : \|\mathbf{1}_F \cos(t\sqrt{\mathcal{B}})\mathbf{1}_E\| > 0\} = \inf\{t>0 : \|\mathbf{1}_F \mathcal{B}^{-1/2}\sin(t\sqrt{\mathcal{B}})\mathbf{1}_E\| > 0\} = d(E,F).$$

This is the sharp finite-propagation front of the chosen wave equation, fixed by its intrinsic metric; it supplies no independent propagation speed, and same-field modular time has onset $|s| = d/b$.

7.7 Interacting readouts on the product

Retain the angular constant and radial profiles $\sqrt{2}\sin r$ and $\sqrt{2}\sin(2r)$ on $(0,\pi)$. The projected local quartic polynomial is

$$W = \tfrac{3}{2}(q_1^4 + q_2^4) + 6q_1^2q_2^2, \qquad H = \tfrac{1}{2}(p_1^2 + q_1^2 + p_2^2 + 4q_2^2) + \frac{g_{\mathrm{eff}}}{24}W,$$

with $g_{\mathrm{eff}} = (\lambda/\pi)\int \mu_{\mathrm{screen}}^2 \sqrt{\det a_{\mathrm{screen}}}\, dx$ for a constant covariant coupling; $g_{\mathrm{eff}} = 1$ is an input. At occupation cutoff 4×4, native source pulses prepare the purification of $e^{-\beta H}/\operatorname{Tr} e^{-\beta H}$ and execute two reads with feedback, retaining all work, reference and record outputs. For $\beta = 1/4$, intervals $1/16$ and 64 split steps, the complete-output distance from an independent exponential is below 1.97×10^{-8}; deleting the interaction or feedback changes the record law by about 7.31×10^{-4} or 1.95×10^{-3}. The worst branch uses 217,110 source pulses, 434,220 load/export events and two native repairs. A separate protocol advances this Hamiltonian on every incident event of two native timing attempts, retaining all 3,600 tapes and 4,900 field-prefix outputs, with timeout probability $35/36$ and complete error below 3.44×10^{-9}. On the refined clock family, incident rate $R_l = 2d4^l$ gives a controlled inverse-count comparison; each metric readout uses its own clock pullback, and no universal physical conversion is inferred.

For $g \ge 0$ the positive quartic form has a Friedrichs Hamiltonian $H \ge H_{\mathrm{harm}}$, Hermite compressions form a core, and

$$Z_d \longrightarrow Z, \qquad 0 < Z \le \frac{e^{-3\beta/2}}{(1-e^{-\beta})(1-e^{-2\beta})}, \qquad \rho_d \longrightarrow \rho \quad \text{in trace norm},$$

so every fixed finite complete adaptive history converges and the limiting Gibbs state has $\sigma_s = \alpha_{-\beta s}$. This proves occupation existence for the two-mode theory, without a small-code certificate: the complete $d=4$ versus $d=8$ output distance is about 0.56045, and $d=8$ versus $d=12$ about 0.24501. An inverse control recovers $(g_{\mathrm{eff}}, \beta, c) = (1, 1/4, 1)$ within 10^{-4} from six noiseless source probabilities and predicts a withheld interval within 10^{-8}, with local Jacobian singular values about $0.264713, 0.001489, 0.000110$. For fixed covariant λ the smoothed metrics give convergent quartic projections. The quartic excites omitted modes; no invariant two-mode band, interacting ultraviolet limit or finite-cutoff microcausality follows [48].

The reference-state boundary. Increasing native-return caps match modular phase magnitudes to the cosine limit, and a fixed cap freezes the low-band phase. On the full configuration algebra the uncapped reference approaches normalized trace with half-trace error at most $3/(8\,2^l)$, its density-wave sector has vanishing weight, and fixed-parameter modular evolution becomes trivial in normalized-trace L^2. If the relevant fluctuation moments converge, any limiting scalar canonical commutator in this bounded-overlap regime is zero. Refinement-rescaled time, other states and other fluctuation regimes are not excluded [12].

8 A common matter metric and its gravitational response

This section is the interface of the executed construction with the gravitational layer of the tower. The conditional scalar description supplies a concrete dictionary for comparing matter and geometry, and the same dictionary yields the static-product control that any responsive geometry must overcome. The counts, ranks and identities quoted here are retained in the receipts of the matter and common-field contracts of the public repository. Write its four-dimensional coefficient density and metric as

$$C = \mu\,\mathrm{diag}(-1, 1, a), \qquad g = \sqrt{-\det C}\,C^{-1} = N^2\,\mathrm{diag}(-1, 1, a^{-1}), \qquad N^2 = \mu\sqrt{\det a},$$

where a is a positive two-dimensional screen tensor and $\mu > 0$. These identities use the complete action density, including its measure. For the measurable native limit, smooth representatives are understood with the stated finite-menu approximation theorem and smearing transport.

8.1 Common principal symbols and finite field operators

Choose an orthonormal coframe e with $g = e^t\eta e$. Transporting Clifford matrices through its inverse gives

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}I, \qquad (\gamma^\mu p_\mu)^2 = (g^{\mu\nu}p_\mu p_\nu)I.$$

The scalar wave symbol, both Weyl principal determinants and the Maxwell symbol therefore have the same characteristic cone. The Maxwell gauge direction is retained, leaving two physical polarizations on a nonzero null covector. A change of clock coordinate rescales all field generators and the reference inverse temperature together, preserving βH. This pointwise dictionary does not select the fields' dynamics or chirality.

For a finite incidence factor C_0 on the declared angular–radial product,

$$D = \begin{pmatrix} 0 & C_0^\dagger \\ C_0 & 0 \end{pmatrix}, \qquad D^2 = \mathrm{diag}(C_0^\dagger C_0, C_0 C_0^\dagger).$$

The node block is the scalar spatial operator $B = C_0^\dagger C_0$. CAR second quantization gives a conditional fermion construction with node propagation $\cos(t\sqrt{B})$ and node-to-edge propagation $-iC_0\sin(t\sqrt{B})/\sqrt{B}$. The full finite construction retains its extra edge modes. A factor matching the finite angular frequency is also tested; its ideal joint generator is not the exact Floquet generator of the finite angular/radial split word. Their difference retains the specified approximation budget.

Completing the actual tetrahedral cone through faces and tetrahedra gives a separate full-cochain control. Its dimensions are $(13, 42, 50, 20)$, its differential ranks are $(12, 30, 20)$, and its Betti numbers are $(1, 0, 0, 0)$. The complete metric $d + d^\dagger$ construction removes the truncated cone's thirty edge-cycle zero modes, preserves its scalar action block, and retains the scalar constant. This

cone is a different global domain from the angular–radial cylinder. Differential-form parity does not establish physical Weyl chirality.

The smooth product admits a spin structure, but scalar radial Dirichlet conditions do not by themselves define a self-adjoint first-order spinor operator. A tested reflecting extension squares to a Neumann component and a Dirichlet component. Spin-connection and curvature terms also enter: a curved spinor square is not a componentwise scalar Laplacian. These domain and connection hypotheses are explicit in the common-field tests.

8.2 Area, volume and the entropy variation

The same product metric gives exact induced measures

$$dA = \mu\, d^2x, \qquad dV = \mu N\, dr\, d^2x.$$

Thus the scalar action measure supplies a whole-screen area dictionary. Fixed counting measure alone does not fix physical three-volume. Radial slices in this product have equal area and are not concentric small geodesic spheres.

Native constrained maximum-entropy completion and its joint counting first law hold on their finite algebraic variation domains. They do not identify fixed cap marginal or fixed total dimension with fixed proper volume. In the exact selected-reference test, forty-five four-port caps have identical counted edge entropy and crossing-seam counts ranging from ten to eighteen. No nonzero universal constant equates that entropy to this boundary proxy. The statement concerns a finite two-dimensional boundary proxy, rather than the whole-screen area above.

The family

$$\mu(z) = \mu_0 e^{-2z/3}, \qquad a(z) = e^{2z} a_0$$

keeps proper volume fixed on the declared radial coordinate interval and decreases its boundary area. At fixed model-coordinate inverse temperature, the cavity thermal entropy also decreases. Hence the trial functional $S_{\text{thermal}} + A/(4G_N)$ is not stationary for positive G_N on these variations. This is a specified thermal-cavity comparison; it is not an identification with local vacuum entanglement or a proof that repair generates those variations.

The distinction between observable algebras matters as well. For an executed symmetric mixture of $\pm\sqrt{\epsilon}$ displacements, the unread work density has positive first-order entropy and energy derivatives at fixed reference Hamiltonian. The retained sign-plus-work density has zero entropy derivative. Both outcomes and the purification are retained. A gravitational entropy identification must specify which region algebra and which records enter its bulk term.

8.3 A fixed-product obstruction and a necessary dynamic response

Proposition 8.1 (Radial null response of the static product)**.** *For a smooth metric*

$$g = f(x,y)\,\mathrm{diag}(-1, 1, h(x,y)), \qquad f > 0,$$

with positive screen metric h, both radial null contractions of its Einstein tensor vanish. With the Gaussian reference and other sources fixed, the same geometry cannot satisfy the Einstein balances for that reference and every coherent scalar excitation at positive gravitational coupling.

Proof. The (t, r) factor has Lorentz symmetry and all metric coefficients are independent of t, r. Its Ricci components obey $R_{tt} + R_{rr} = 0$ and $R_{tr} = 0$, also verified by direct differentiation for

arbitrary smooth f, h. For $k = \partial_t \pm \partial_r$, $g(k,k) = 0$, so trace and cosmological metric terms vanish and $G(k,k) = 0$. The restored angular-constant scalar sector admits

$$\phi = A \sin(\pi r/L) \cos(\pi t/L).$$

It obeys the product wave equation and the declared radial Dirichlet condition. Its action-derived coherent relative stress satisfies

$$\Delta T(k,k) = (k\phi)^2 = \frac{A^2\pi^2}{2L^2} > 0 \quad \text{at } t = 0,\ r = L/4.$$

Subtracting the reference and excited Einstein equations contradicts $\Delta G(k,k) = 0$. This uses relative stress and supplies no absolute vacuum-energy prescription. □

A coherent amplitude changes stress quadratically; this argument is not a claimed failure of the linear entropy first law. A symmetric mixture of opposite displacements supplies a first-order state direction, with its observable algebra specified separately. Independently changing compensating sources and geometries outside the static product class are outside the proposition.

For the smooth spherical extension

$$ds^2 = -2du\,dv + R(u,v)^2 d\Omega^2,$$

direct curvature calculation gives $G_{uu} = -2R_{uu}/R$. The null Einstein balance consequently requires

$$R_{uu} = -4\pi G_N R T_{uu}, \qquad \theta_u = -\frac{\theta^2}{2} - 8\pi G_N T_{uu}, \qquad \theta = \partial_u \log(4\pi R^2).$$

Here u is affine along the declared null curves. These are necessary area-response equations in this symmetry class, not a selected repair law or a solution of the full Einstein system. The alternative finite nine-null theorem also requires its independent balances; a pointwise tetrad map alone does not supply them.

The exact block, null-stress and focusing algebra is formalized in the Lean library of the public repository, and the source matrices, curvature identities, complete finite records and approximation controls are executed with the observer-particle model. Their conditional realization establishes neither observed Standard Model masses nor generalized-entropy stationarity on a physical refinement family.

9 Prediction protocols

A response without a fitted coefficient. In the stationary binary number sector, disclose local occupations and apply one extra fair native repair on seam (a,b) exactly when $n_a = 1$. Detailed balance gives, for any fixed future record event F whose continuation does not consult the earlier pulse mark,

$$\Pr_{\text{feedback}}(F) - \Pr_{\text{passive}}(F) = \tfrac{1}{2}\mathbb{E}_{\text{passive}}[(n_b - n_a)\mathbf{1}_F].$$

Complete capped clock and field forecasts and the native spatial Green response satisfy this identity exactly, with every trigger, coin and current label retained [12]. No matched public intervention dataset exists for these forecasts.

A controlled continuum response. For a disclosed port profile h, a uniform seam (a, b) executes its unchanged fair repair with probability $[1 + \epsilon(h_b - h_a)(n_a - n_b)]/2$, and the clock counts executed repairs, including zero currents. The exact linear occupation response is $2\chi K(I - K^r)h$, and on the refining source

$$\partial_\epsilon \mathbb{E}\eta_l(g)|_0 \longrightarrow 2\langle g, (I - e^{-\tau A})f\rangle.$$

At $\tau = 1$ with four selection slots per target repair, stopped likelihood bounds control the nonlinear mean error by $36\epsilon^2\|g\|$, and for a unit-axis flat coordinate at $\epsilon = 10^{-6}$, $\liminf_l \mathbb{E}_\epsilon \eta_l(f) \geq \epsilon\, 9/32792 - 36\epsilon^2 > 0$ [12]. In the selection-slot clock the derivative converges to $\langle g, R_\tau f\rangle$ with $R_\tau = 2(I - e^{-\tau \bar{A}})$, and positivity improvement of the heat semigroup gives, for nonzero nonnegative Lipschitz profiles at distance $r > 0$,

$$0 < -\langle g, R_\tau f\rangle \leq 2e^{-r^2/(2\tau)}\|f\|\|g\|, \qquad \tau > 0.$$

This derivative has no exact finite support cone. Reading its control, support and slot parameter as physical local interventions and time would contradict an exact relativistic cone; that identification is not derived, and a causal microscopic theory with diffusive effective observables is consistent with it.

9.1 The phase experiment

The classical repair law fixes the diagonal of every admissible record law and none of its coherences (Theorem 2.3). A fixed interference experiment reads the multiplier g that separates the members, and its count protocol, comparing the executed laws C and E, is fixed before any data [74, 75]. All members are ordinary quantum instruments, so the test separates record laws and makes no claim to separate OPH from quantum mechanics.

Write $x = \{2, 3\}$, $z = \{1, 2\}$ and $w = \{0, 2\}$. Prepare $|\phi_\theta\rangle = (|x\rangle + e^{i\theta}|z\rangle)/\sqrt{2}$ for $\theta \in \{0, \pi, \pi/2\}$, or the incoherent mixture $\frac{1}{2}(|x\rangle\langle x| + |z\rangle\langle z|)$, apply one repair on the seam $(0, 1)$, retain its record, and measure $P_+ = |\phi_0\rangle\langle\phi_0|$ and $P_- = I - P_+$ inside each committed record block.

Proposition 9.1 (Phase law of the two record instruments)**.** *For the preparations above:*

(i) the nonzero recorded source densities of C *are* $\frac{1}{2}|\phi_\theta\rangle\langle\phi_\theta|$ *for* $(c, j) = (0, 0)$, $\frac{1}{4}|x\rangle\langle x|$ *for* $(1, 0)$ *and* $\frac{1}{4}|w\rangle\langle w|$ *for* $(1, -1)$*; those of* E *are* $\frac{1}{4}|x\rangle\langle x|$ *for* $(c, d) = (0, 0)$ *and* $(1, 0)$, $\frac{1}{4}|z\rangle\langle z|$ *for* $(0, -1)$ *and* $\frac{1}{4}|w\rangle\langle w|$ *for* $(1, -1)$*;*

(ii) the plus probabilities are $p_{\mathsf{C}}(\theta) = \frac{3}{8} + \frac{1}{4}\cos\theta$ *and* $p_{\mathsf{E}}(\theta) = \frac{3}{8}$*, and the incoherent preparation gives* $3/8$ *for both; the phase-reversal contrast is* $1/2$ *for* C *and zero for* E*;*

(iii) all four preparations have identical classical load probabilities, the readout preserves the probability of every old record, and no record is erased and no outcome postselected.

Proof sketch. On the seam $(0, 1)$ the configuration x has contrast zero for both coins, and z has contrast -1, current 0 for coin zero and current -1 for coin one, which moves it to w. The coherent sum survives only in the C branch $(0, 0)$, where both currents vanish; E separates x from z by their contrasts [75]. □

For every admissible record law the plus probability is $p_\Phi(\theta) = \frac{3}{8} + \frac{1}{2}\operatorname{Re}(g\, e^{-i\theta})$, since the empty seam of x stays in place and z waits with probability $\frac{1}{2}$ while keeping coherence g with x. The settings 0, π and $\pi/2$ therefore estimate $\operatorname{Re} g$ and $\operatorname{Im} g$ for whichever law is realized; laws with equal g, such as E and T, are separated by the same construction with the full seam $\{0, 1\}$ in place of z, which gives $\frac{1}{2} + \frac{1}{2}\operatorname{Re}(k\, e^{-i\theta})$ [49].

Table 7: Plus probabilities of the four fixed settings [74, 49].

Preparation	C_0	C	E	T
Phase 0	$\frac{3}{8}+\frac{\sqrt{2}}{4}$	5/8	3/8	3/8
Phase π	$\frac{3}{8}-\frac{\sqrt{2}}{4}$	1/8	3/8	3/8
Phase $\pi/2$	3/8	3/8	3/8	3/8
Incoherent mixture of x and z	3/8	3/8	3/8	3/8

The count protocol fixes four settings with 8,192 fresh independent attempted trials each and no optional stopping. Each attempt falls in one of twelve common bins of coin, current and outcome, or aborts. At each setting it scores the twelve bin indicators, the plus indicator and the abort indicator, $m = 4(12 + 2) = 56$ statistics, with sampling half-width $b = 1/32$, independently established apparatus probability budget $\eta = 1/64$ and familywise sampling requirement $\delta = 0.001$.

Proposition 9.2 (Sampling certificate)**.** *If the attempted trials of each setting are independent, then with* $n = 8192$ *and* $b = 1/32$ *the probability that any of the 56 observation intervals* $[\hat{p}-b, \hat{p}+b]$ *misses its actual probability is at most* $112e^{-16} < 112/S_{32} < 0.000012606 < \delta$*, where* $S_{32} = \sum_{k=0}^{32} 16^k/k!$*.*

Proof. Hoeffding's inequality [117] gives $2e^{-2nb^2} = 2e^{-16}$ for each statistic, a union bound over 56 statistics gives $112e^{-16}$, and $S_{32} < e^{16}$; the final comparison is exact rational arithmetic [75]. □

A candidate is rejected when some observation interval is disjoint from its ideal probability widened by η. Under the apparatus contract, rejection occurs only when an observation interval fails, and both candidates use the same intervals; if the calibration guarantee fails with probability at most β, the combined bound is $112/S_{32} + \beta$. An abort has ideal probability zero, and a resolved excess returns the verdict that the apparatus budget is exceeded. Without supplied counts the verdict is "no data", and software demonstrations are labelled synthetic [74]. One survivor discriminates the conditional instruments, two survivors do not separate them, and two rejections mean neither specified apparatus model explains the counts; in every case the same counts estimate g.

9.2 The source-executed implementation

Both complete recorded instruments, the four preparations and the readout also run as emitted twelve-port control words [72, 73]. A configuration is written as the seam pair and the ten spectators: the six endpoint pairs use three bits and the spectators six, placing the 78 configurations injectively in a nine-qubit register with 434 unused addresses. A local permutation with blank record bits realizes each repair from phase-exact transpositions, and the fair coin is prepared by a source pulse and recorded. The encoding is a mathematical isometry, not a physical interface from occupation states.

Proposition 9.3 (Exact source realization of the two instruments)**.** *In the nine-qubit encoding* J*:*

(i) the encoded recorded channels intertwine the native ones,

$$\mathcal{E}_{\mathrm{code}}(J\rho J^*) = (I_{\mathrm{record}} \otimes J)\, \mathcal{E}_{\mathrm{native}}(\rho)\, (I_{\mathrm{record}} \otimes J^*),$$

on all 6,084 matrix units of the 78-state carrier and therefore with every untouched reference; the complete Gram identity has 156 nonzero Kraus entries;

(ii) E *is* C *followed by a three-pulse source controlled-NOT (CNOT) that copies the seam-total parity* p*, with contrast decoded as* $d = 2j + (2c-1)p$*; relative to the retained coin and current labels one additional binary record is necessary and sufficient, and forgetting it leaves parity dephasing without restoring the coherence of* C*;*

(iii) the readout is the Lüders instrument [121] of the rank-one plus projector and its complement, realized by coupling the plus projector to a blank pointer and measuring only the pointer; the minus outcome is one rank-77 projector whose internal coherences survive, and reading the full computational basis would implement a different instrument.

Proof sketch. For fixed current the endpoint map is an injective translation, and for fixed contrast the spectators and endpoint sum reconstruct the input, so every input column of the joint coin, record and source isometry has two entries $2^{-1/2}$ with disjoint supports; an independent enumeration checks all 6,084 matrix units. For (ii), $\{2,3\}$ and $\{1,2\}$ share the C record $(0,0)$ and have contrasts 0 and -1 [73]. □

The complete attempted trial costs 32,006 to 32,010 native pulse requests, with one proper-code load and one export per pulse; a readout costs 26,632 pulses, a local C repair 5,370 and a local E repair 5,373, with peak live width 11 qubits for C and 12 for E [72]. Executed terminal states reproduce Table 7, and on synthetic counts drawn from executed branch norms the evaluator keeps the generating candidate and rejects the other. The compositional error contract

$$N_p u + N_l\, l + N_e\, e + p + z + c + a + r + s + m + g + n \le \frac{1}{64},$$

with pulse, load and export counts and their channel errors and whole-trial allowances for preparation, initialization, coin, randomizer, records, storage, measurement, coefficient compilation and numerical execution, gives sufficient conditions for $\eta = 1/64$; an equal allocation at the largest schedule is $1/6146496$ per term. The same source alphabet, code services and classical transitions admit two quantum instruments whose phase-zero observable differs by $1/4$, so these ingredients alone do not select the record law.

9.3 Retrospective audit of the closure coupling output

The detector of Section 3.4 evaluates the structured-running branch of the closure map; the flagship synthesis reports a second, self-consistent closure map with a much smaller residual [3], and this construction consumes the bare source map only. Its output, approximately 136.994835177413, falls short of the 2022 CODATA value $\alpha^{-1} = 137.035999177(21)$ [113] by about 300.388218 parts per million, and exact rational arithmetic shows its narrow numerical interval disjoint from the band of five reported standard uncertainties [74, 75]. The band is a declared comparison allowance, not a confidence interval. The audit rejects the additional hypothesis that this output is the complete physical endpoint with zero missing correction; no fit is performed and P is unchanged.

9.4 A sequential design on one carrier

The phase test above assumes fresh preparations. A sequential design runs it on one continuing carrier in its binary total-two sector without resets or independent trials. Each trial applies 379 unread native attempts to the state the carrier holds, performs the fixed-seam native current herald, whose useful yield is 5/33, and on either useful branch executes the source mixer, three unread native attempts, a second mixer and the native coin and current pointer; a failed herald counts as a trial. After exactly 3,927 trials the two locked likelihoods are compared on all six counts,

and an impossible record returns a verdict outside both models. Conditional-law bounds on the dependent transcript give a wrong-candidate probability of at most 3/400 under the exact-source model; a uniform service allowance of 1/1570800 per complete trial would suffice for a one-percent bound, and that allowance is not authenticated [26]. Two consecutive trials per candidate and their reference controls are executed.

9.5 Native records against public data

Published records test precise attachments of native records without a laboratory realization of the carrier; each test is retrospective and states its readout hypotheses.

The binary endpoint map is an identity or a local exchange, so diagonal preparations stay diagonal under recorded operations and local classical feedback. Under independent settings and local factorized responses the diagonal branch obeys $P(\text{win}) \leq 3/4$ in the CHSH game, that is $S \leq 2$; a control that lets one side read both settings wins every trial, and the native example on the 66-state carrier, with all 1,056 histories, gives $S = 26/33$. Replaying the deposited loophole-free records of Hensen et al. [29] reproduces 245 eligible trials, 196 wins and $S = 16036094/6619647 \approx 2.42250$, with a published fixed-count tail bound near 0.03917 [27]. This rejects the local classical reading of the diagonal branch under that experimental contract; it does not test the quantum C/E instruments.

Two native ports, or two regions of one carrier, counting incident repairs under the uniform scheduler have independent interval increments with exact count laws. Attaching the published 300-second blocks of the Ar^{13+}/Yb^{+} frequency ratios of King et al. [30] to the normalized count ratio, through a gain of at least one, a declared bracket on the Yb^{+} frequency, an absolute error of 10^{-12} per block and direct phase-count readback, gives conditional probability bounds below 10^{-104} for two ports and below 10^{-22} over all ordered pairs of the 3,433 regional event sets, after charging every choice of subwindow, gain and retained blocks [28]. This rejects the direct attachment with its readout assumptions; the native event clocks are unaffected as clocks of the model, and a clock calibration needs an explicit model of the measuring apparatus.

Published trapped-ion records test a supplied experimental model executed through the same source interface: two calibrated preparation parameters predict 100 reserved complex readouts with per-spin-mean root-mean-square error 0.06647, and a changed-potential transfer predicts 125 derived observations with root-mean-square error 0.2876427868 in two-point position-estimator units [11]. They test the supplied oscillator model, whose action and apparatus do not follow from native repair.

10 The premise map

The premise map relates the construction to the conditional theorems of the library [52, 53] and is the starting ledger of the joins of Section 11. It applies one rule: a result "H implies C" discharges an input I only when C establishes I on the same domain with the required errors and H has independent support; a change of regulator, state or code, or a calibration relabelled as a derivation, discharges nothing. An antecedent is either discharged for the finite objects of the construction by execution or proof, or supplied, and discharge on the finite domain differs from discharge of the physical premise on a physical refinement tower. Six of the seven claim records of Table 3 consume no canonical physical premise; the detector record consumes the accepted closure inputs P and N and the selected coupling map, and lists the same-quantity bridge for P and the physical Thomson readback contract as unsupplied [2]. The crosswalk represents all 84 canonical premise identifiers

in 27 groups and inventories 343 registered claims and 661 Lean modules, without individually instantiating every listed claim [52]. Table 8 summarizes seven theorem routes.

Finite entropy and the gravitational premise. For positive sector counts d_i, normalized probabilities p_i and fixed $M = \sum_i d_i$, the counting entropy obeys

$$S_{\text{count}} = H(p) + \sum_i p_i \log d_i = \log M - D(p\|d/M) \le \log M.$$

The maximizer is $p_i = d_i/M$. In the positive-real relaxation, joint fixed-M variations have zero first derivative there and second derivative $-\sum_i (p_i' - d_i'/M)^2/p_i$. This is no operation varying integer dimensions, and with bulk factors full saturation also requires maximally mixed bulk blocks. Uniform native preparation realizes counting weights. Fixed count is not identified with fixed physical volume, nor the logarithmic sector term with a geometric area term of universal coefficient [13]. For every cap of the wave reference's return-length law, an absolute bound holds on the full $D = \binom{V}{V/3}$ configuration space:

$$\Delta = \log D - S(\sigma) \le \frac{2025}{64V}, \qquad I_\sigma(R : R^c) \le \log J + \Delta,$$

where J counts the allowed regional particle numbers. A complete base-face patch has $V/20$ ports and $3\,2^l$ crossing seams. Its stationary reduced entropy is extensive in port count, while the return reference's mutual information divided by crossing-seam count vanishes. Neither observable therefore has a nonzero universal coefficient times that boundary count. This does not exclude interpreting a whole spherical screen as a bulk boundary; that physical region map is a premise [13].

Table 8: Theorem routes, antecedents discharged on the declared construction, and supplied antecedents [52, 48].

Route	**Derived from native records (N) or constructed on declared inputs (D)**	**Supplied**
Consensus and repair [6]	Committed-record preservation; the exact gap of the record-discarded channels; schedule-independent repair of the completed common-source history; the exact law of two overlapping observers; federation output bounds below 0.000382 at $s = 1/4$ and 6.656×10^{-9} after four incident ticks.	Repair completeness, local diamonds and transactional descent for an unrestricted observer menu; conflicting updates; global history repair by native hardware; all-level federation routing.
Quantum records [7]	(D) All-input CPTP maps with coherent write isometries and complete binary instruments; the exact correctable algebra, optimal recovery and active encoded memory; the compiled joint native/work channel; GNS representations of the actual observation state; the quantum correspondence on 4,356 matrix units and the coherent screen observer within 7.779×10^{-10}.	Hilbert representation, subsystem split, Born readout, coherent preparation and readout services (not available from diagonal native resources, Section 2.5), a physical encoding of loads in the code.
Record capacity [7]	M_0 for each declared record interface with a carrier projector certificate; $\ln M_0 \le N$ on a declared carrier.	Complete-fiber closure, a physical selector and the horizon identification behind $\Lambda \ell_\star^2 = 3\pi/N$.

Table 8 continued

Route	Derived from native records (N) or constructed on declared inputs (D)	Supplied
Gauge structure [8]	Branch charge covariance of $Q = g(P)N$; a four-state CAR and link code with operator Gauss constraints and even parity.	Physical current and action for the complete response algebra, coupling attachment, nonabelian and chiral matter.
Scalar continuum [7, 3]	Refinement transfer and occupation limits with complete-output bounds; the infinite-occupation interacting signal. (N) The native H^1 form of spectral dimension two, Gaussian density paths, incident-clock parametrizations, the precision $-\partial_\tau^2 + A^2$ and active incident-clock waves [12]. (D) A thermal CCR field on the declared radial product with a shared active clock; a quartic projection with interacting Gibbs readouts and a fixed-band occupation limit.	The physical scalar action, quantization and vacuum; inertia; a calibrated clock; a spatially refining interacting family.
Modular clocks [9]	Finite modular flow of actual observation states; $t = -\beta s$ for a fixed reference; a source-prepared finite reference with its positive-log clock; an active native-transfer clock whose quadratures four movement-bit covariances predict [20, 21]. (D) Faithful encoded product-field states sharing the active clock.	A physical reference and its complete modular dynamics; Bisognano-Wichmann identification; common null translations; clock calibration; the relation to causal depth.
Gravity [9]	A rank-three response and axial circulation from committed records; native causal cones; the common spherical continuum with density and incident-clock limits [12]; exclusion of a bounded energy identification with a smooth compact three-dimensional Dirichlet bulk. (D) A measurable Lorentzian density of the product wave with its recovered angular tensor; the exact measure fails C^1; radial-average observations do not select the radial architecture.	Physical region, metric and area attachment; geometric meaning of counting-entropy stationarity; small-ball geometry; conserved stress; universal coupling and normalization; a responsive metric.

Four retained audit controls fix exact boundaries of the construction [48]. Each rejects a named identification or an overly broad inference; none rejects the theorem premises.

Complete-carrier center. The centers of M_{66} and M_{78} consist of scalars, so the unextended working algebra cannot supply twelve nonzero primitive central ports. Binary occupations are noncentral, overlap and sum to $2I$, and total-two occupations also admit the eigenvalue two. The native matrix algebra is kept as a sector; a separate twelve-block central-address extension supplies a central algebra with port-controlled operations, and occupation measurement is a separate contract.

Marked triplets. The slow Gram matrix satisfies $G^2 = 4G$, $\operatorname{Tr} G = 12$ and $(5I - A)G = (5-\sqrt{5})G$. One fixed order-five symmetry has trace $(1-\sqrt{5})/2$ on the slow frame and $(1+\sqrt{5})/2$ on the lower response triplet, which excludes an invertible intertwiner for the same action, so spatial and internal response domains are kept distinct. The comparison separates these two specified triplets only; it neither shows every internal triplet inequivalent to the frame nor identifies carrier rechartings with physical rotations. The local rank-three carrier band and the refining global low band are distinct. The conditional Spin–matter construction has independent commuting factors, and its locked diagonal action fails to give a uniform fermionic central sign. A physical separation

requires the spatial and internal actions to be identified on the same clocked interacting response and preserved by its dynamics.

Feedback class. With the two endpoint loads in the uniform mixture of their one-particle configurations, a rational number-preserving exchange gives endpoint probabilities 769/1250 and 481/1250 for the two private preparations, a difference of 144/625. Four Kraus operators certify that it is CPTP, and its private commutant has dimension one, which excludes every single-product exponential representation, alternate logarithm branches included. The no-feedback theorems for reduced loads and single products hold within their stated classes, and this channel marks the class boundary.

Configuration equilibrium. For $A = I + L_{\mathrm{path3}}$ the controls $\varepsilon = \pm 1/16$ keep the normalized configuration Gibbs marginal and change the full reference and the vacuum. For a stiffness response $Y = \gamma(A + \varepsilon A^2)$ one has $Y_{13} = \gamma\varepsilon$ and a third-mode fit residual $6\gamma\varepsilon$, while the actual acceleration is $-Yq$. The theorem identifying inertia with the same capacity keeps that premise, and equal equilibrium configurations do not select the kinetic law.

11 Ordered joins, acceptance conditions and stopping tests

The completion target is one native repair and memory process whose observers' compatible records and interventions admit the observed 3+1-dimensional quantum-field description, with its matter and interactions, and a common classical-gravity limit, each as a controlled effective description in its regime of validity [31, 48]. Observer clocks, the defect and maintenance-energy mechanism, the interacting field and particle description and the realized Einstein premises must come from that same process. Component benchmarks keep their own contracts and are not pooled into that realization. Hardware, protocol and implementation choices are design variables, and no uniqueness theorem for a chosen rule is a prerequisite; closure-derived P and N are accepted inputs that discharge no other interface.

Every comparison of observable laws uses one complete-output criterion. For every protocol p of a declared menu,

$$\sup_{p} \tfrac{1}{2} \left\| D_{\mathrm{native}}(\rho_{\mathrm{native}}(p)) - D_{\mathrm{eff}}(\rho_{\mathrm{eff}}(p)) \right\|_1 \le \epsilon,$$

with failures, unresolved waits and admitted references retained in both outputs and decoders that keep the claimed observations. An acceptance names the protocol menu, decoders, refinement family and error budget before the comparison [48]. Exact equality of limiting observable laws is the mathematical target; finite simulations and experiments are assessed with declared approximation and measurement errors, and agreement must be constrained by measurement, since a consistent world model can describe physics different from that of nature.

Finite constructive joins. A separately versioned campaign applies existing finite theorems to four executed component joins [33]. Its clock inventory covers all 66 transported two-port cores at the declared modular parameter, with supplied coherent services and complete failure accounting. A joint instrument advances the transported modular detector and a protected field walk on the same native incident history; a control preserving both separate marginals changes their joint law. This operational comparison does not require equal generators. The positive native-return wave reference remains available even though the tested three-charge reference fails direct substitution for it.

On the binary two-load configuration complex, 300 movement edges and 495 declared triangle/square cells give integral first homology $\mathbb{Z}/2$. Actual C/E records realize an odd exchange loop, but its native branch amplitude is positive: a history class alone supplies no fermionic phase or protected particle. Complete native currents also give the positive form $D(v) = v^T Lv/396$. In the existing declared product-field symbol, rest gap and independent radial curvature agree in the ideal limit; the executed split words have rest/inertia ratio $\chi \cot \chi$ with a controlled subdivision error. This does not assign mass to the history class. Reference-like localization, the physical momentum and clock maps, and one defect action supplying rest energy, inertia and stress remain open. These finite joins add no physical comparison or integrated acceptance.

A compiled Ramsey source service. On the finite density representation of the existing native coherent walk, one additional Ramsey arm gives a 16-qubit controlled source program. Its walk uses 123,646 source pulses; the full-register reversible shift is checked on all 65,536 basis addresses. The 720-arc mean-zero density sector has an exact isometry into the original 3,960-arc binary-source walk, but the full binary-arc unitary is not compiled. Source arrays execute the Ramsey preparation, controlled walk and terminal analysis, and complete two-attempt C/E joint laws retain clock-export aborts, both field outcomes and untouched references. One selected four-attempt continuation keeps the wave register and three FIFO exports without fresh field preparation. It is a selected unnormalized history, not an exhaustive four-attempt distribution. Numerical routine guards and counted proper-code loads/exports are separate from physical apparatus accuracy and calibrated seconds [36].

A bounded repair/reference alternative. A separately frozen finite family replaces the fair placement coin on a seam (a, b) by probability $\theta_a/(\theta_a + \theta_b)$ of placing the one endpoint load at a, for declared positive port weights θ and a fixed source-independent seam selector. Complete C and E record instruments then fix the faithful binary-shell product-pair reference $\pi(x) \propto \prod_i \theta_i^{x_i}$ and covary with its modular action. This is a changed local law, with the inherited native branch support; its full branch algebra has no nontrivial reducing source charge. The earlier transported three-charge reference cannot be made reversible by this endpoint-only rule: its per-configuration weights obey $w_1^2/(w_0 w_2) = 125/64$, whereas this product family requires one [34]. Neither its weights nor a physical clock are selected by these finite statements.

Reference balance and relational charge. For the unchanged uniform-seam C/E law, the complete binary-shell population kernel is symmetric on 300 movement edges and has the unique uniform reversible measure. Holding the transported clock reference fixed produces a one-attempt total-variation drift of 17/750 or 17/600 for the two tested cores. A recorded two-port core that follows every nonzero current instead gives a reversible joint source/core measure on 66^2 states. Its overlap charge $q = |x \cap c|$ is preserved by every complete record block; the three sectors are connected separately and have stationary weights 2/5, 5/9, 2/45. The charge's native gradient form is zero. These are source-plus-controller sectors, not a localized intrinsic particle or an energy splitting. The reduced diagonal distribution converges to the joint stationary measure from the stated fixed-core preparation under independent uniform seams; adaptive feedback and the full archived record are outside that stationarity result [33].

Sharp sectors and active memory. An exact test of the same binary source/core charge finds that every state in a sharp $q = 0, 1, 2$ sector differs from the declared transported reference on the full ten-port exterior by trace distance at least 3/5, 4/9, 43/45, respectively; conditioned references

attain these bounds. For an uninterrupted complete native C/E history, every positive closed odd exchange flag has a rank-one source input domain. The result prevents that flag from carrying an arbitrary unknown coherent qubit through a final decoder, but intermediate active recovery is a different operation grammar. The existing source-executed $q = 1$ code attains the all-seam exact-correction dimension bound two. Its recovery leaves a moved core with $q = 1$ probability $7/8$ unless an explicit classical core reset is appended. The code has native movement probability $1/6$ per uniform-seam attempt, an excess of $1/66$ over the uniform-source baseline, while coherent recovery work is outside that statistic. Neither the code nor this activity identifies a localized reference-preserving particle or its rest energy [35].

A joint readout of the incident count. The compiled modular detector and density-wave Ramsey service also support an operational comparison when the native tape is hidden from the readout. Conditional on a successful preparation and terminal read in the declared two-attempt policy, let $N \in \{1, 2\}$ count port-0 incidents. Rational enclosures give a reduction in optimal squared-error risk from the joint detector/field output, relative to the detector alone, exceeding 0.0027325 for C and 0.0027329 for E. Independent resampling of the field bit with both separate marginals preserved removes the gain. The finite preparation and complete source-abort allowances retain a gain above 0.00272 under the declared numerical routine guards. This is inference of one dimensionless shared count. The stochastic readout does not instantiate a strictly monotone reading, a two-event affine calibration or its distinct third-event check from the finite clock interfaces, and supplies no calibration in seconds [37].

Optimal small-cap screening for the corrected qubit. For every exterior cap size $1 \le k \le 10$, hypergeometric averaging gives the minimum over sharp-q states of the worst trace distance over all k-port exterior caps to the transported reference. The conditional reference attains the bound. At $q = 1$, the unrestricted one-port value is $8/225$. Exact all-seam correction imposes a stronger restriction: the finite flagged-code theorem forces nonedge source support and scalar port-occupation compressions. An exterior port adjacent to either core endpoint can then carry no code occupation. Edge cores have four safe exterior ports, distance-two cores have two, and opposite cores have none. A crossed four-port qubit attains worst one-port distance $61/450$, optimal among exact all-seam sharp-$q = 1$ qubits on an edge core and improved from the earlier code's $82/225$. All complete raw C/E branches satisfy its correction conditions; the compiled recovery described below supplies its source pulse service. Its full exterior differs by at least $4/9$. Separately, a sharp-$q = 2$ pair starting on any native edge exits that edge set with hazard $1/15$ per uniform-seam attempt, excluding an invariant edge bond in this finite candidate [38].

Same-code activity and unselected sector action. The improved code and its predecessor both have native moving-current probability $1/6$ per uniform-seam attempt. The stationary co-moving $q = 1$ ensemble has rate $5/33$, but averages over core labels, so their difference $1/66$ is a specified diagnostic comparison. On the joint source/core graph, each connected q sector has a uniform stationary measure. Any positive reweighting of those three sectors remains stationary for the same reduced diagonal source/core population law under uniform seams. Doubling the $q = 1$ weight leaves its conditional record law unchanged while shifting a separately declared Gibbs modular contrast by $-\log 2$. Thus native stationarity and the current form do not select the rest gap. The earlier code's mean repair plus recovery costs, 439409 and 863483/2 source pulses for C/E, belong to the distinct fixed-seam $(0, 5)$ protocol; they are not a measured uniform-seam energy. The finite action-measure uniqueness theorem applies within each connected sector, and

the proper-time/inertia and Ward theorems require the same candidate's action, kinetic and stress maps [39].

Compiled screened recovery and the same-code transport test. The screened encoder and all 208 positive complete C/E recovery branches have a source execution on the 512-row work register. Their ideal identity $R_h K_h V = \sqrt{p_h} V$ preserves arbitrary logical/reference states through any finite corrected history, with immediate supplied recovery and an explicit archived core-label reset. Initial encoding uses 65,484 source pulses. Uniform-seam mean native-plus-recovery costs are 37499681/60 for C and 36870437/60 for E per attempt; retained records use 23 or 24 classical bits. Numerical source composition has a stated 100-attempt bound under its routine guards, separate from exact ideal composition and physical apparatus error [40].

On this same code, the native current second moment is a rank-eleven weighted port Laplacian with trace 1/3. For the earlier source eigenprofile, its quadratic form is 13/120, while the two logical basis states have variances 1/2 and 5/2. The ratios 13/60 and 13/300 therefore do not supply one wave rate for this code. Summing every corrected branch gives identity logical transfer, with four unit superoperator eigenvalues, and the core resets to $(0, 1)$ at each attempt boundary. Transient native current is present, but continuing coherent dispersion and a rest gap are not identified. A common global phase is invisible to the channel and is not a measured zero rest energy [41]. The finite conserved-sector theorem additionally proves stationary reweighting, conditional cancellation and zero sector-potential gradient for every positive sector factor. Exact enumeration binds its premises to the 4,356 native source/core states; the general implication is checked in Lean, while the actual finite kernel is checked in Python [42].

Four shared clock events. A continuation compiles four transported-core modular words at parameter one and source pair preparation/analyzers. Four terminal horizons on copies of the same C/E protocol retain all stopping outcomes and source aborts, with up to three FIFO exports. Conditional clock and Ramsey ensemble means are strictly increasing at the four shared checkpoint types. A raw affine fit on events one and two fails the reserved fourth event by approximately 0.0105927 in field probability; the source-error allowance leaves a positive cross-product margin. These are ensemble readings with nonlinear detector response. Model-conditioned response inversion recovers a common count but does not establish a physical clock rate, individual-shot monotone reading or timelike atlas [43].

Conditional SI attachment. The unit utility takes an explicit T seconds per model-clock unit and signal speed v in model lengths per clock unit. Identifying that signal with c fixes $L = cT/v$. If the same candidate supplies a coherent rest angular frequency ω_0 and the physical identification $E = \hbar\omega$, its conditional rest mass is $m = \hbar\omega_0/(Tc^2)$. The SI defining values of c and h do not supply T or ω_0 [44]. The independent small-momentum check is $\omega_0\omega''(0)/v^2 = 1$ for a massive relativistic dispersion. The corrected screened code supplies no such dispersion or rest gap, so no particle-mass or GeV prediction is assigned by this conversion.

11.1 The joins in dependency order

The joins below are ordered by acceptance dependency [32]. Each states what must be shown on the same process, the results of this paper it starts from, and the test that would stop it. A theorem premise assigned to a join must be proved on the actual model and retained domain, with its state, algebra, operation, clock and refinement maps bound in the premise map of Section 10, before the theorem's conclusion is used.

The admissible class. A versioned class of carrier and refinement, repair and memory laws, record instruments, resources, preparations, clocks and interpretation maps is fixed before validation, together with the observable domains, Standard Model, neutrino and gravity targets, allowed calibrations, input ancestry, uncertainties, budgets and decision thresholds. It distinguishes the fixed C/E law, the permitted quantum record implementations of Theorem 2.3, and any bounded alternative-law family. *Stopping test:* if the declared class is empty or jointly inconsistent, that exclusion is proved. An unbounded unspecified escape class leaves a whole-program exclusion unarmed, and an unfinished freeze is neither confirmation nor falsification.

One continuing native observer. One replayable native repair and memory process prepares, acts, reads, retains records and continues on overlapping accessible regions, with coherent resources, ancillas, initialization, read and write, retention, feedback, timeout and recovery services each constructed or counted, and complete classical and quantum outputs including untouched references, failures and continuation. It starts from the finite observers of Sections 3 and 6, the compiled native attachment and active encoded memory of Section 6.5, and the overlapping observers and matched screens of Section 4; several of these run on separate schedules with separate error budgets, and their joins need their own implementation and error analysis. In parallel the theorem library is mapped onto the model: every theorem the chain needs receives a native interface and an owning join, and results that no join needs leave the active chain with their provenance. *Stopping test:* a source-access, coherence, retention or resource impossibility rejects the tested implementation class. The coherence floor of Section 2.5 is such a result for diagonal native resources: it makes coherent services a necessary resource of that class.

Continuing regime, faithful reference and local modular clocks. The same process prepares a reference with support and faithfulness on the accessible observer algebras, keeps a continuing regime, executes the modular actions and clock readouts, and derives compatible local clock comparisons and the common normalization used by energy and propagation. Passive repair, dimensionless modular flow, event counts and physical duration stay distinct, and tracial and compressed-algebra controls are retained. It starts from the fixed-reference relation and modular representations of Section 6.6, the finite reference and positive-log clock of Section 3.7, the event clocks of Sections 3.8 and 4.2, and the incident clocks and faithful encoded-field states of Sections 7.4 to 7.6. Two retained limits bound it: passive repair cannot act as a nontrivial modular automorphism of an invariant observer algebra, and the uncapped native-return reference tends to the trace. *Stopping test:* a proof that the frozen preparation and access grammar forces trivial modular action, incompatible regional flows or no admissible physical clock map; a failed chosen reference alone is a candidate obstruction.

A localized native defect and its protecting invariant. An actual native state or observable sector has a localized core, a reference-like exterior and a boundary charge invariant under the full admitted repair law, defined independently of a target mass, with autonomous protection or a controlled lifetime, every seam and branch retained, and quantum and record sectors checked separately. *Stopping test:* an exhaustive invariant and localization no-go over every sector and allowed refinement of the frozen defect class stops this mechanism. The classical and collision exclusions of Section 11.2 do not exhaust arbitrary quantum or history sectors.

Maintenance energy, kinetics and conserved stress. One intrinsic action and energy, and one kinetic response, are derived for the same defect, with reference subtraction, clock normalization

and an independently justified stress and Ward identity. Action, energy, entropy production, event counts and host work stay distinct; moving and rest responses share one signal speed with no species-dependent conversion, and interaction energy and reference or cutoff dependence are accounted. *Stopping test:* a proved inability to define a common positive energy, kinetic and stress dictionary in the frozen class, or a mandatory incompatibility between maintenance and propagation. A scalar code cost or a failed ansatz has narrower scope.

Transport, particle identity and currents. The same excitations carry charges, spin and statistics, family structure, gauge-current couplings and allowed creation, fusion, annihilation and decay, with transport and charge conservation shown on the native process, leakage and recovery bounded, extra modes and chirality retained, and stable defects, metastable excitations and massless collective modes distinguished. It starts from the charged realization of Section 3.4 and the flux-dressed code of Section 6.1. *Stopping test:* incompatible required charge, spin, statistics, current or transport structures exclude the covered defect realization.

An interacting chiral 3+1 field description. A refining same-process observation law has a nontrivial interacting chiral 3+1-dimensional effective description, compared through the complete-output criterion with locality, quantum correlations, positivity and controlled errors. Every taste, chirality, dark mode and residue is retained until a derived decoupling map removes it, kinetic, Higgs and current responses are obtained jointly, and the photon-containing theory stays gapless, with massive thresholds, confinement observables and any pure Yang–Mills gap stated separately. It starts from the occupation limits of Section 6, the continuum operator and its density, clock and wave limits of Section 7, the declared radial product with its smoothing theorem and the fixed-band quartic projection. *Stopping test:* a covered chirality, locality, positivity, continuum or interaction no-go stops the frozen realization class. A supplied scalar product, finite free symbol or pole list does not pass this join.

The physical dictionary and the matched spectrum. A source-selected spectrum, mixing and interaction packet in compatible physical conventions uses the same defects, energy, kinetics and clocks. The charged benchmark of nine running Yukawa couplings and four Cabibbo–Kobayashi–Maskawa parameters is derived together, with source and decoder revisions, empirical ancestry, matching scheme and scale, inputs, covariance and theory uncertainty bound before the comparison; neutrino, unstable and confined sectors keep their own obligations. *Stopping test:* a wrong unavoidable matched spectrum with an independently justified error budget rejects the fixed candidate, and the class only if every permitted nuisance and source choice is excluded. Calibration on known data is not independent validation.

Responsive geometry and causal clocks. A source-derived event, causal and geometry dictionary has an effective Lorentzian domain, compatible observer directions, local clock comparisons and count, volume and length maps with controlled errors, and responds to admitted matter changes. It starts from the rank-three response and causal cones of Section 7.1 and the common matter metric of Section 8, and it keeps the finite rank-three carrier geometry distinct from physical 3+1 event geometry. *Stopping test:* a mandatory incompatibility between source geometry, clocks and nonzero matter stress excludes the covered common realization. The static radial-product failure is a scoped control, not a universal gravity no-go.

Realized Einstein premises. A premise-by-premise same-tower witness supplies geometrically normalized local modular flow whose directional charges define the matter stress used in propagation, its Ward identity, the edge-center entropy split and area normalization, the modular first law, generalized-entropy stationarity at fixed proper volume with controlled small-ball remainders and enough null directions, universal coupling and one positive Newton constant [9]. A stationary Markov law, a Gibbs reference or a stationary conditional record entropy is not this variation; the counting-entropy and boundary-proxy controls of Section 10 and the cavity comparison of Section 8 are retained. *Stopping test:* a proof that a necessary entropy sign, modular or stress normalization, or stationarity variation is incompatible across every allowed same-process realization. An unmet premise is an open join, not a theorem failure.

The composed gravitational response. The conditional Einstein theorem is applied with a checked map for every premise, giving the Einstein equation up to its cosmological term with one positive Newton constant and universal coupling of the same defect energy and stress, and finite and refinement errors are propagated into gravitational observations. A quantum spin-two graviton lies outside the classical theorem. *Stopping test:* a counterexample satisfying every theorem premise refutes the implication; a necessary failure of joint gravitational response excludes the frozen program only with coverage of the whole admissible class.

Joint independent tests. One frozen physical dictionary is compared with independent field, particle and gravity observations through complete joint and conditional laws and intervention responses, with predeclared rejection rules, eligible data, nuisance ranges, covariance and theory errors. It starts from the prospectively fixed count protocol, the sequential design and the retrospective replays of Section 9; the phase test needs a realized preparation and observable and an independently established service budget, and it measures where the realized record law lies in the classified family. *Stopping test:* the fixed formulation is rejected when an unavoidable prediction fails a valid frozen comparison; a whole-class empirical exclusion requires every admitted realization and nuisance choice to fail.

Across all joins an obstruction review records, for every exclusion, the target version, mandatory premises, quantified class, evidence, independently checked coverage, surviving escape routes and consequence. A mathematical error, a candidate failure, a class no-go, an empirical rejection, an inconclusive search and a budget stop are kept distinct. Whole-program falsification requires a verified contradiction in indispensable premises, a proof excluding every frozen realization of a required join, or an unavoidable independently excluded prediction; a finite unsuccessful search, an unconstructed map or an exhausted budget proves no impossibility.

11.2 Negative controls on the tested law

Four results reject specified mechanisms of the tested law. Each is a fact about that law and its stated class.

The native double-load defect is transient. On the twelve-port carrier, the only configuration functions invariant under every admitted seam and coin are constants, and the binary shell is the unique closed sector. A collision state carries one double load above that shell. Under the uniform seam probe, the complete C and E instruments both satisfy $\sum_h K_h^\dagger P_{\mathrm{coll}} K_h = \frac{5}{6} P_{\mathrm{coll}}$ for the projector onto the twelve collisions, so after n attempts the collision probability is $(5/6)^n \operatorname{Tr}[(P_{\mathrm{coll}} \otimes I)\rho]$ for arbitrary quantum inputs and untouched references. The mean number of attempts to split is six, and the double load never hops before splitting. Load differences telescope around all twenty faces

for all 78 states, so this gradient map carries no holonomy. Holding one port fixed creates sectors only by disabling its five repairs, which changes the law. The record costs of the corrected memory code have expectations independent of the logical state, so they cannot split logical-state masses [45]. No persistent quantum subspace supported on the collisions exists under this evolution; other quantum and record-encoded sectors are untested.

The static radial product fails null balance. For the smooth static product metrics of the conditional scalar description, both radial null Einstein contractions vanish, while a coherent scalar excitation in the restored angular-constant Dirichlet sector has positive relative null stress. That geometry cannot satisfy the Einstein balances for the reference and the excitation at positive coupling with other sources fixed (Proposition 8.1). A responsive geometry is required.

Passive repair has no wave sector. The record-discarded native channels are Hilbert–Schmidt self-adjoint with spectrum in $[0, 1]$ and scalar fixed algebra (Theorem 3.4). The executed quantum channel strictly attenuates a coherent mode, which rules out unitary conjugation on this carrier, and the half-step map of its principal logarithm is not positive, so it generates no completely positive semigroup [11]. The closed mean law is damped (Proposition 7.3(iv)), ballistic rescaling freezes the density paths, and the native decay rates grow like Laplacian eigenvalues, so no common clock factor gives a wave dispersion (Section 7.3). Every wave in this paper is declared record processing or a compiled apparatus step on supplied codes; other dilations or generators are not excluded.

The tested mass laws miss the charged hierarchy. The declared linear Weyl–Higgs candidate gives charged-lepton coefficient ratios $1 : 1.1903 : 1.4982$ against the observed $1 : 206.77 : 3477.23$, and the whole complex affine class bounds the largest-to-smallest mass ratio by 543.55 against an observed lower bound of 3470 [46, 47]. These are conditional diagnostics of unmatched coefficients under a direct pole-ratio identification; no maintenance-energy map underlies them.

12 Reproducibility

Every executable result has a replayable receipt. Package manifests bind source, document and receipt artifacts by SHA-256. The repository's verification entry point recomputes these hashes, replays retained receipts and compares canonical bytes. Exact certificates must match exactly, and floating diagnostics carry declared replay tolerances. Independent test suites check the executable statements against separately written oracles, including adversarial mutations. Analytic limit proofs state their domains, normalizations and order of limits; a finite receipt does not replace those proofs. Model sources and the premise crosswalk are available at the pinned revisions cited with their results [12, 52]; the flagship synthesis and basis document are cited on the main branch [3, 4].

Statements and declarations

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Data availability. The model documents, executable sources, retained receipts and claim records supporting this study are available in the public OPH repository at the baseline revision [2] and the revisions cited with individual results.

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Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- Aharonov_Bohm_loop_phase [108].
- analytic_vector_self_adjointness_and_core_strong_resolvent_convergence [173, 182].
- binary_Kraus_Stinespring_instrument [120, 189, 177].
- Bose_Hubbard_lattice_action [155].
- canonical_lattice_scalar_quantum_simulation [157].
- Caputo_Liggett_Richthammer_exclusion_spectral_gap [111].
- classical_CHSH_local_response_bound_and_published_Bell_null_contract [138, 29].
- classical_controlled_quantum_channels [199].
- classical_Weyl_eigenvalue_counting_law [183].
- closed_quadratic_form_Galerkin_and_Gibbs_trace_convergence [188].
- coherent_state_evolution_under_positive_quadratic_action [145, 139].
- coisometry_involution_quantum_walk_spectral_mapping [193, 151].
- conditional_Hoeffding_Azuma_concentration_for_randomized_Gram_witnesses [117, 131].
- conditional_law_sequential_coupling [165].
- conditional_probability_and_Brier_decomposition [133, 172].
- convex_Schrodinger_spectral_gap_comparison [128].
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- finite_Gibbs_reference_modular_Hamiltonian_identity [150, 110].
- finite_Kraus_Stinespring_construction [189, 163, 136].
- finite_Lindblad_master_equation [147, 168].
- finite_matrix_block_elimination_and_stopped_instrument_resolvents [203, 175, 142].
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